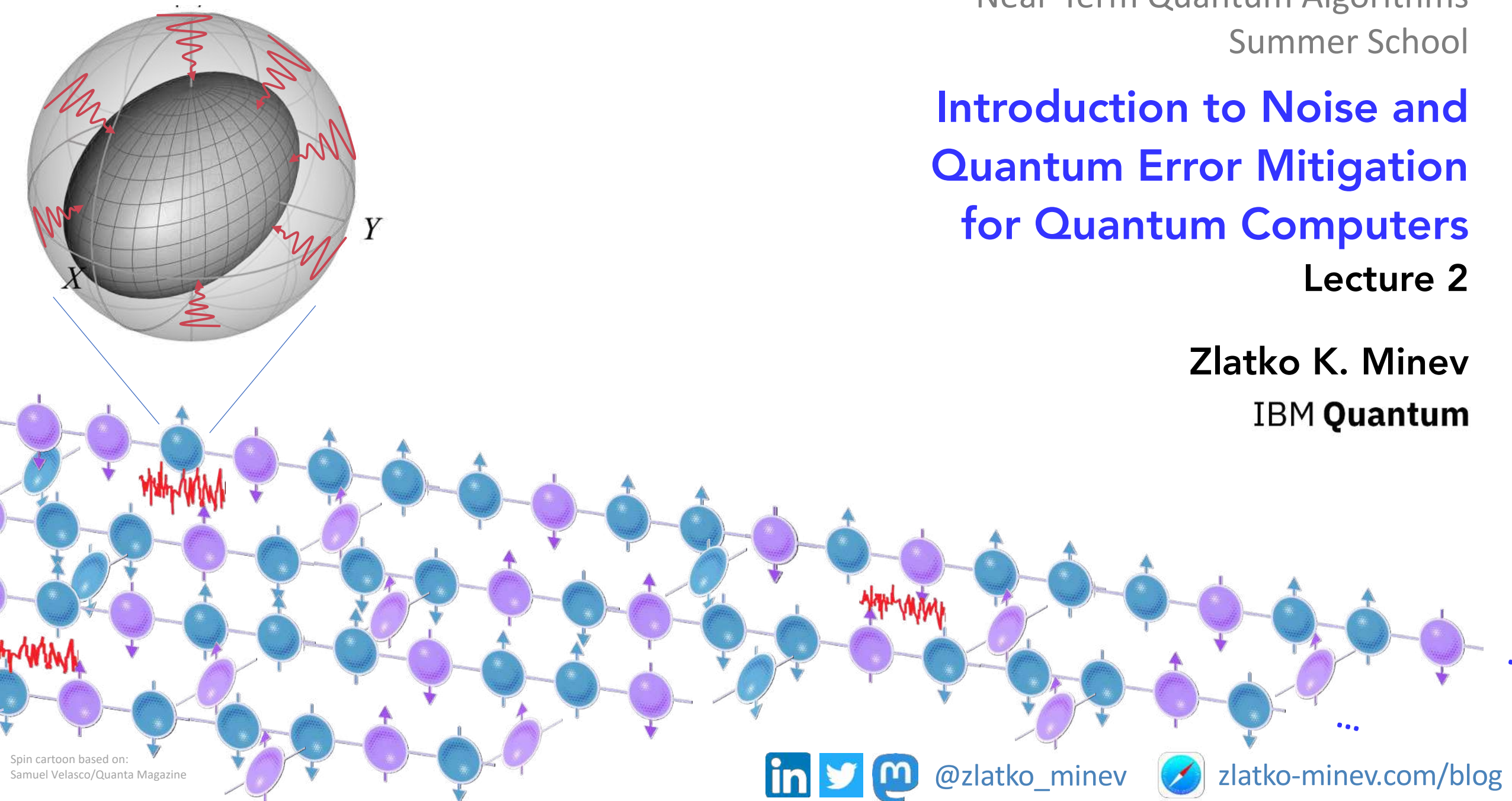


Introduction to Noise and Quantum Error Mitigation for Quantum Computers

Lecture 2

Zlatko K. Minev
IBM Quantum



Spin cartoon based on:
Samuel Velasco/Quanta Magazine



@zlatko_minev



zlatko-minev.com/blog

What is one thing you learned in Lecture 1?



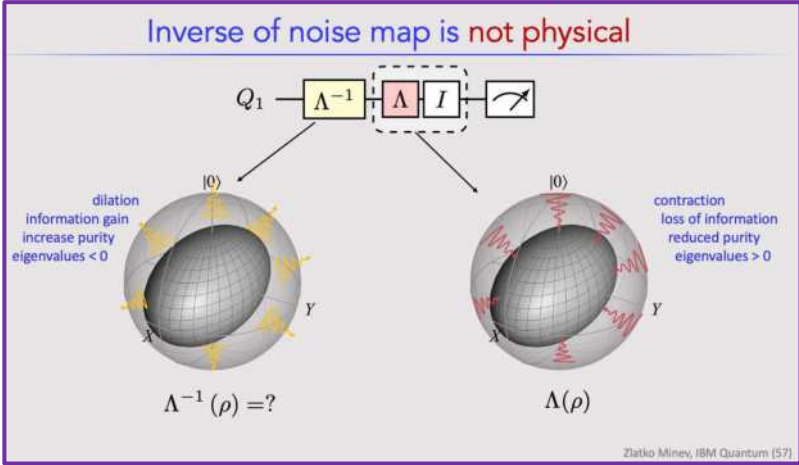
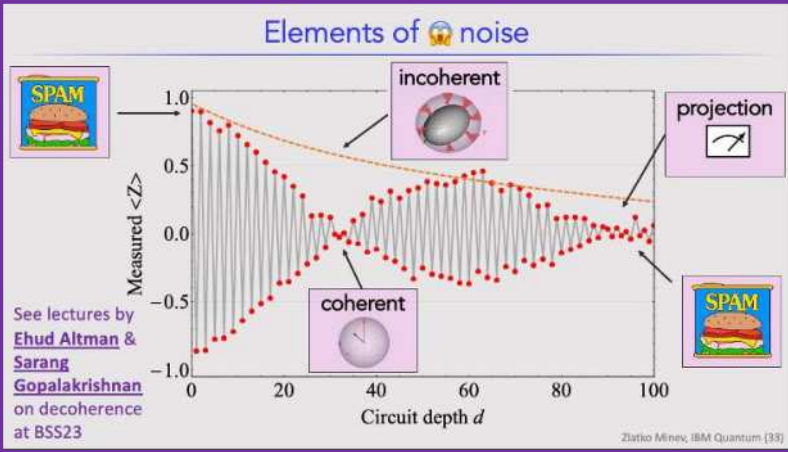
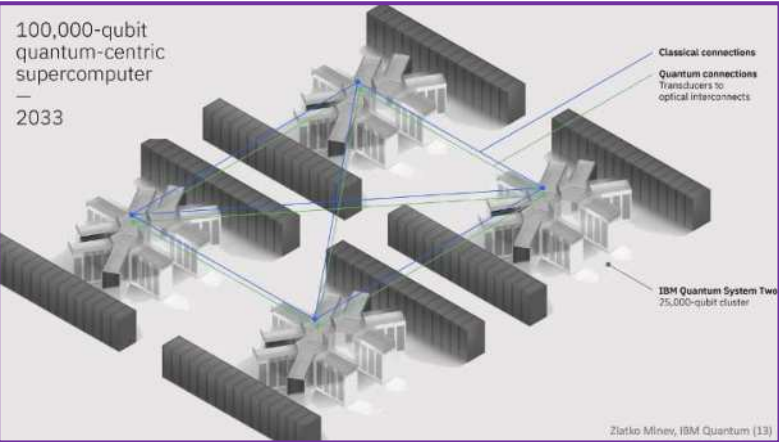
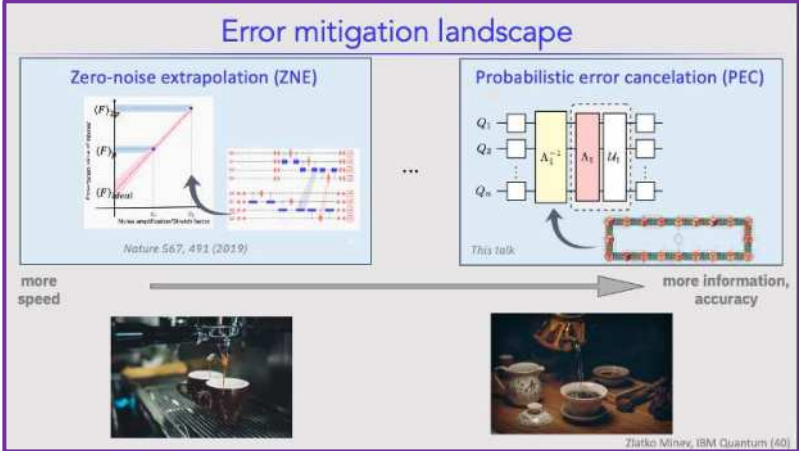
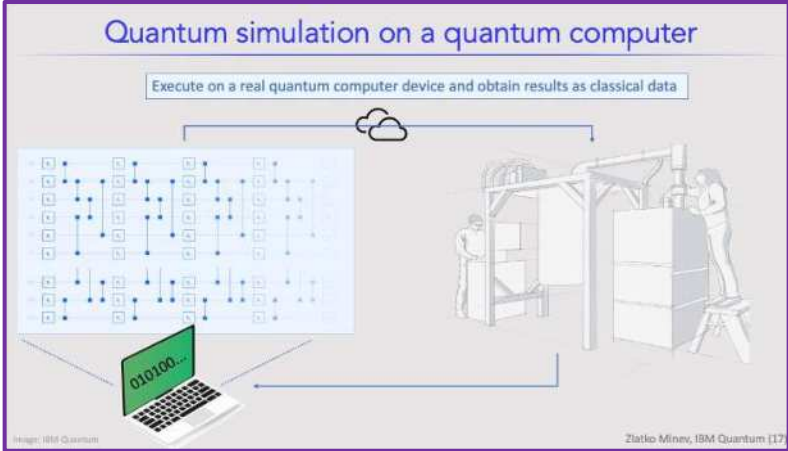
Review of Lecture 1



Quantum computers

Biggest Problem: Noise

Mitigation



Quantum Error Mitigation

Lecture 2

Mitigation fundamentals

Probabilistic error cancelation (PEC)

- Summarize one qubit example
- Analogy to random walks
- Error bars & confidence

Learning quantum noise

- Challenge
- Overcoming
- Sparse model

Putting it together

- Experiments – Ising model
- Consequences for the big picture

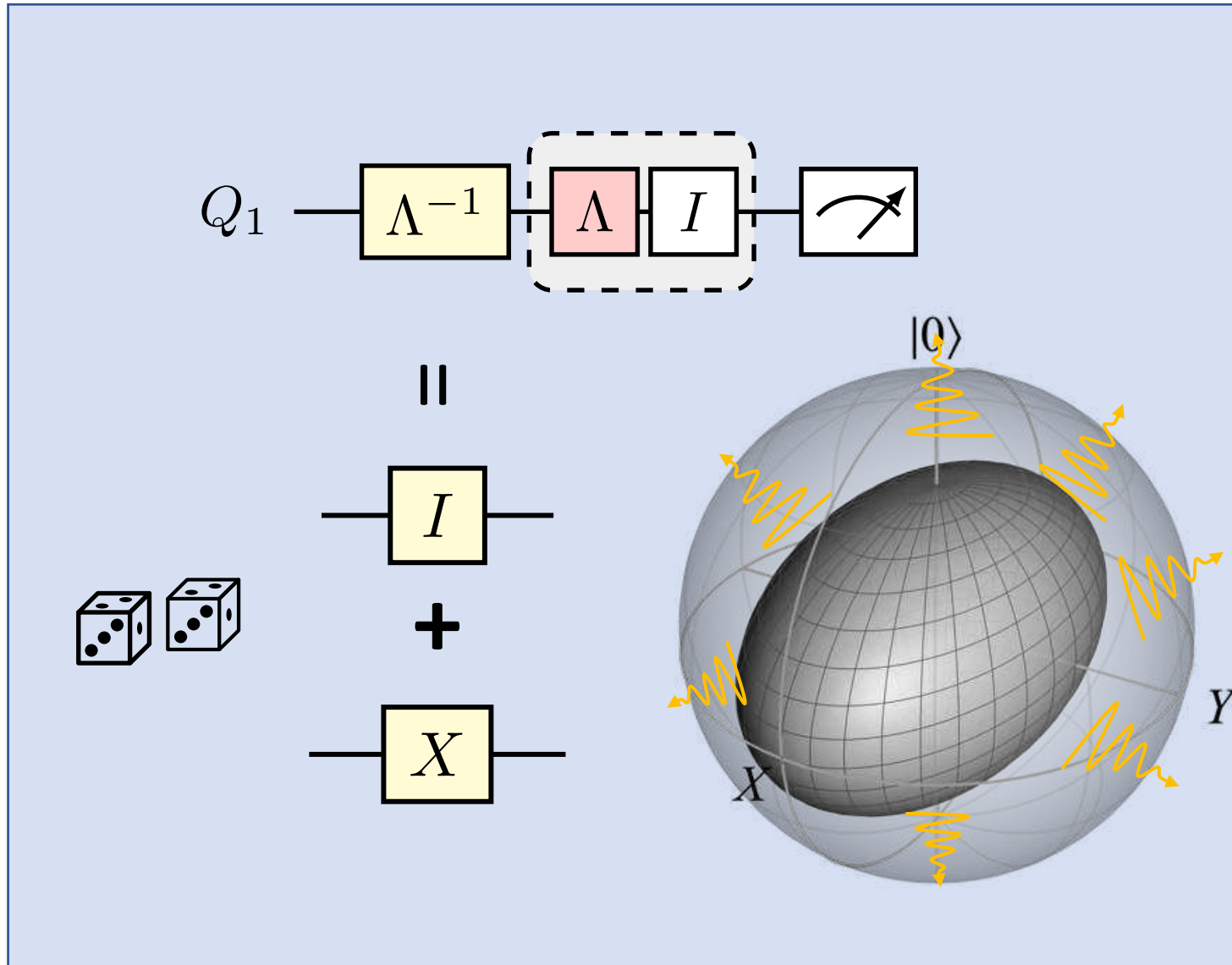
Appendix:

- Generalized
- Show unbiased estimator

Hardware progress

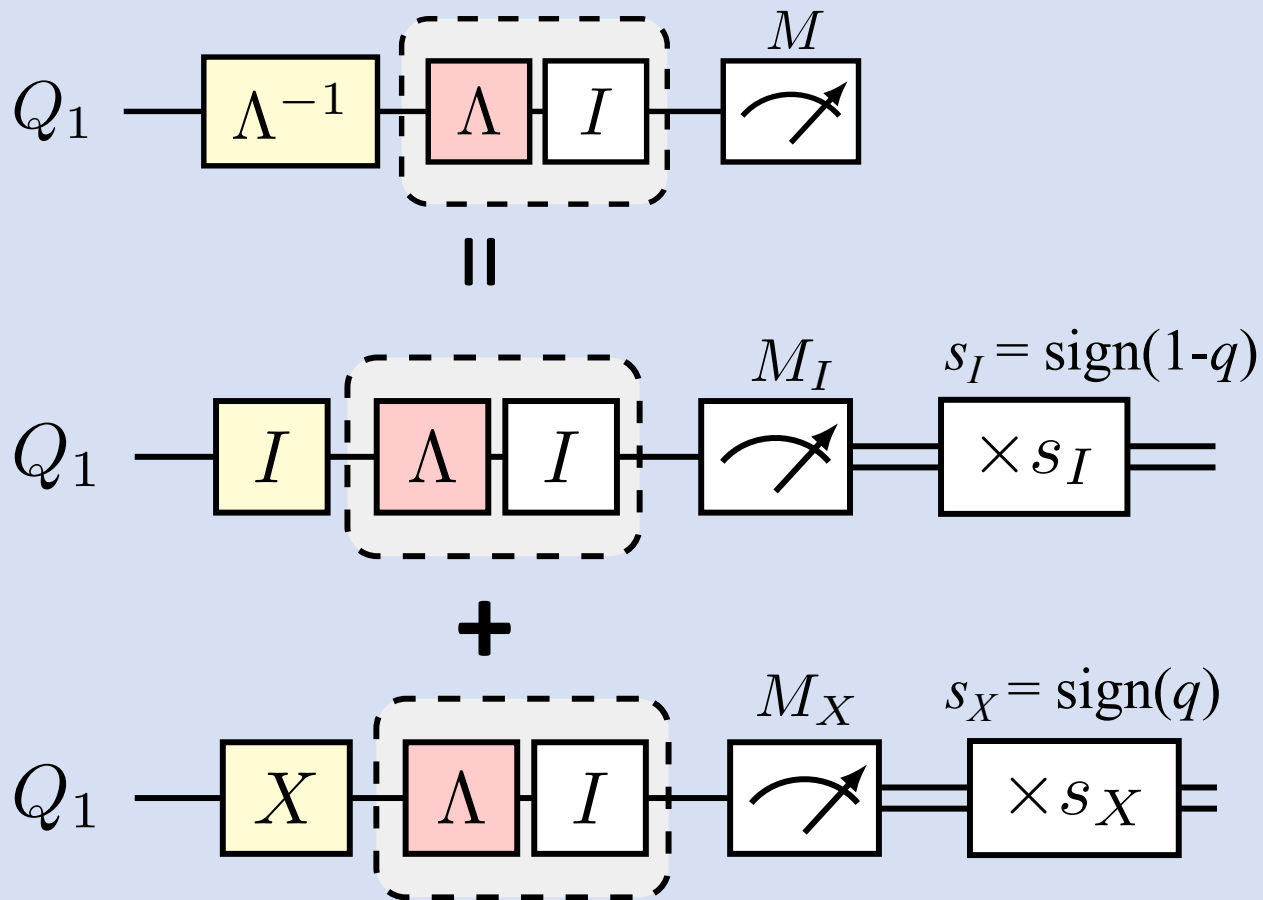


Canceling noise with noise



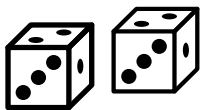
How to implement?

random circuits
(quantum trajectories)



probability

$$P_I = |1-q|/\gamma$$

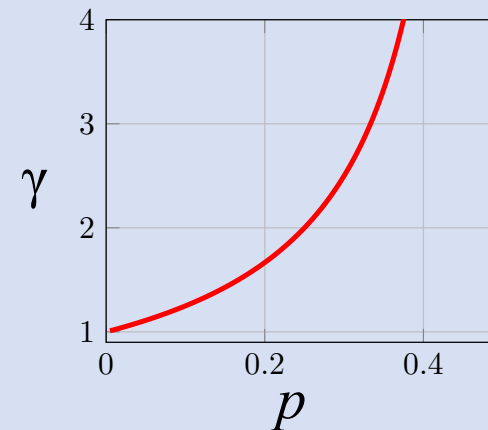


$$P_X = |q|/\gamma$$

$$P_I + P_X = 1$$

sampling overhead

$$\gamma = |1-q| + |q|$$



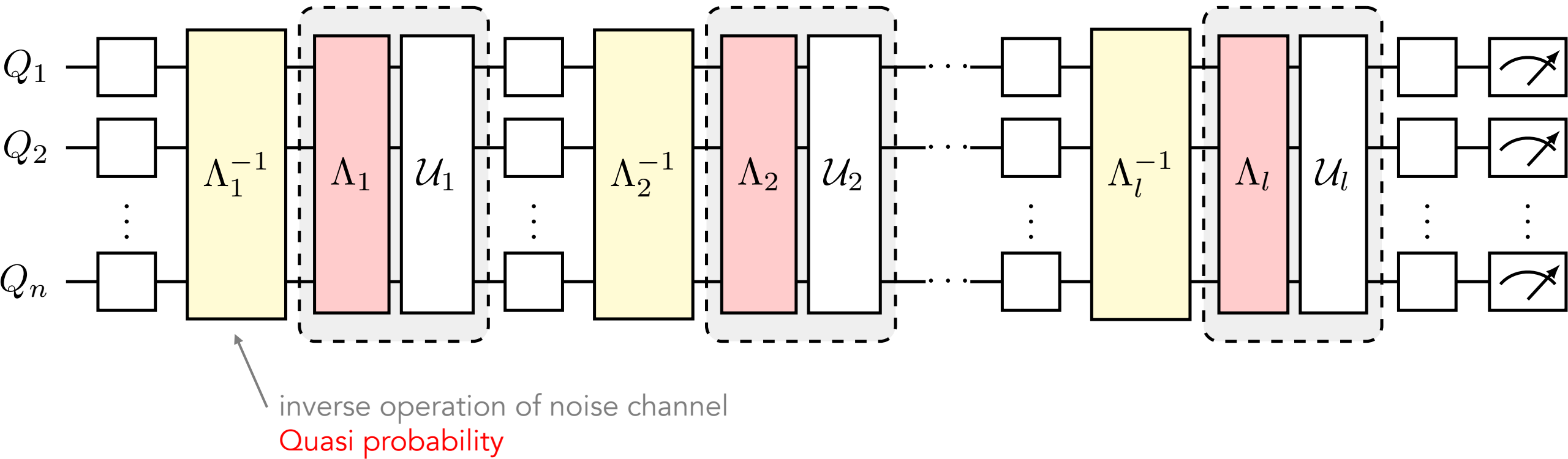
mitigated expectation

$$\langle M \rangle = \gamma (s_I P_I M_I + s_X P_X M_X)$$

Gain: Bias-free estimate!

Cost: Variance

More qubits, gates, and channels



Generalizing: Raveling trajectories with quasiprobabilities

Channel we *want*
to implement

CPTP operation we
can implement

$$\mathcal{C}(\cdot) = \sum_i a_i \mathcal{F}_i(\cdot)$$

Real coefficients, turn
into *quasi-probability*

Putting the following techniques all on the same footing

Technique

Prob. error cancelation (PEC)

Circuit cutting (knitting) of gates

Circuit cutting of wires

Classical sim. algorithms (QP)

Channel $\mathcal{C}$

noise inverse

non-local gate

large unitary

unitary

Deep dive:

Probabilistic error cancellation (PEC)

To learn and cancel quantum noise

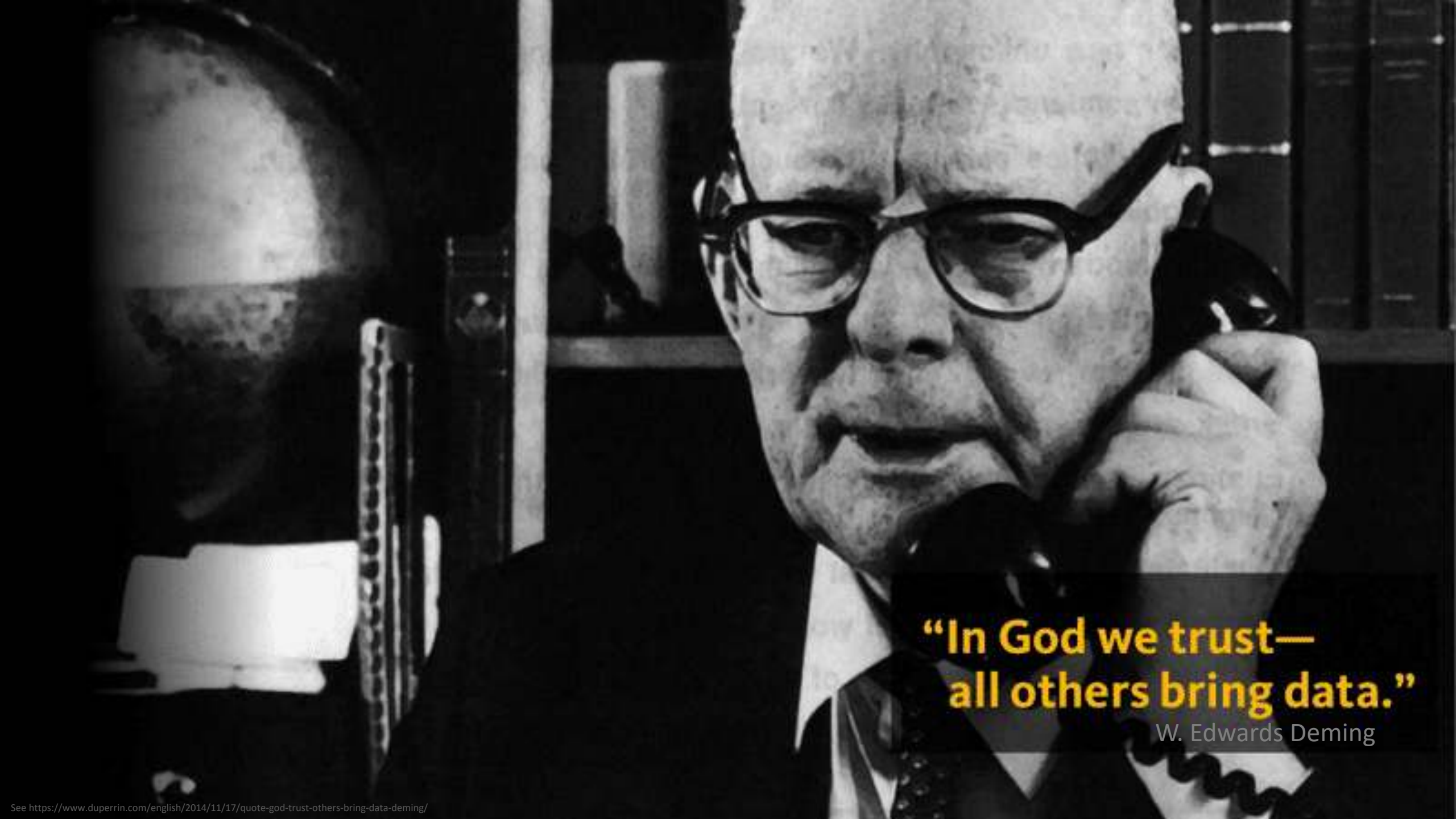


Got Slides?



Paper: [Nature Physics \(2023\)](#)

Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, Kristan Temme

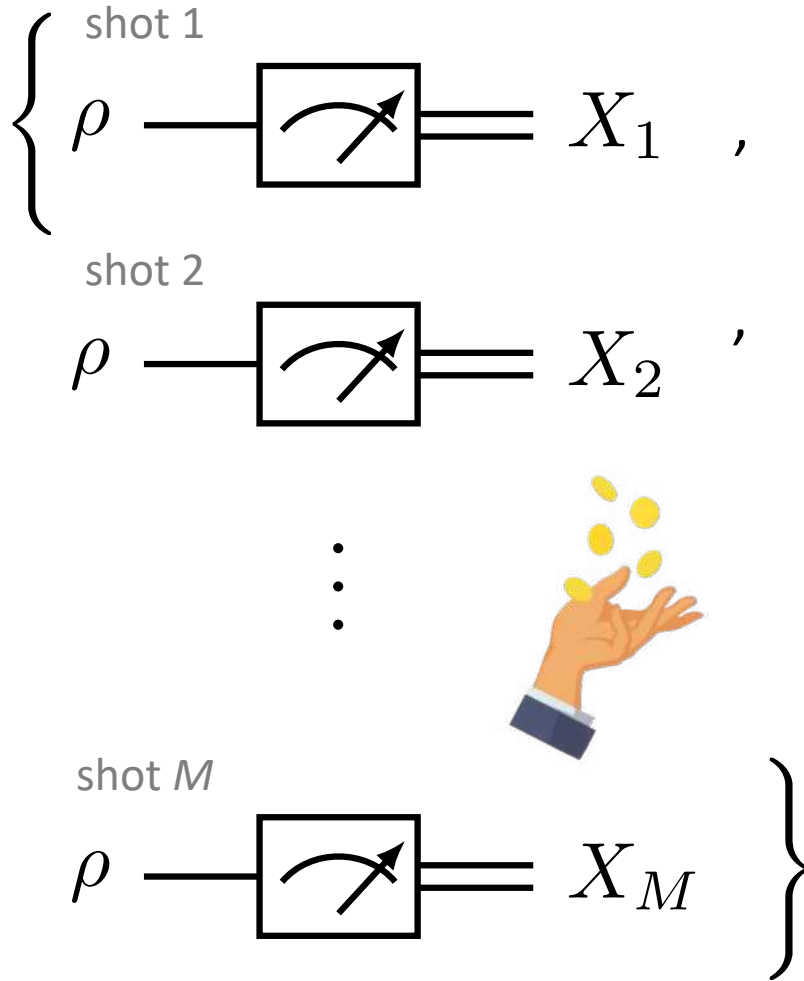
A black and white photograph of W. Edwards Deming, an older man with glasses, wearing a suit and tie, holding a rotary telephone receiver to his ear. He is looking slightly to the side with a thoughtful expression. The background shows a bookshelf with several books.

**“In God we trust—
all others bring data.”**

W. Edwards Deming



Ideal single qubit measurement with M shots



M shots with IID distribution

M outcomes: independent and identically distributed (iid) random variables

$$X_1, X_2, \dots, X_M \in \Sigma$$

$$X_1, X_2, \dots, X_M \sim \Pr[X = x] = \langle \hat{\mu}(x) \rangle$$

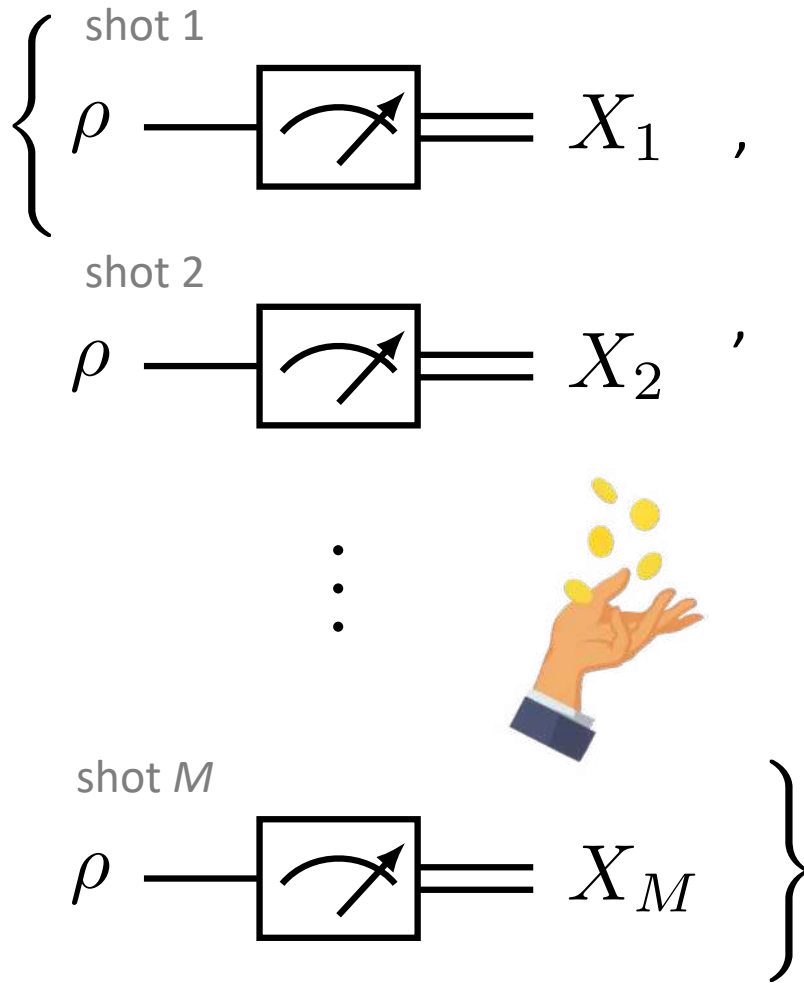
Empirical mean random variable (sample mean statistic)

$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

Find the expectation value and variance of the empirical mean



Empirical mean: an unbiased estimator



$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

Find the expectation value and variance of the empirical mean



$$\mathbb{E}[S] = \mathbb{E}\left[\frac{1}{M} \sum_{m=1}^M X_m\right]$$

$$\mathbb{E}[X_m] = \mathbb{E}[X] = p \quad \forall m \in \{1, \dots, M\}$$

$$\mathbb{E}[aX_m + bX_n] = a\mathbb{E}[X_m] + b\mathbb{E}[X_n]$$

$$\forall m, n, \quad a, b \in \mathbb{C}$$

$$= \frac{1}{M} \sum_{m=1}^M \mathbb{E}[X]$$

linear functional

$$= \mathbb{E}[X]$$

expectation value of a single shot

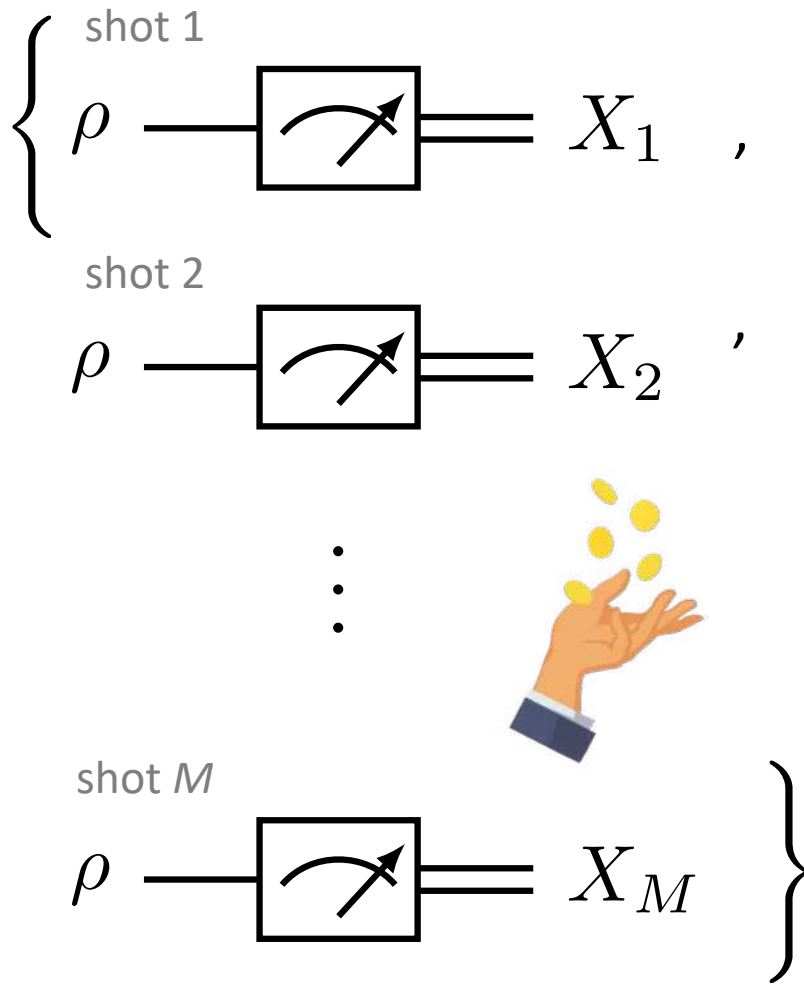
$$= \langle \hat{M} \rangle$$

relation to quantum operator we derived earlier (unbiased estimator)

$$= p$$

relation to probability to measure 1

How noisy is our estimate of the empirical mean?



$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

$$\begin{aligned} \mathbb{V}[S] &= \mathbb{V}\left[\frac{1}{M} \sum_{m=1}^M X_m\right] \\ &= \frac{1}{M^2} \sum_{m=1}^M \mathbb{V}[X] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{M} \mathbb{V}[X] \\ &= \frac{p(1-p)}{M} \end{aligned}$$

$$\begin{aligned} \sigma_S &= \sqrt{\mathbb{V}[S]} \\ &= \sqrt{\frac{\mathbb{V}[X]}{M}} \\ &= \sqrt{\frac{p(1-p)}{M}} \end{aligned}$$

Find the expectation value and variance of the empirical mean



Use key identity for variance

$$\mathbb{V}[aX_m + bX_n] = a^2 \mathbb{V}[X_m] + b^2 \mathbb{V}[X_n]$$

(you can derive this from the definition)

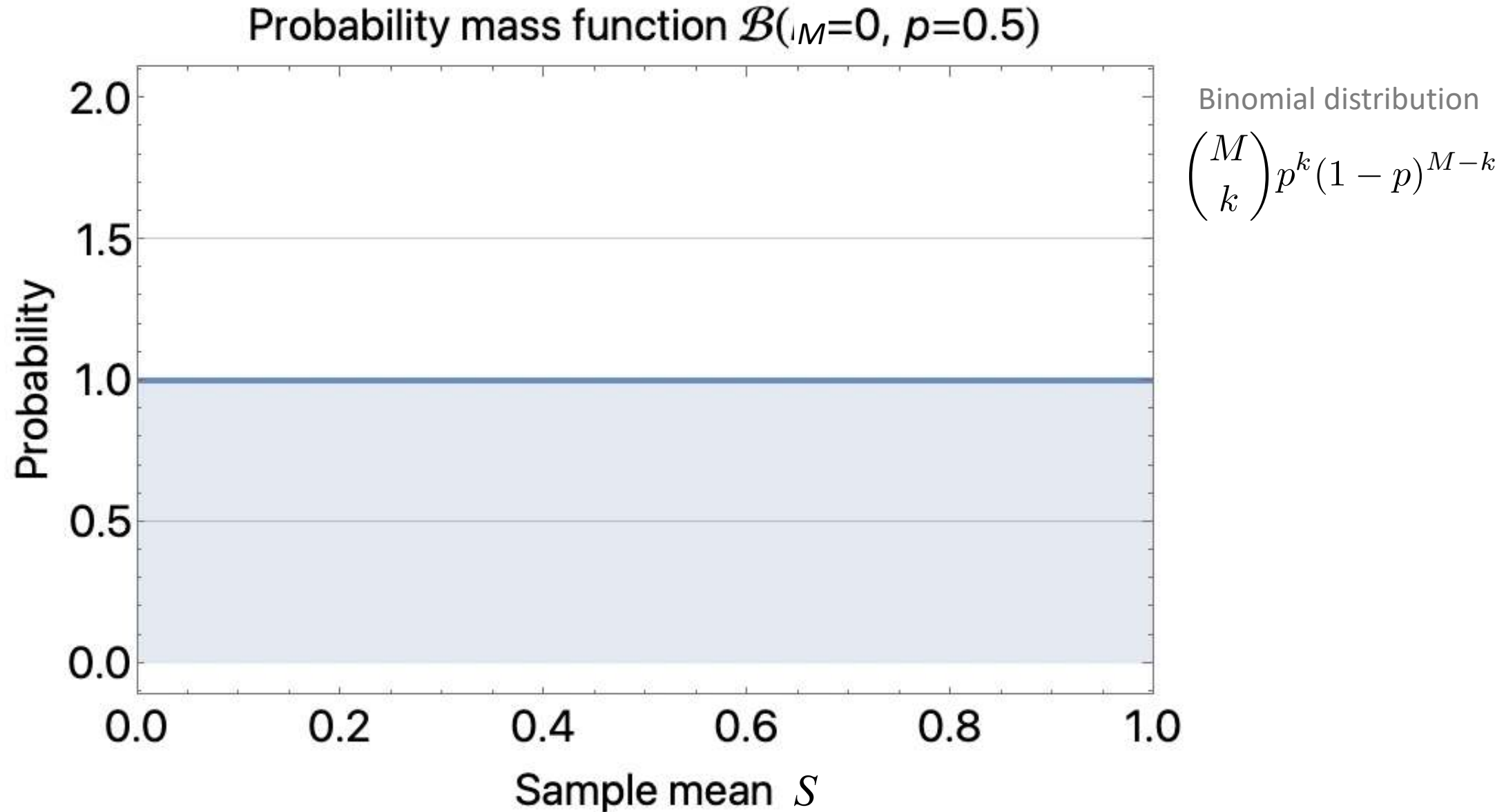
$$\mathbb{V}[X_m] = \mathbb{V}[X] = p(1-p) \quad \forall m \in \{1, \dots, M\}$$

The variance is reduced by the number of sampler we take!

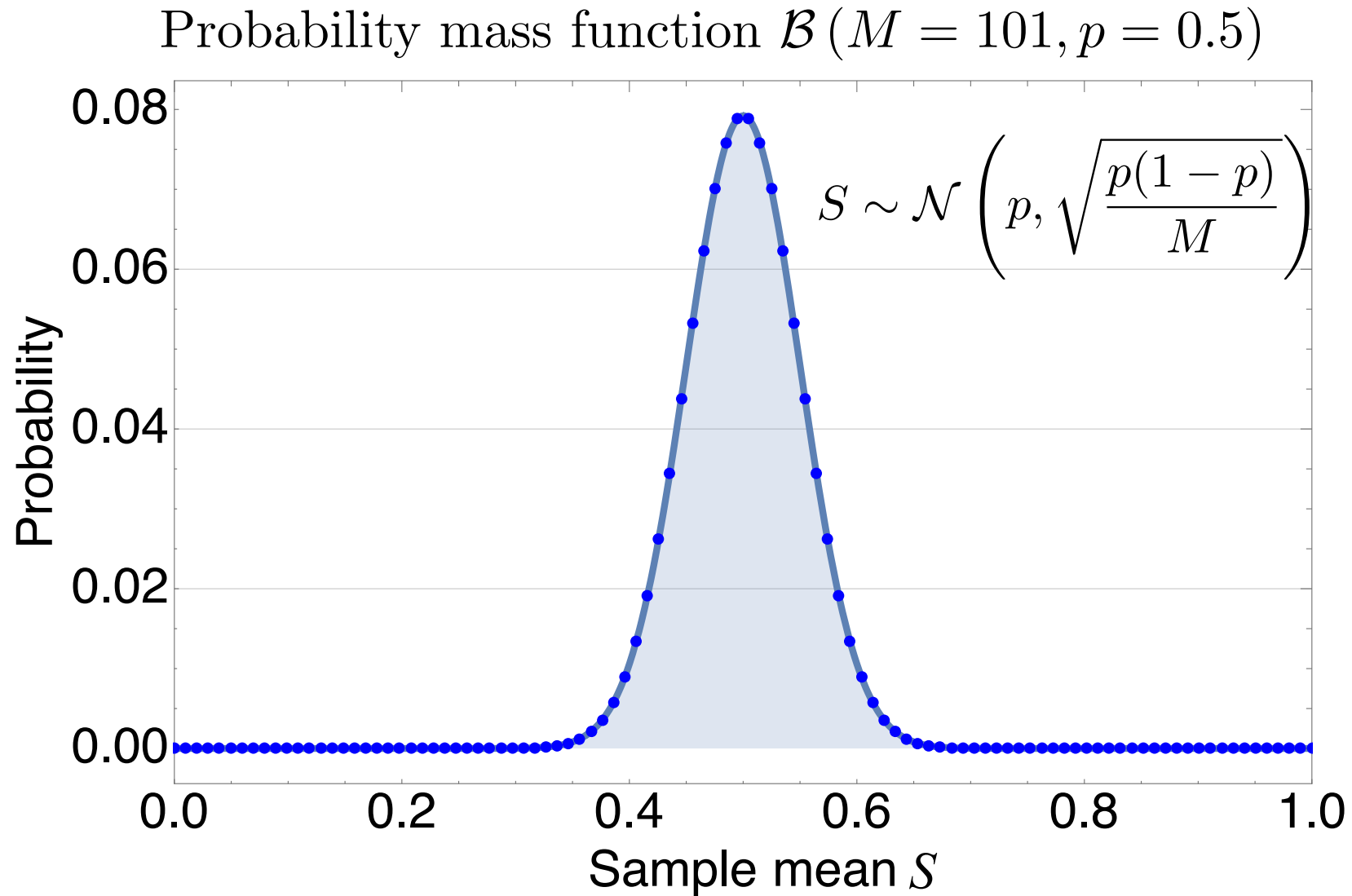
Thus we can suppress the quantum projection noise with enough shots.

The standard deviation drops as one over square root of the number of shots.

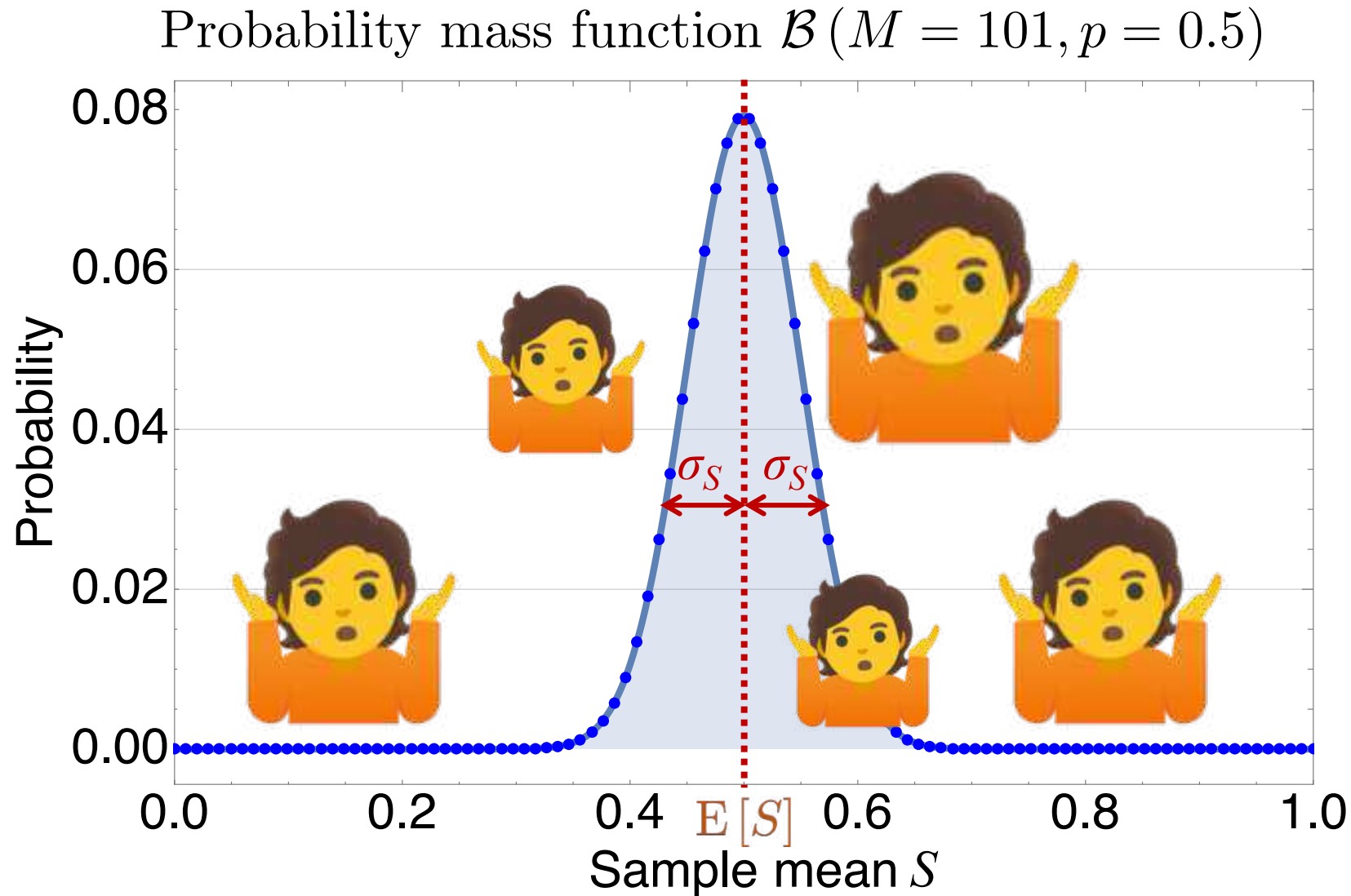
Animation of convergence of shots expectation value and mean



Sampled output distribution



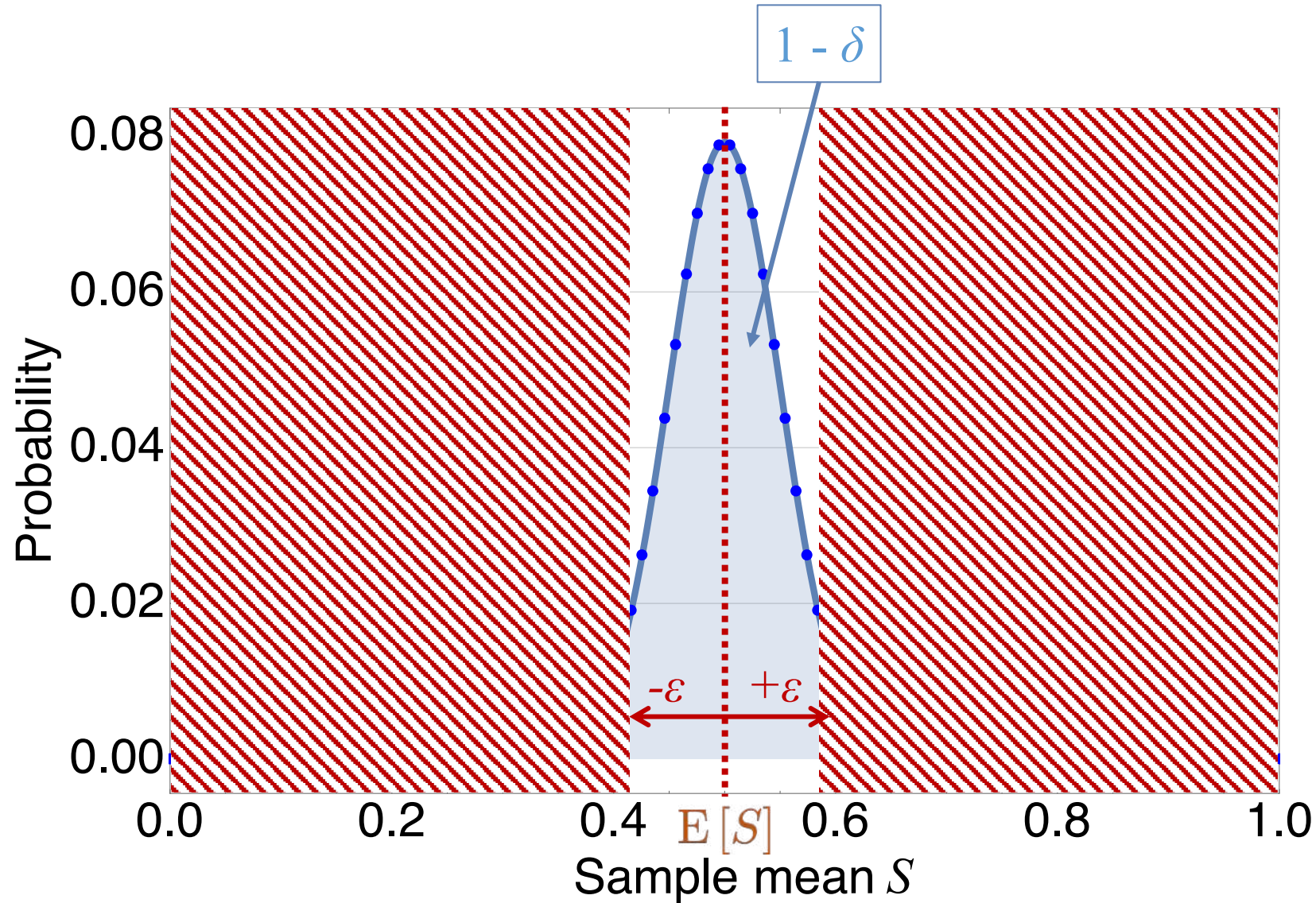
Properties of the output distribution



$$E[S] = p$$

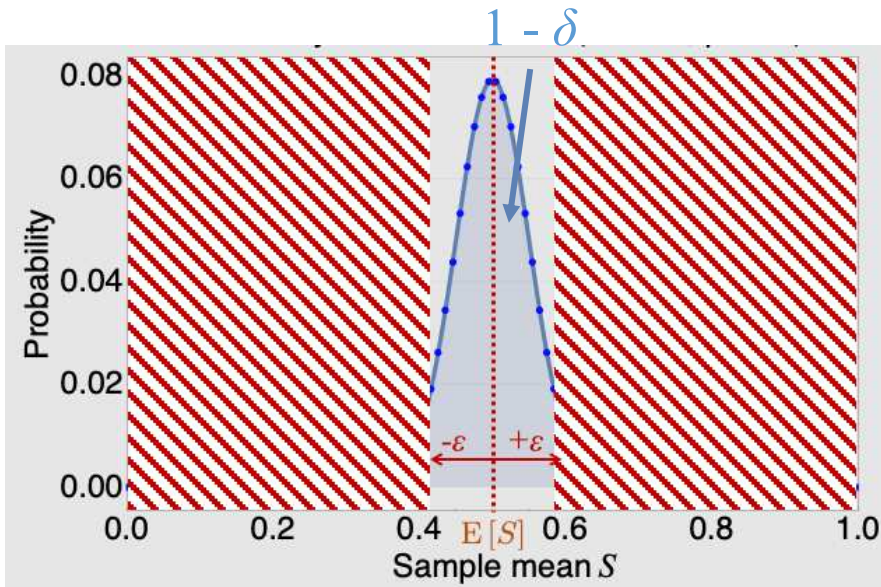
$$\begin{aligned}\sigma_S &= \sqrt{\text{Var}[S]} \\ &= \sqrt{\frac{p(1-p)}{M}}\end{aligned}$$

Concentration Measure for Sampling Expectation Values



Error Bound on Quantum Expectation Values

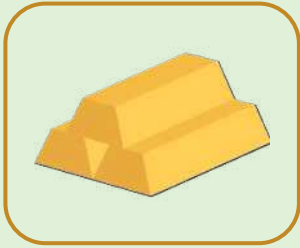
Concentration Measure for Sampling Expectation Values



Chernoff-Hoeffding two-sided tail bound, given $|O(x)| \leq 1$ for ϵ specified (additive) precision (worst case additive error) for S with success probability at least $1 - \delta$.

$$\Pr \left[\left| S - \langle \hat{O} \rangle \right| > \epsilon \right] \leq \delta := 2 \exp \left(-\frac{1}{2} M \epsilon^2 \right)$$

↑ δ specific failure probability for meeting precision ϵ empirically.



Empirical mean & sample properties

$$S = \frac{1}{M} \sum_{m=1}^M O(X_m)$$

Very general scaling for sampling problems!

Tf. required number of shots is at least [or with high probability (greater than 2/3)]

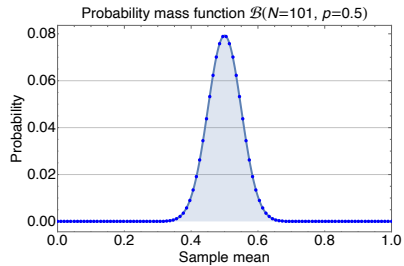
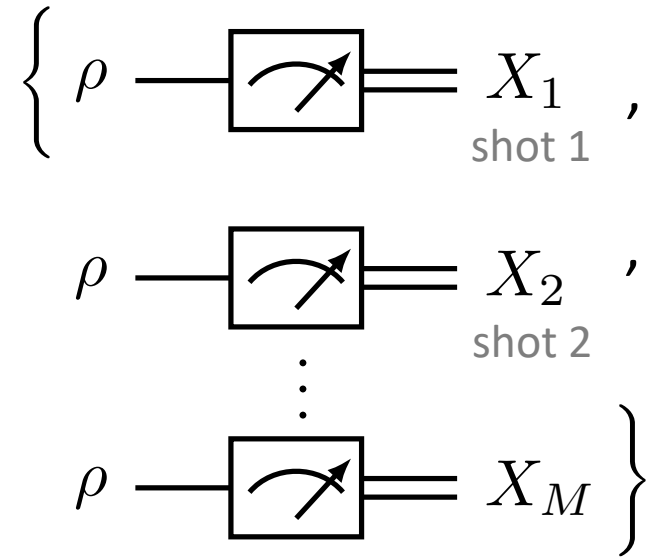
$$M \geq 2\epsilon^{-2} \log(2\delta^{-1}) \quad \left[M \gtrapprox 4\epsilon^{-2} \right] .$$

* Can find even tighter bound here owing to smaller $[0,1]$ range

Note that this scales same way (mod δ) as the variance bound with $\epsilon = \sigma$:

$$M \geq \frac{1}{4} \epsilon^{-2}$$

Ideal single qubit measurement with M shots



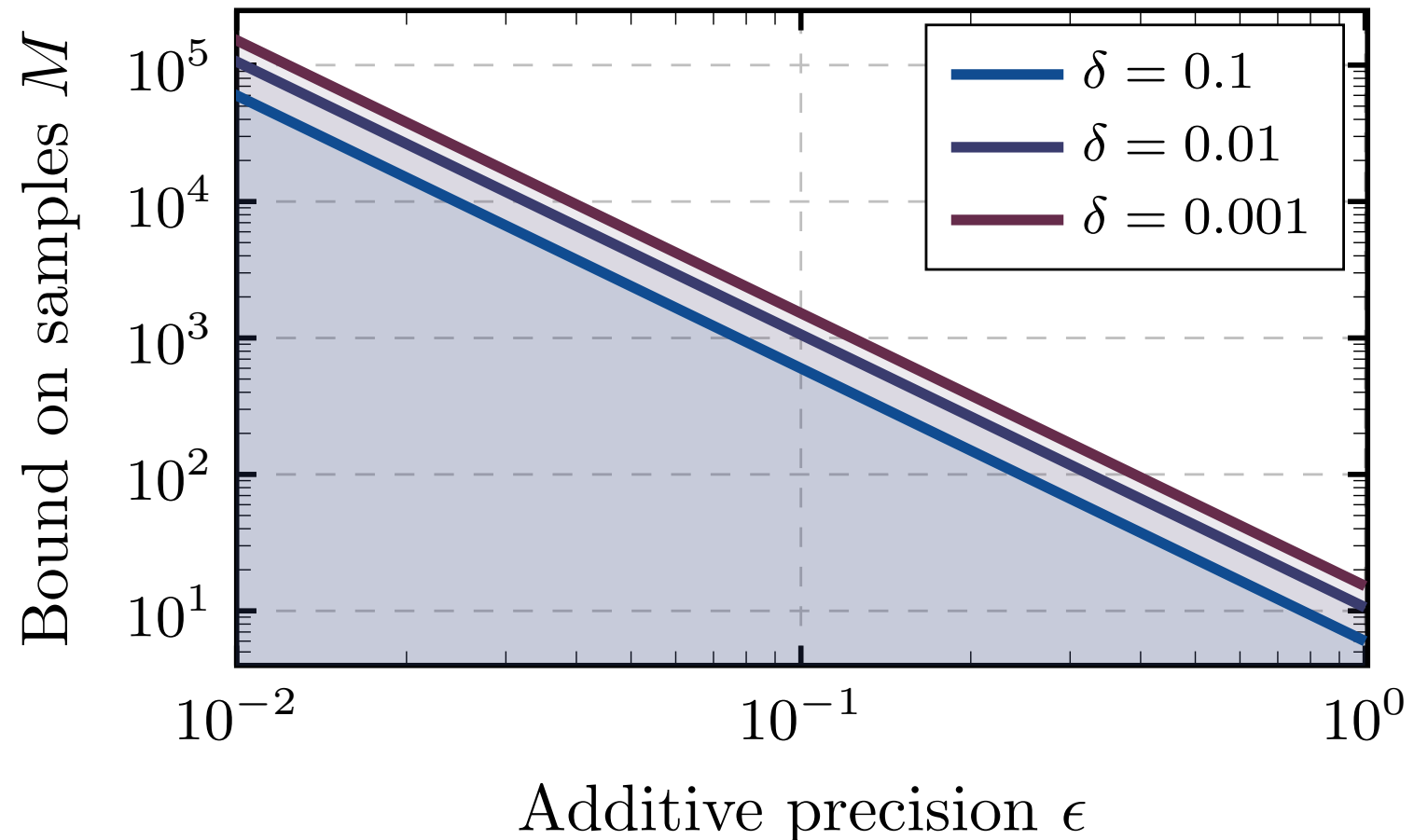
Observe that the **probability** δ is much cheaper than the **precision** ϵ .

Observe, n is not part of the equation.

“Knowing you are *not* wrong is cheaper than knowing you are right.”

Tail bound on sample complexity

$$M \geq 2\epsilon^{-2} \log(2\delta^{-1}) \quad [M \gtrsim 4\epsilon^{-2}]$$



Concentration inequalities and tail bounds



Making a list,
checking it twice,
going to see
which inequality
is nice!

Markov? Hoeffding?
Jensen? Chebyshev?
Chernoff?

<https://www.zlatko-minev.com/blog/inequalities>

1. Probability (Technical note 11.9 v0.6)

1A. Concentration inequalities and tail bounds

Unless otherwise specified, all variables are real $\mathbb{R}$. Inequalities come as one-sided $\Pr(\dots \leq \dots)$ and two-sided $\Pr(|\dots| \leq \dots)$. Notation: X is a random variable, $\mu := \mathbb{E}[X]$, $\sigma^2 := \text{Var}[X]$, $S_n := X_1 + \dots + X_n$.

Inequality	Conditions	Common form	Notes / Alternate form
Single random variable			
Markov ¹	Non-negative $X \geq 0$	$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$	$\forall a > 0$ $\Pr[X \geq k\mathbb{E}[X]] \leq \frac{1}{k}$ $k > 1$ [3, Sec. 5.1][6, Thm 1.13]
extension	+ non-negative, strictly increasing func Φ $\Phi(X) \geq 0$ increasing	$\Pr[X \geq a] = \Pr[\Phi(X) \geq \Phi(a)] \leq \frac{\mathbb{E}[\Phi(X)]}{\Phi(a)}$	$\forall a > 0$ Wiki
Reverse Markov	upper-bounded by U (can be positive) $\max X = U$	$\Pr[X \leq a] \leq \frac{U - \mathbb{E}[X]}{U - a}$	$\forall a > 0$ [1, Sec. 3.1]
Chebyshev ²	Finite mean and variance $\mathbb{E}[X]$, $\text{Var}[X]$ finite	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{a^2}$	$\Pr[X - \mathbb{E}[X] \geq a \cdot \sigma] \leq \frac{1}{a^2}$ $\forall a > 0, \sigma^2 = \text{Var}[X]$ [1, Sec. 3.2] [3, Sec. 5.1][2, Thm 18.11]
Cantelli	Improved Chebyshev (same; but one-sided)	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{\sigma^2 + a^2}$	$\forall a > 0, \sigma^2 = \text{Var}[X]$ Wiki
Chernoff ³	Generic	$\Pr[X \geq a] = \Pr[e^{tX} \geq e^{ta}]$	$\forall t > 0, a \in \mathbb{R}$ [1, Sec. 3.3]
Jensen	$f: \mathbb{R} \rightarrow \mathbb{R}; f$ convex	$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$	[3, Prob. 5.3][6, Thm 1.14]
Hoeffding's lemma	$\mathbb{E}[X] = \mu$ $a \leq X \leq b$	$\mathbb{E}[e^{\lambda X}] \leq e^{\lambda \mu} e^{\frac{\lambda^2 (b-a)^2}{8}}$	$\lambda \in \mathbb{R}$ [1, Sec. 3.4]
Sum of random variables			
Chernoff-Hoeffding (one-sided)	n independent random vars $X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$ $X_i \in [a_i, b_i] \forall i$	$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp\left(\frac{-2t^2 n^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	[1, Sec. 3.5]
(two-sided) ⁴	(same as above)	$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq 2 \exp\left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	$\forall t \in (0, \frac{1}{2})$ [5, Thm.1.1]
(two-sided iid)	same plus iid, range, mean μ for each $X_1, \dots, X_n \in [0, 1]$ $\mathbb{E}[X_i] = \mu$ iid	$\Pr\left[\left \frac{S_n}{n} - \mu\right \geq \epsilon\right] \leq 2 \exp(-2n\epsilon^2)$	$\forall \epsilon > 0$ [6, Thm 1.16]
Thm 1.3	n independent random vars $X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$	$\Pr[S_n - \mathbb{E}[S_n] > \epsilon] \leq 2 \exp\left(\frac{-\epsilon^2}{4 \sum_{i=1}^n \text{Var}[X_i]}\right)$	$\epsilon \in (0, 2 \text{Var}[S_n] / (\max_i X_i - \mathbb{E}[X_i]))$ [5, Thm. 1.3]
Azuma			
Weak law of large numbers	n independent iid random vars $X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = \mu$ iid	$\lim_{n \rightarrow \infty} \Pr\left[\left \frac{1}{n} S_n - \mu\right \geq \epsilon\right] = 0$	$\forall \epsilon > 0$ [3, Sec. 5.2][6, Thm 1.15]
Strong law of large numbers	(same)	(same)	$\Pr\left[\lim_{n \rightarrow \infty} \frac{1}{n} S_n = \mu\right] = 1$ [3, Sec. 5.5]
Advanced			
Bennett	n independent zero-mean $X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = 0$ iid	$\Pr[S_n > \epsilon] \leq \exp\left(-n\sigma^2 h\left(\frac{\epsilon}{n\sigma^2}\right)\right)$	$\sigma^2 := \frac{1}{n} \sum_{i=1}^n \text{Var}[X_i], \forall \epsilon > 0,$ $h(a) := (1+a) \log(1+a) - a$ for $a \geq 0$ [1, 4.1]
Bernstein	(same)	(same)	(same) [1, 4.2]
Efron-Stein	scalar func of vars $f: \mathcal{X}^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set $\mathcal{X}$	$\text{Var}[Z] \leq \sum_{i=1}^n \mathbb{E}[(Z - \mathbb{E}_i[Z])^2]$ $Z := g(X_1, \dots, X_n)$ $\mathbb{E}_i[Z] := \mathbb{E}[Z X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n]$ [1, 4.3]
McDiarmid's	scalar func of vars $f: \mathcal{X}^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set $\mathcal{X}$	$\Pr[f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right)$ condition: c -bounded difference property $\forall \epsilon > 0$ $ f(X_1, \dots, X_i, \dots, X_n) - f(X_1, \dots, X'_i, \dots, X_n) \leq c_i$ [1, 4.4]

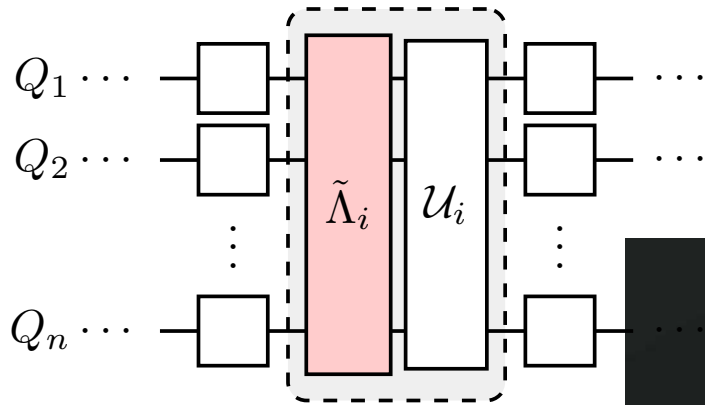
¹Markov's inequality bounds the first moment of random variable. Use it when a constant probability bound is sufficient [1, Sec. 3.3].

²Chebyshev is derived from Markov. It bounds the second moment. It is the appropriate one when the variance σ is known. If σ is unknown, we can use the bounds of $X \in [a, b]$.

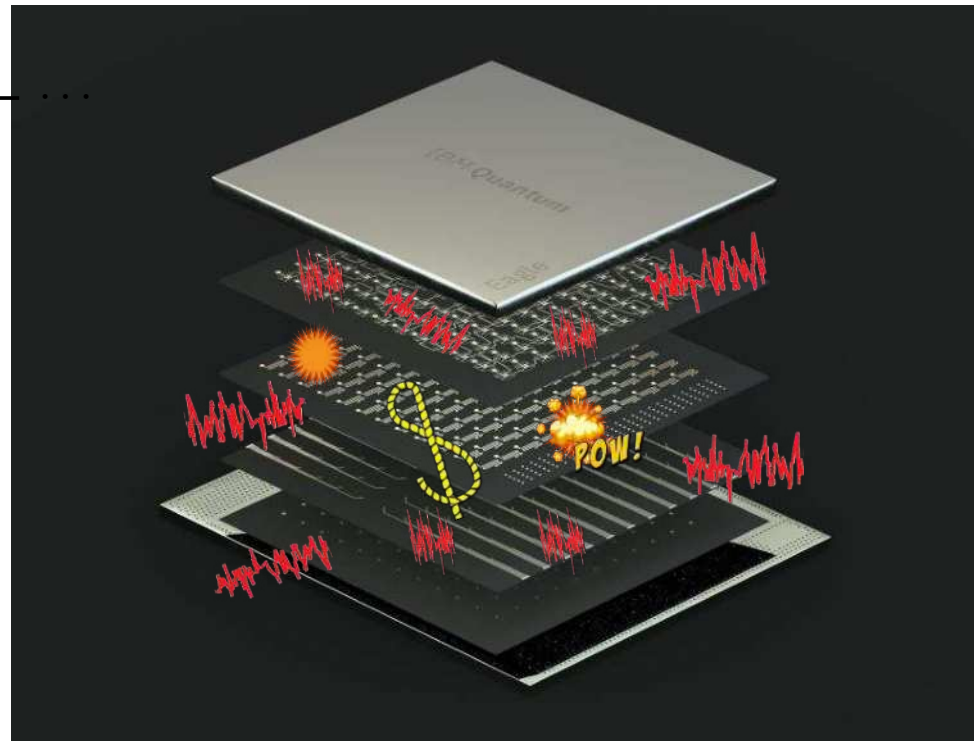
³Chernoff bound is used to bound the tails of the distribution for a sum of independent random variables. By far the most useful tool in randomized algorithms [1, Sec. 3.3].

⁴This probability can be interpreted as the level of significance ϵ (probability of making an error) for a confidence interval around the mean of size 2ϵ . Therefore, we require at least $\log(2\alpha)/2\epsilon^2$ samples to acquire $1 - \alpha$ confidence interval $\mathbb{E}[X] \pm t$.

Is it possible to learn the noise with accuracy, efficiency, and scalability?



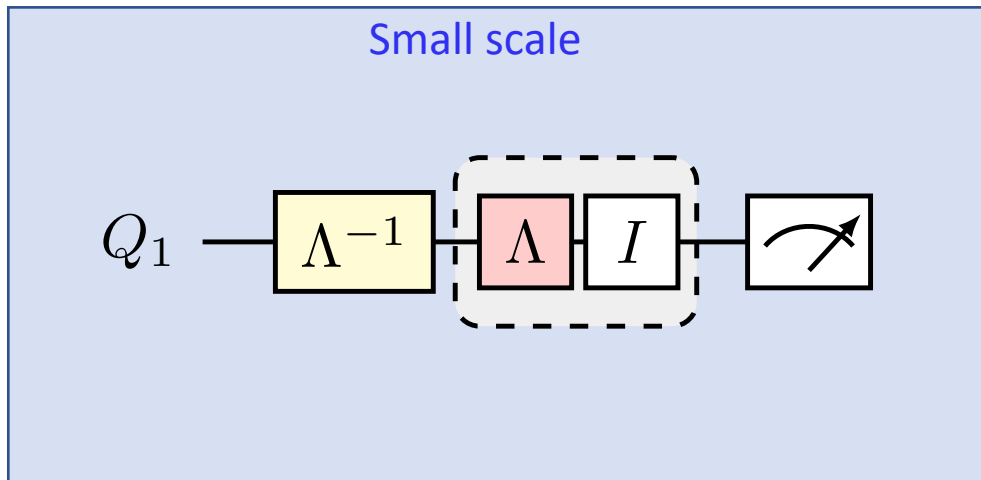
Energy relaxation T_1
Dephasing T_2
Coherent errors ZZ
Classical crosstalk
Quantum crosstalk
State preparation error
Measurement correlated errors
...



Control errors
Photon shot noise
1/f charge noise
1/f flux noise
Nonequilibrium quasiparticles
Leakage
Cosmic rays
...

PEC: Nice, **but** why hasn't worked so far for experiments?

Practical challenges



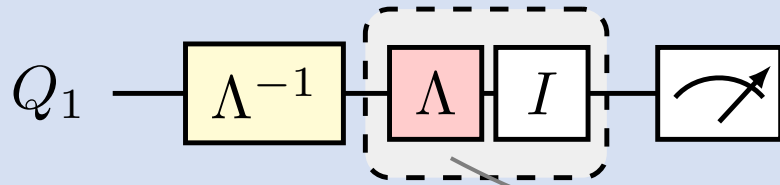
Critically hinges on knowing the full noise near perfectly

Despite the method's theoretical appeal (1-10), practical challenges have limited its demonstration to the one and two-qubit level (2, 3)

1. S. Endo, S. C. Benjamin, Y. Li, Physical Review X 8, 031027 (2018).
2. C. Song, et al., Science Advances 5, arXiv:2109.04457(2019).
3. S. Zhang, et al., Nature Communications 11, 587 (2020).
4. C. Piveteau, D. Sutter, S. Woerner, arXiv:2101.09290 (2021).
5. S. Endo, et al., J. Phys. Soc. of Japan 90, 032001 (2021).
6. C. Piveteau, et al., arXiv:2103.04915 (2021).
7. R. Takagi, Phys. Rev. Research 3, 033178 (2021).
8. R. Takagi, S. Endo, S. Minagawa, M. Gu, arXiv:2109.04457 (2021).
9. Y. Guo, S. Yang, arXiv preprint arXiv:2201.00752 (2022).
10. ...

PEC: Nice, **but** why hasn't worked so far? Challenges

Small scale



2 qubits

255 parameters

10 qubits

10^{12} parameters

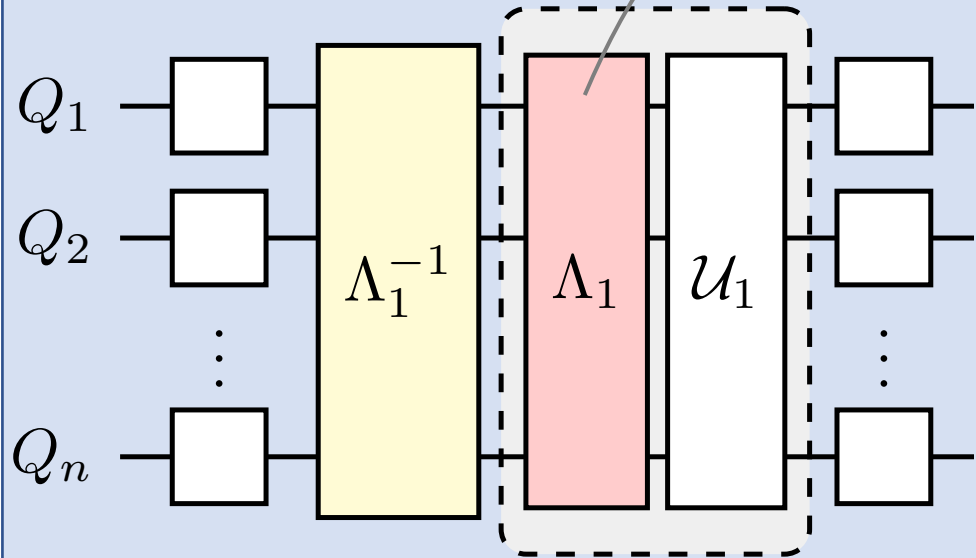
50 qubits

10^{60} parameters

noise param values $10^{-2} - 10^{-5}$

additive error sampling cost ($>10^2 - 10^{10}$)

Large scale



Challenges

learning complexity

- efficient
- scalable
- accurate
- compact, tractable representation

noise in full device

- cross-talk
- correlated errors
- parallel gates

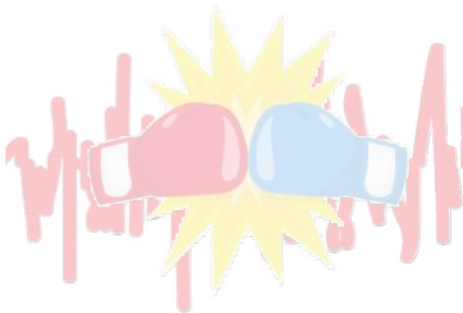
Outline



Idea

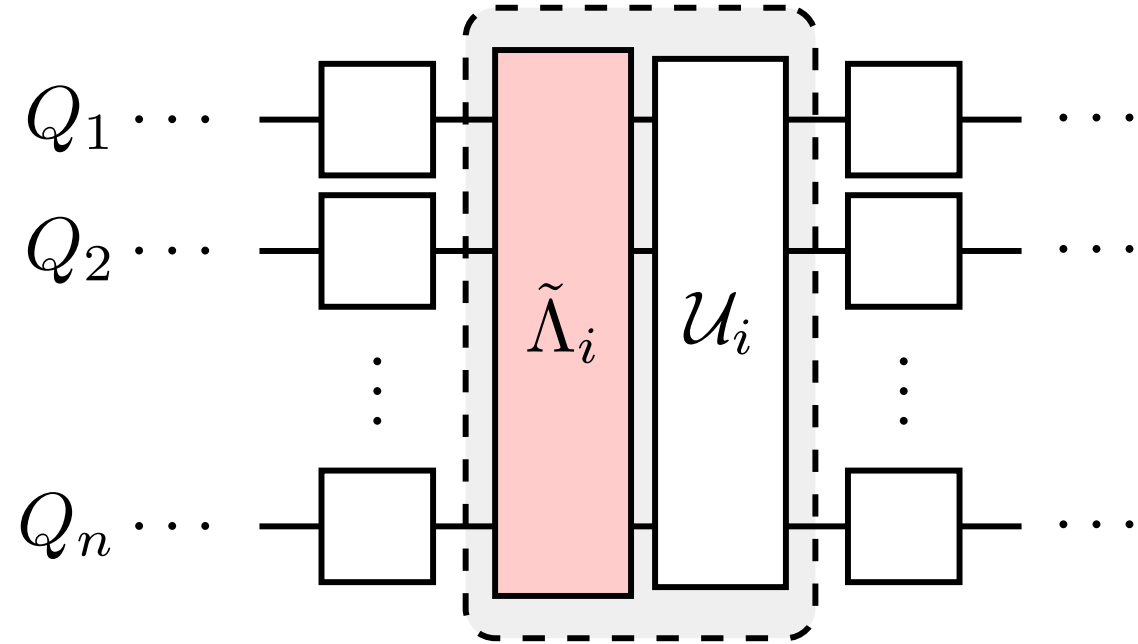


Learn



Cancel
(realization)

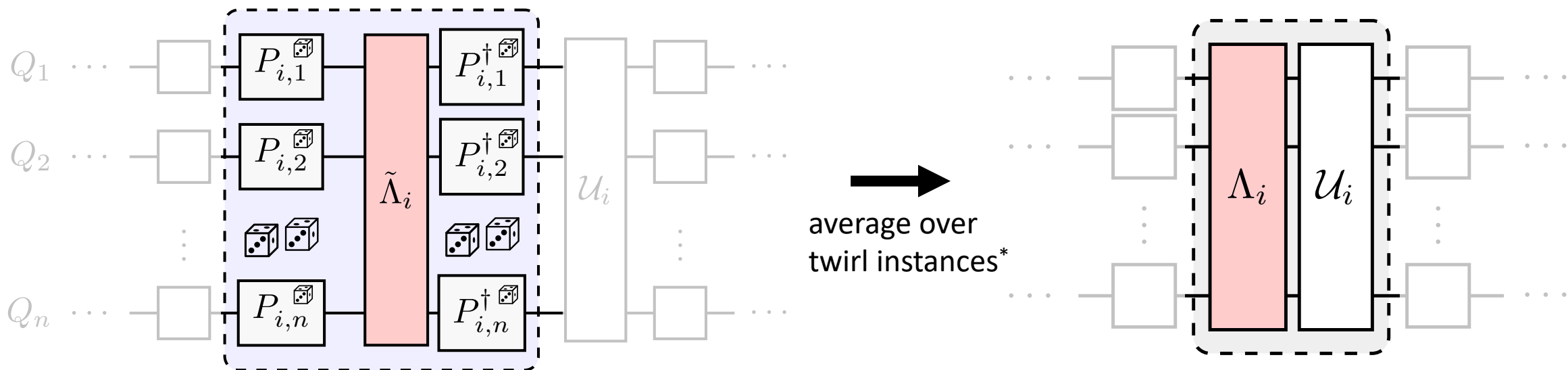
Step 1: Simplify the noise



noise that includes cross-talk errors, etc.
characterized by some $4^n \times 4^n$ matrix

Step 1: Simplify the noise: twirl

twirl reduces to noise $4^n \times 4^n$ matrix to diagonal one with 4^n entries in Pauli basis



Twirling references

1. C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, **Tutorial** W. K. Wootters, et al., Phys. Rev. Lett. 76, 722 (1996).
2. E. Knill, arXiv:0404104 (2004).
3. O. Kern, G. Alber, D. L. Shepelyansky, EPJ D 32, 153 (2005).
4. M. R. Geller, Z. Zhou, Physical Review A 88, 012314 (2013).
5. J. J. Wallman, J. Emerson, Physical Review A 94, 052325 (2016)
6. Hashim *et al.*, Phys. Rev. X 11, 041039 (2021)
7. Tutorial: zlatko-minev.com/blog/twirling (2022)
8. ...



Stochastic Pauli channel

$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

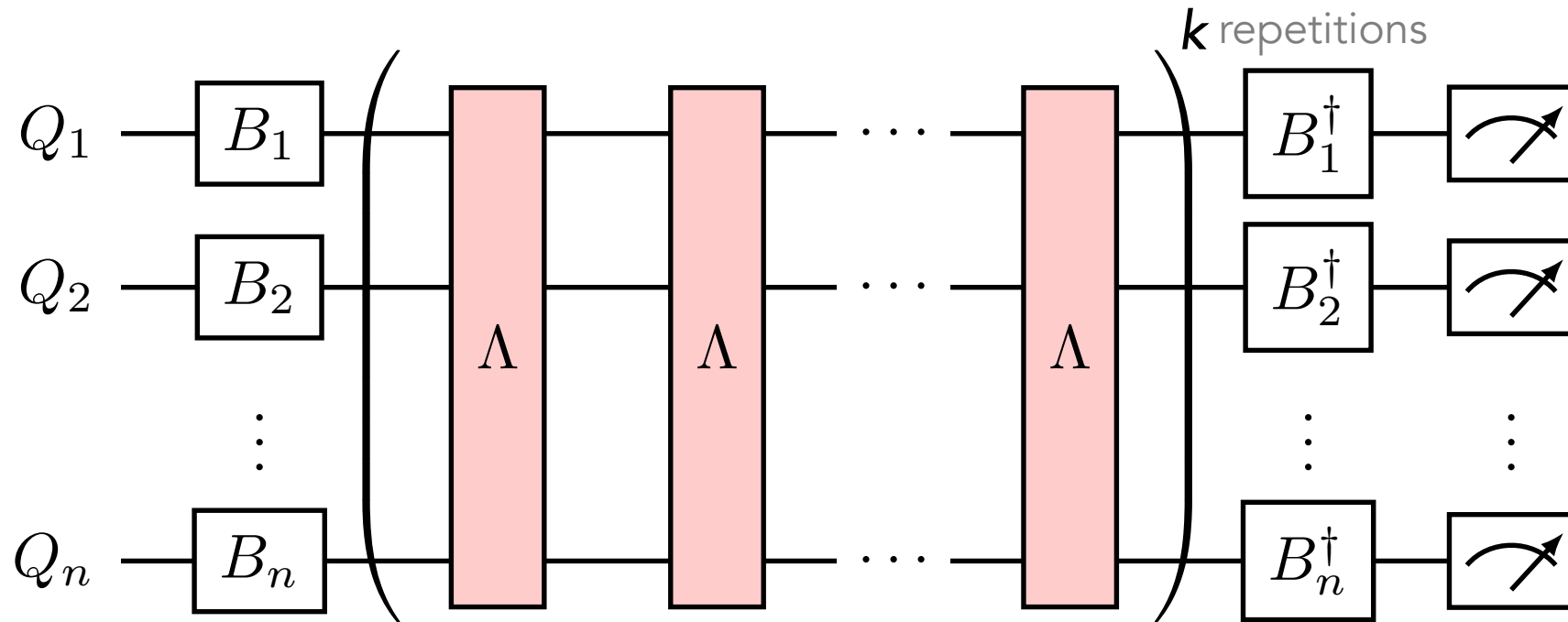
$$\Lambda(P_a) = f_a P_a \quad \text{eigenvecs are Paulis}$$

* some sub-Clifford twirl group (use Paulis)

Zlatko Minev, IBM Quantum (30)

Step 2 wish: amplify noise

for the i -th layer



prepare circuit in pre-determined Pauli basis

Since diagonal channel will amplify eigenvalues learn with multiplicative precision

measure circuit in same pre-determined Pauli basis

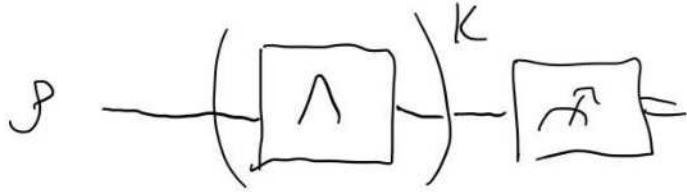
Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

For something of a review of protocols, see Helsen, *et al.*, *A general framework for randomized benchmarking* (arXiv:2010.07974)

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

For a qubit

$$\rho = \frac{1}{2} \left(\hat{I} + \rho_X \hat{X} + \rho_Y \hat{Y} + \rho_Z \hat{Z} \right)$$

In general:

$$\rho = \frac{1}{2^n} \sum_b \mathcal{P}_b \hat{P}_b$$

$$\Lambda(\rho) = \rho'$$

$$\rho' = \frac{1}{2^n} \sum_b \mathcal{P}'_b \hat{P}_b$$

Pauli decomposition of a density matrix is a powerful tool - offering a versatile representation. It expresses a density matrix as a linear combination of Paulis, often referred to as the Pauli basis.

Posterior state: Action of the channel on the input state

Pauli decomposition of posterior state

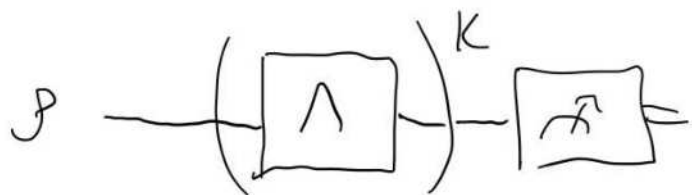
Since the channel is linear

$$\Lambda(a \hat{A} + b \hat{B}) = a \Lambda(\hat{A}) + b \Lambda(\hat{B})$$

$$\Lambda(\rho) = \Lambda\left(\frac{1}{2^n} \sum_b \mathcal{P}_b \hat{P}_b\right)$$

$$= \frac{1}{2^n} \sum_b \mathcal{P}_b \underbrace{\Lambda(\hat{P}_b)}$$

Just need to know action on P_b !



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

or written another way:

$$\begin{aligned} f_c' &= \sum_b f_b \frac{1}{2^n} \text{Tr}(\hat{P}_c^\dagger \Lambda(\hat{P}_b)) \\ &= \sum_b f_b \frac{\langle\langle P_c | \Lambda | P_b \rangle\rangle}{\langle\langle P_c | P_c \rangle\rangle} \end{aligned}$$

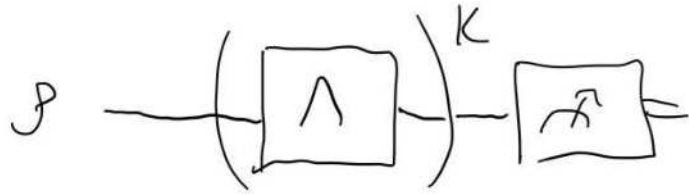
$$\begin{aligned} \Lambda(\rho) &= \Lambda\left(\frac{1}{2^n} \sum_b f_b \hat{P}_b\right) \\ &= \frac{1}{2^n} \sum_b f_b \underbrace{\Lambda(\hat{P}_b)}_{\text{just need to know action on } P_b} \\ &= f' \\ &= \frac{1}{2^n} \sum_c f_c' \hat{P}_c \end{aligned}$$

since $\text{Tr}(\hat{P}_a^\dagger \hat{P}_b) = \delta_{ab} 2^n$
orthogonal

Solve for f_c' by equate & orthogonal case:

$$\begin{aligned} \text{Tr}\left(\hat{P}_c^\dagger \frac{1}{2^n} \sum_b f_b \Lambda(\hat{P}_b)\right) &= \text{Tr}\left(\hat{P}_c^\dagger \frac{1}{2^n} \sum_{c'} f_{c'}' \hat{P}_{c'}\right) \\ \sum_b f_b \text{Tr}(\hat{P}_c^\dagger \Lambda(\hat{P}_b)) &= \sum_{c'} f_{c'}' \text{Tr}(\hat{P}_c^\dagger \hat{P}_{c'}) \\ &= \sum_{c'} f_{c'}' \delta_{cc'} 2^n \\ \frac{1}{2^n} \sum_b f_b \text{Tr}[\hat{P}_c^\dagger \Lambda(\hat{P}_b)] &= f_c' \quad \checkmark \end{aligned}$$

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

The measurement outcome then is (say measured using $\hat{P}_c$ basis):

$$\langle \hat{P}_c \rangle = \text{Tr}(\hat{P}_c \rho')$$

For a single step

$$= \text{Tr}(\hat{P}_c \sum_{c'} \frac{1}{2^n} \rho_{c'} \hat{P}_{c'})$$

$$= \sum_{c'} \rho_{c'} \frac{1}{2^n} \text{Tr}(\hat{P}_c \hat{P}_{c'}) \rightarrow 2^n \delta_{cc'}$$

$$= \rho_c$$

now sub in

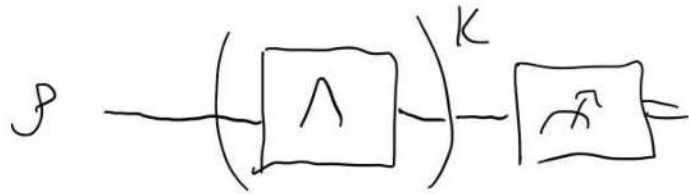
$$= \sum_b \rho_b \frac{\langle\langle P_c | \Lambda | P_b \rangle\rangle}{2^n}$$

→ for a Paul channel $\langle\langle P_c | \Lambda | P_b \rangle\rangle = f_c \delta_{bc} 2^n$

$$= f_c \rho_c \leftarrow \text{initial state coefficient}$$

fidelity c of Λ

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

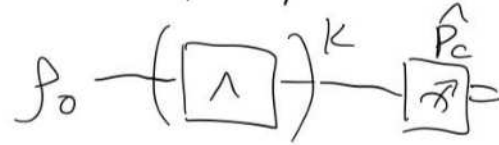
$$\Lambda(P_a) = f_a P_a$$

What is ρ_c for an eigenstate of $\hat{P}_c \rightarrow 1$.

e.g. for $c=z$: $\hat{P}_c = \hat{Z}$ and $\rho = |0\rangle = \frac{1}{2}(I + Z)$

$$\Rightarrow \rho_{c=z} = \text{Tr}(Z \rho) = +1$$

For repeating the channel K times



$$\rho' = \Lambda^K \rho$$

$$\Rightarrow \rho'_c = \sum_b \rho_b \frac{\langle\langle P_c | \Lambda^K | P_b \rangle\rangle}{2^n}$$

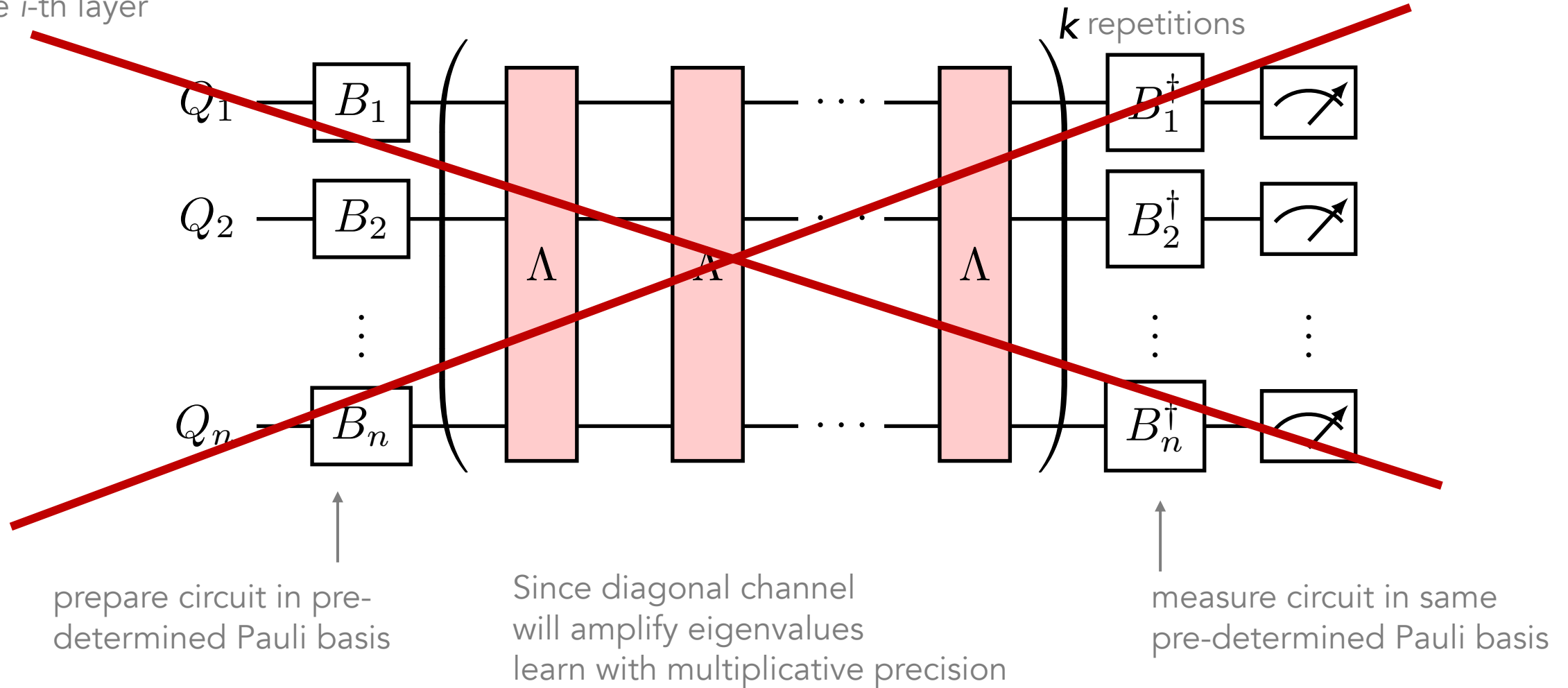
$$= \sum_b f_b^K \rho_b$$

$$\Rightarrow \langle \hat{P}_c \rangle = f_c^K \rho_c$$

Can include all diagonal SPAM

Step 2 wish: amplify noise

for the i -th layer



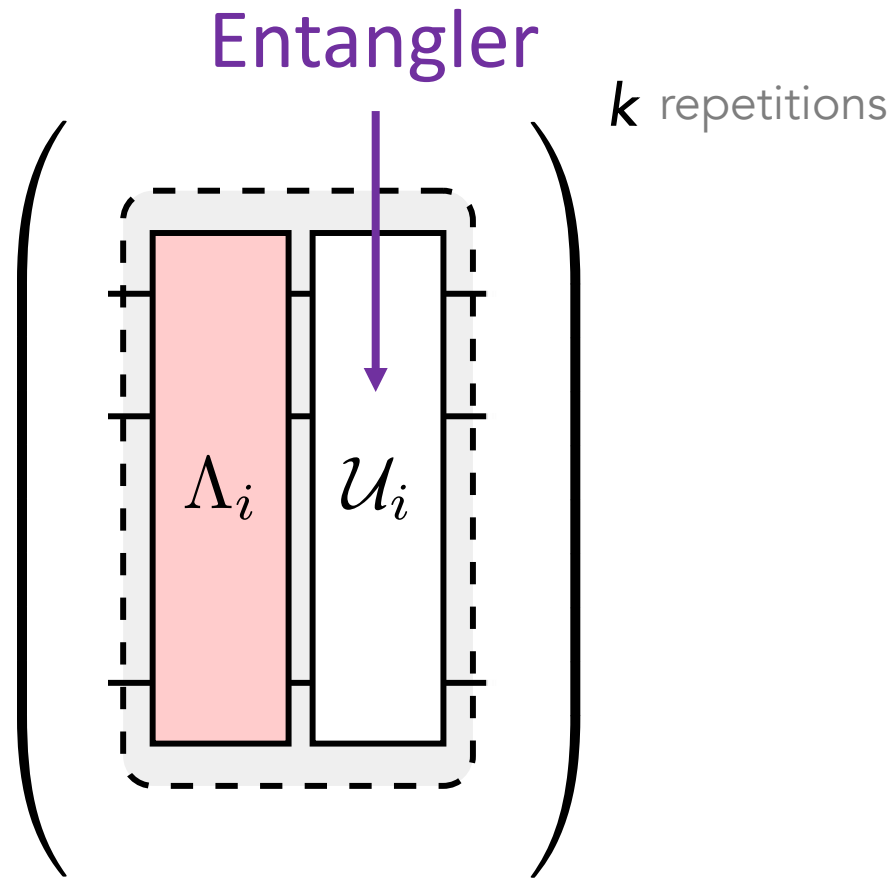
Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

For something of a review of protocols, see Helsen, *et al.*, *A general framework for randomized benchmarking* (arXiv:2010.07974)

Step 2: Ideally, amplify the noise and learn



Ideally wish

~~$$\Lambda_i^k(P_a) = f_{ia}^k P_a$$~~

Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

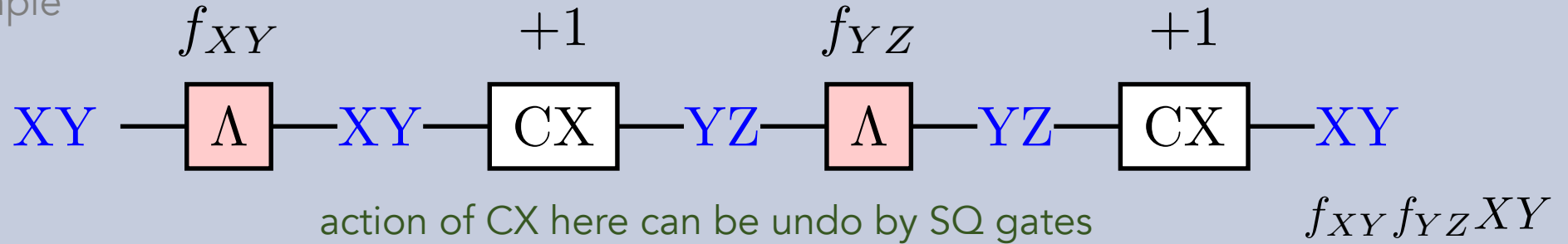
Erhard *et al.*, arXiv:1902.08543; Ferracin *et al.*, arXiv:2201.10672, ...

For something of a review of protocols, see Helsen, *et al.*, arXiv:2010.07974

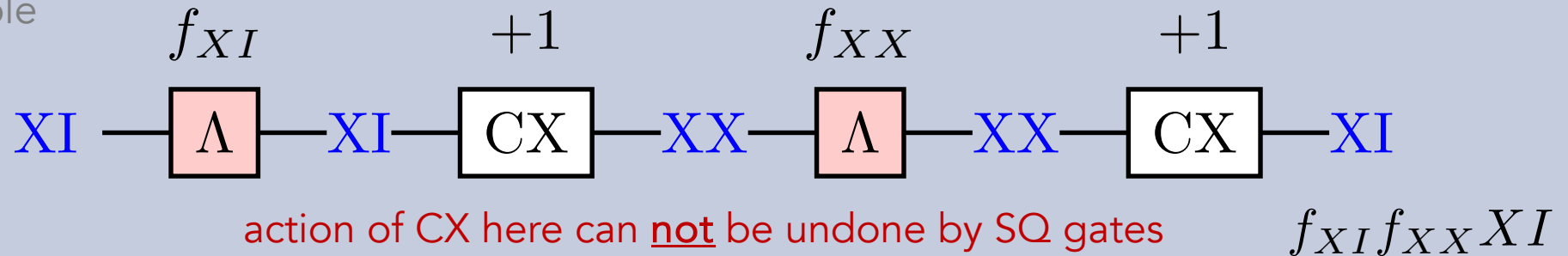
Let's see how the amplification works with gates: no-go theorem

$$\Lambda(P_a) = f_a P_a$$

2Q example



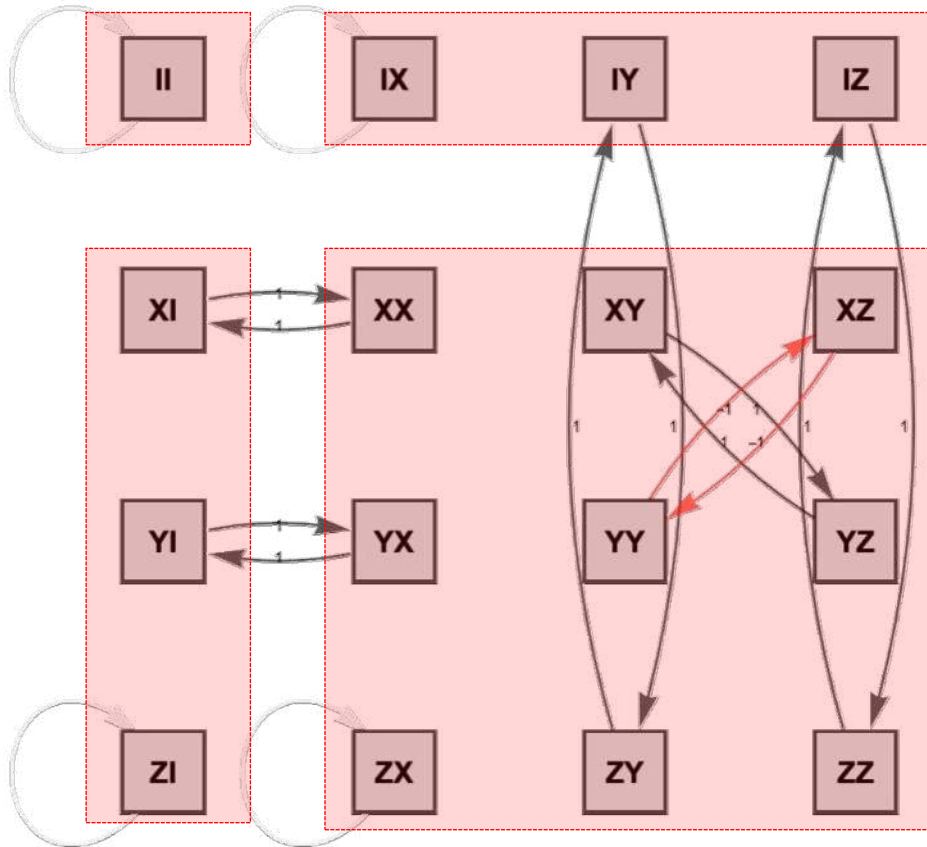
2Q example



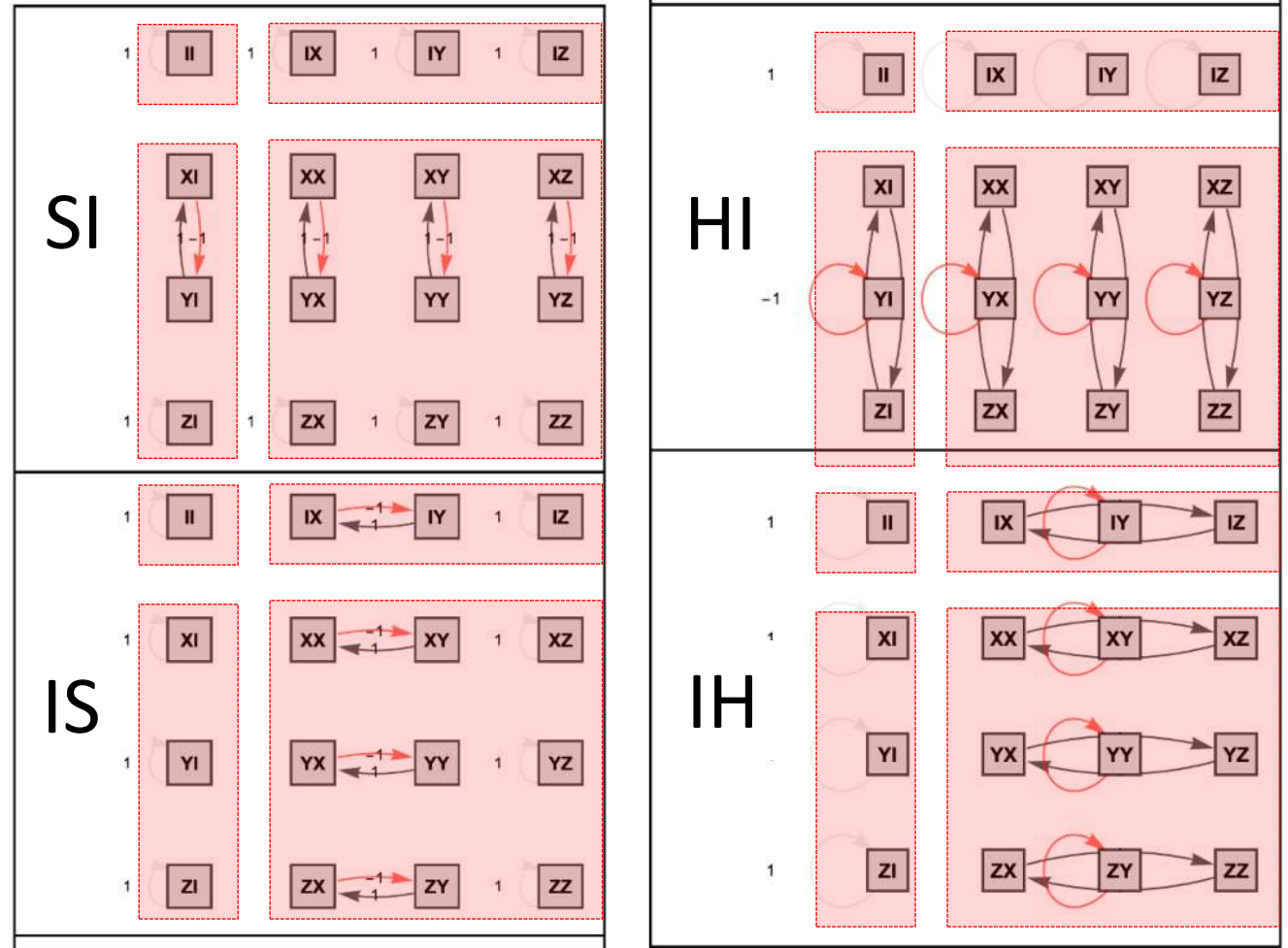
fundamental degeneracy – can not undo some non-local – need entangling operation

How to gates move state Paulis around?

cNOT

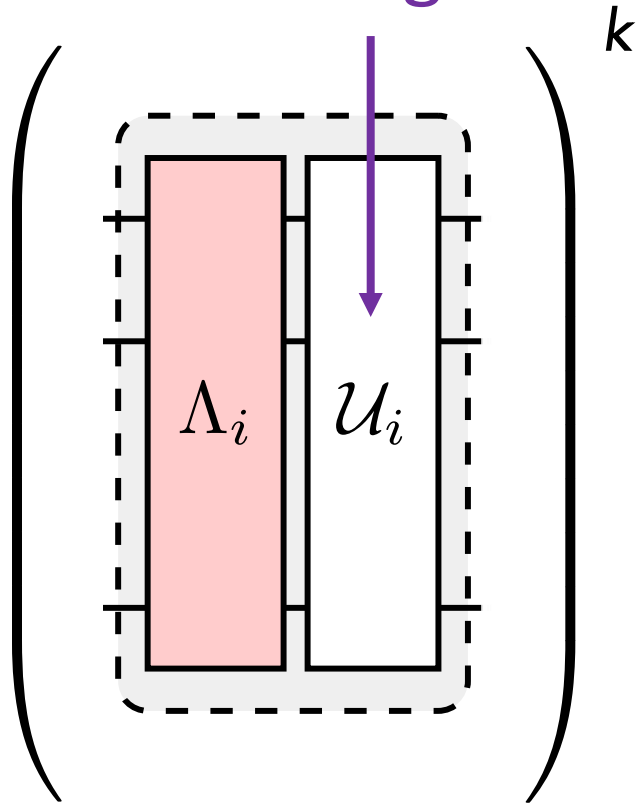


Example single qubit gates



Not so simple to learn noise of entangling gates

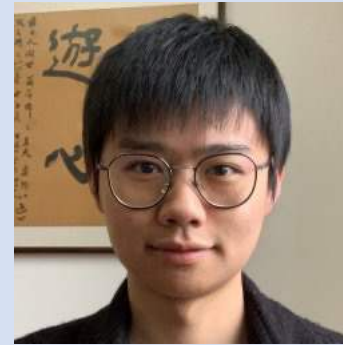
Entangler



Ideally wish

~~$$\Lambda_i^k(P_a) = f_{ia}^k P_a$$~~

Fundamental **no-go theorem on learning**



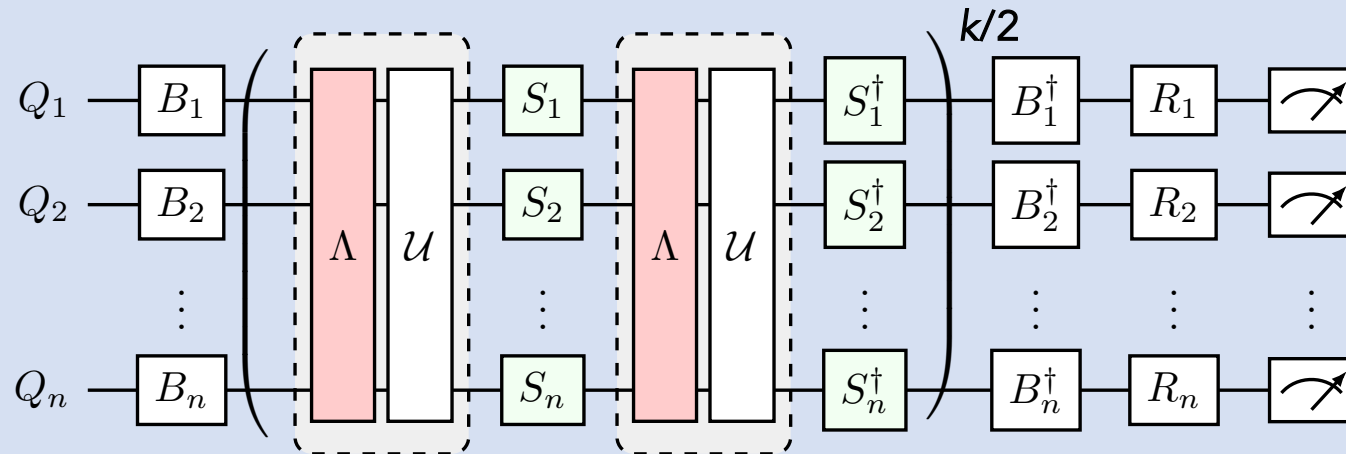
For general and in-depth

Senrui Chen, Y. Liu, M. Otten, A. Seif, B. Fefferman, L. Jiang
arXiv:2206.06362 (2022)

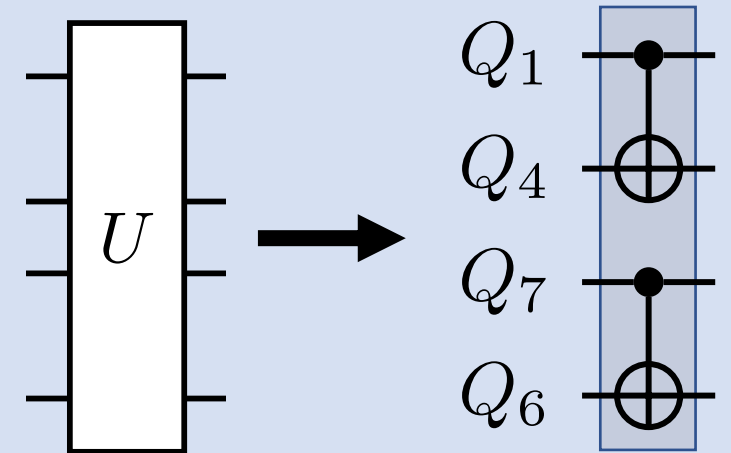
or supplement of our paper for qubit version and work by
S. Flammia, S Benjamin, and teams.

Solution: Custom protocol + weak assumption

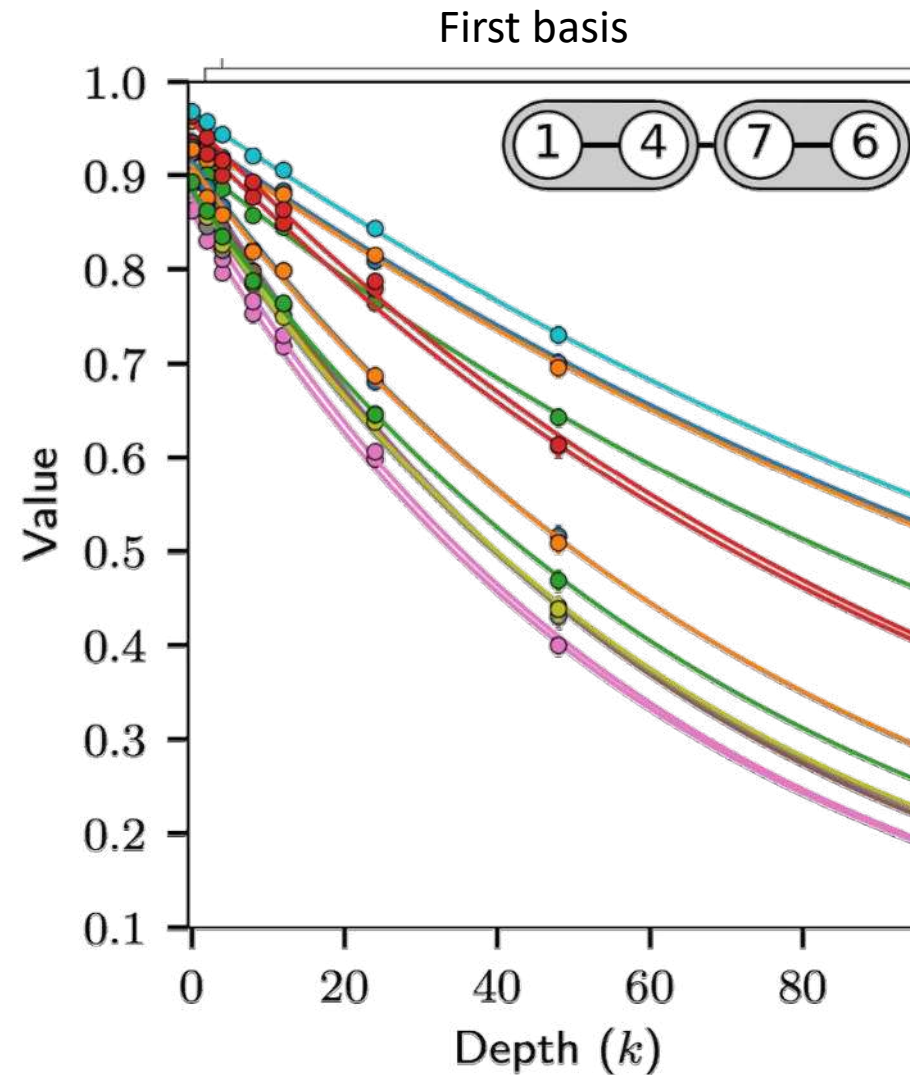
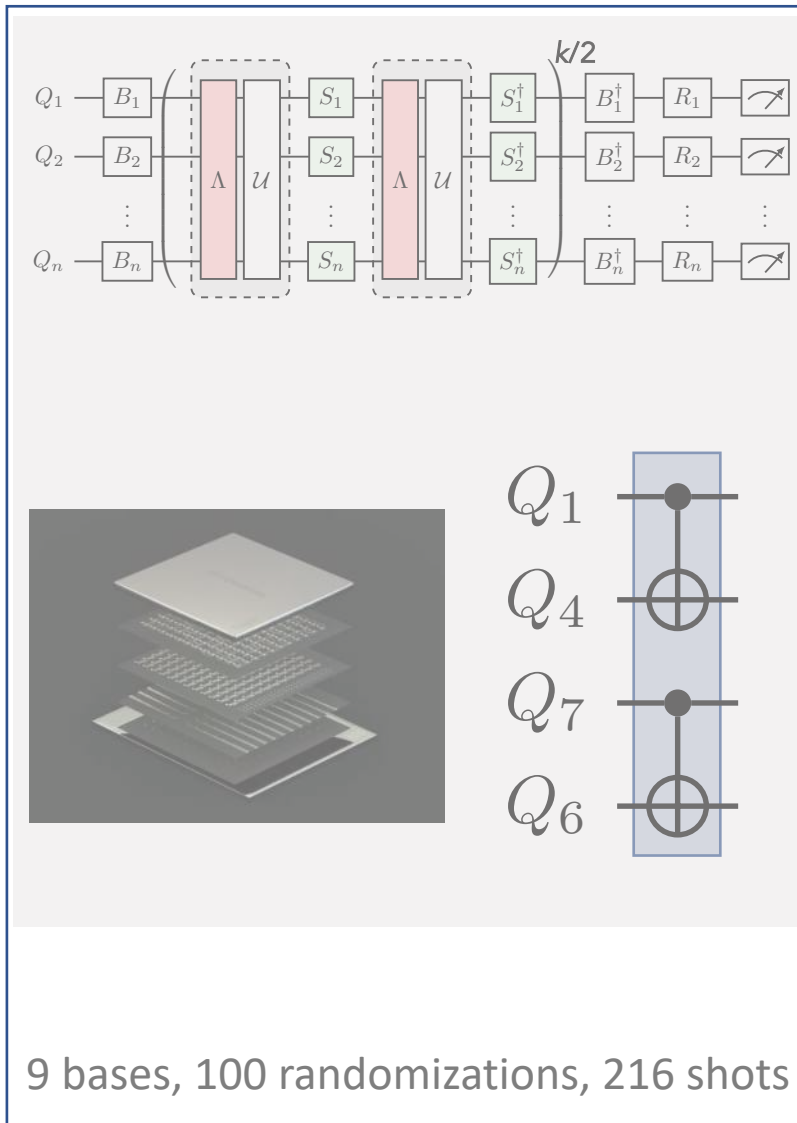
Learning circuits



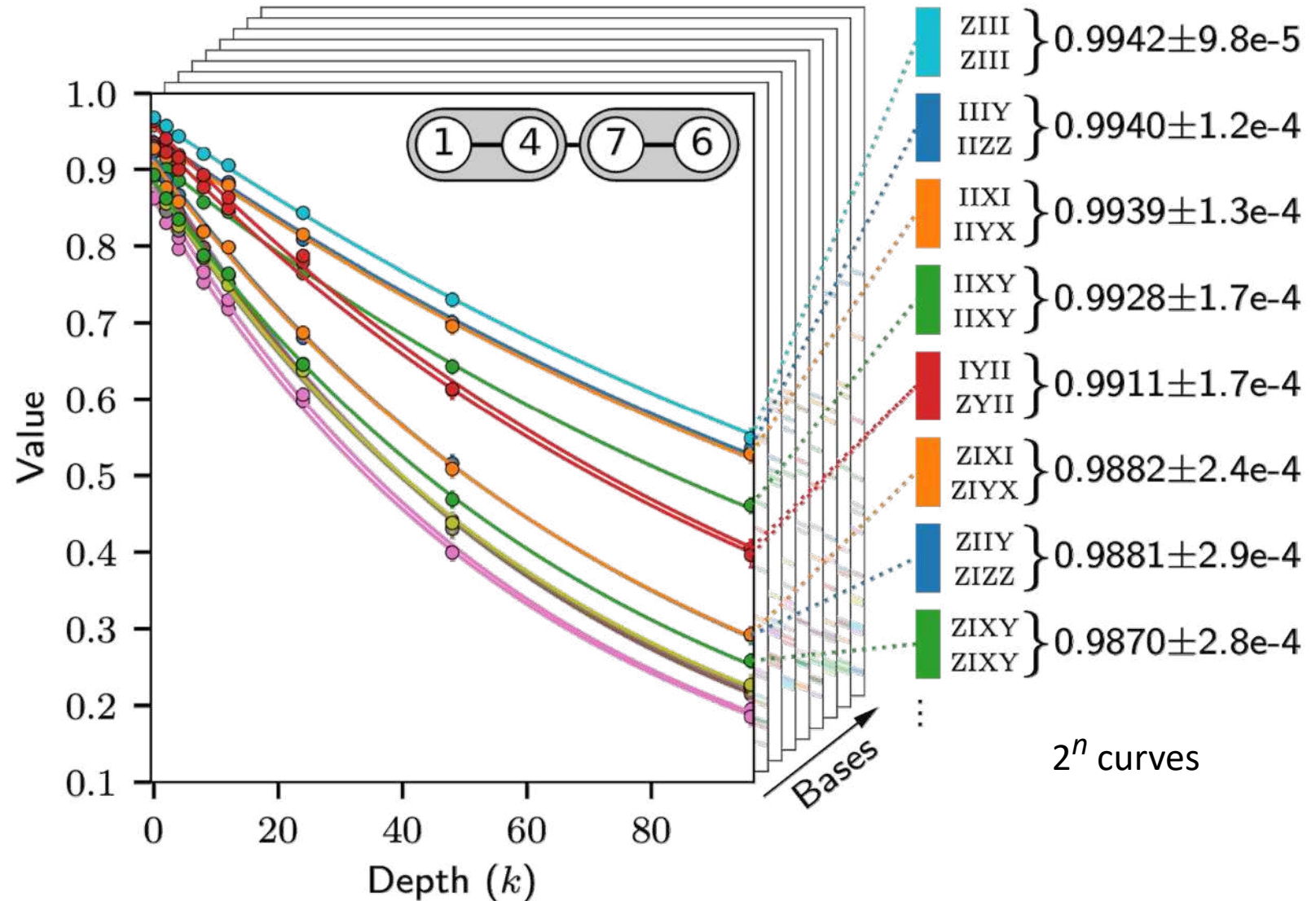
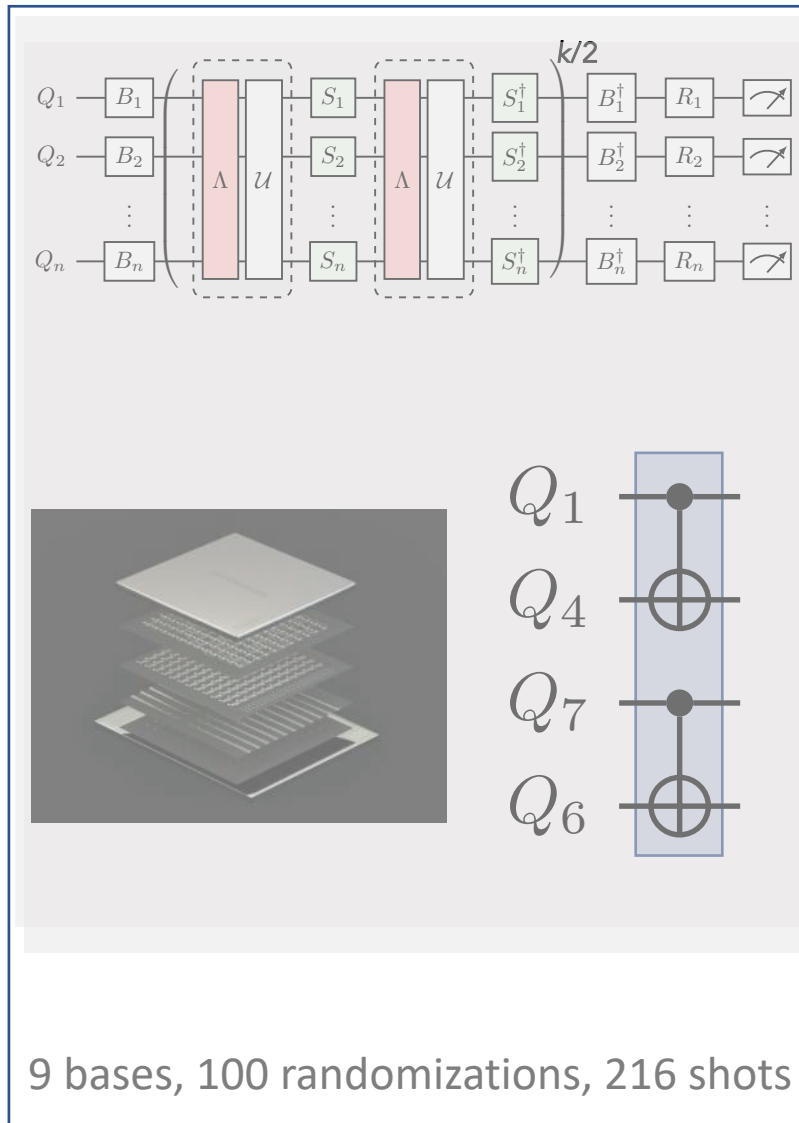
Example



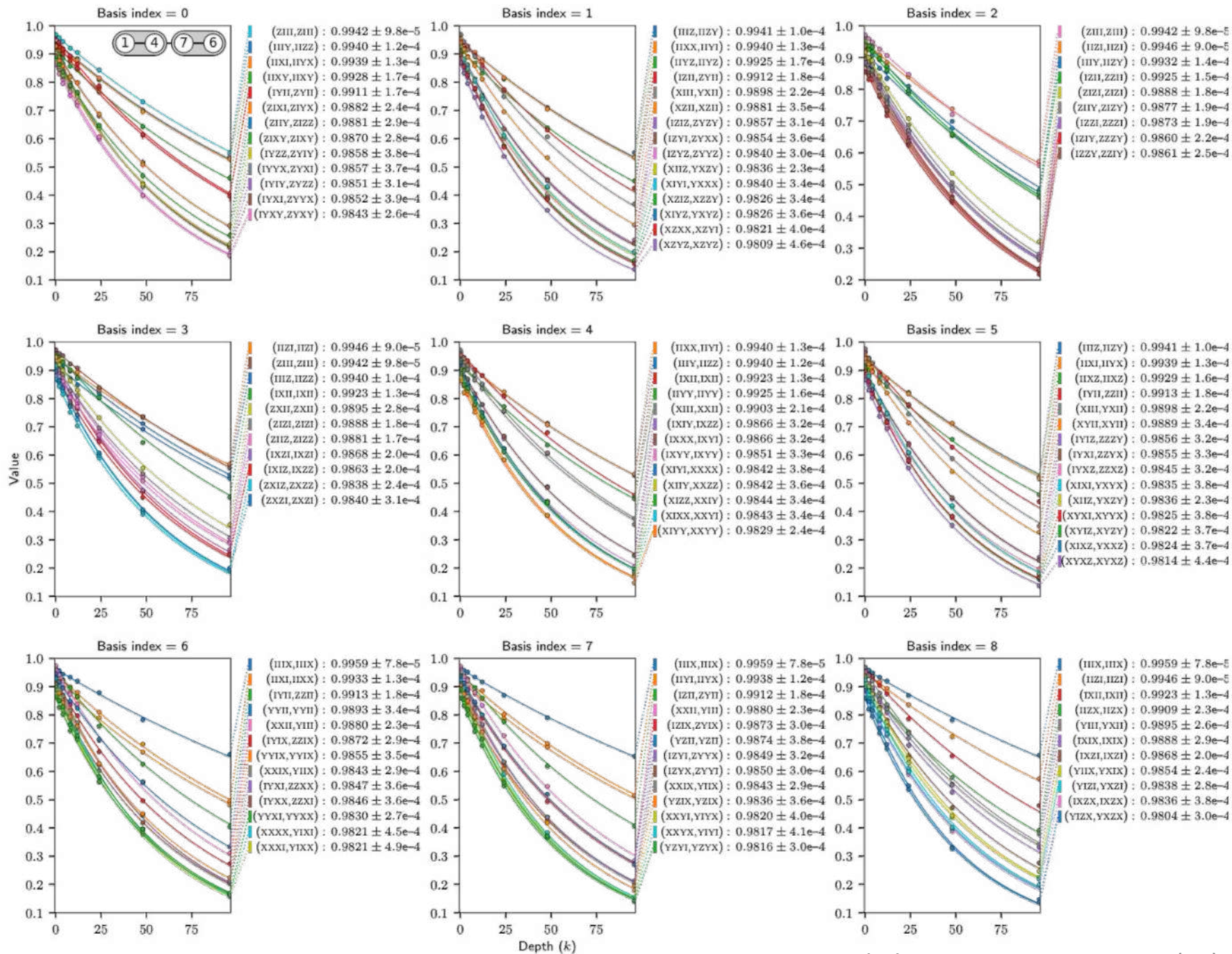
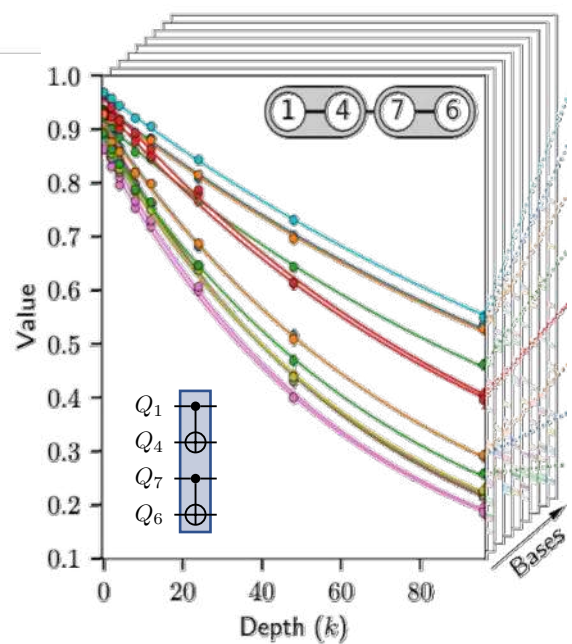
Learning the noise: raw data



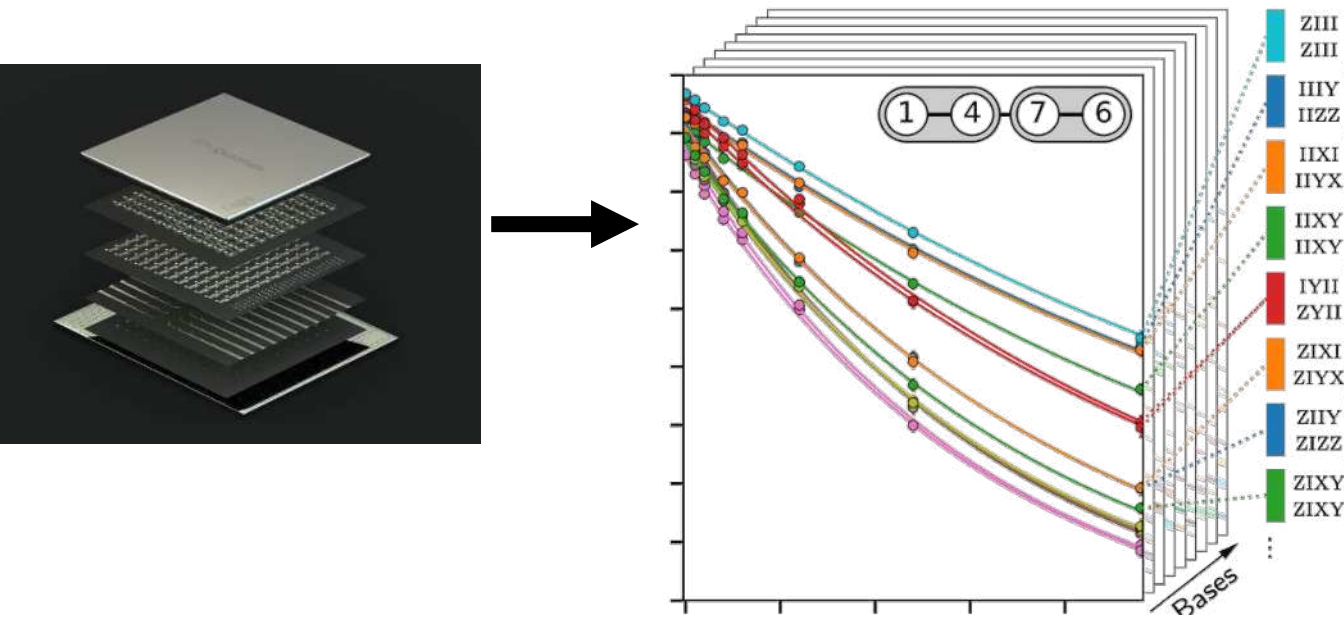
Learning the noise: raw data



Raw data



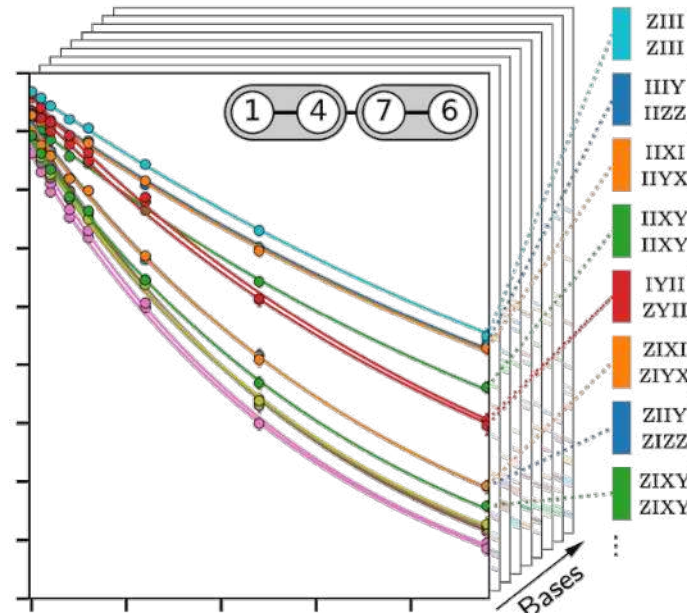
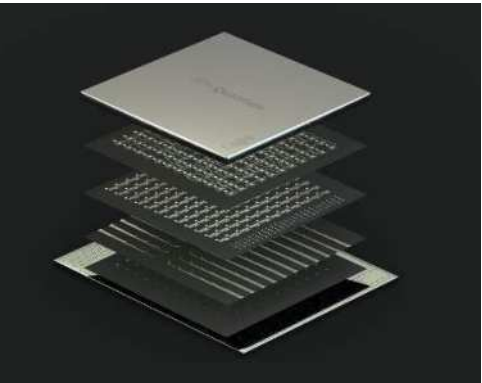
Reconstructing quantum channel from measurement data



$$\Lambda(\rho) = \sum_{a=0}^{4^n-1} c_a P_a \rho P_a^\dagger$$

Still 4^n

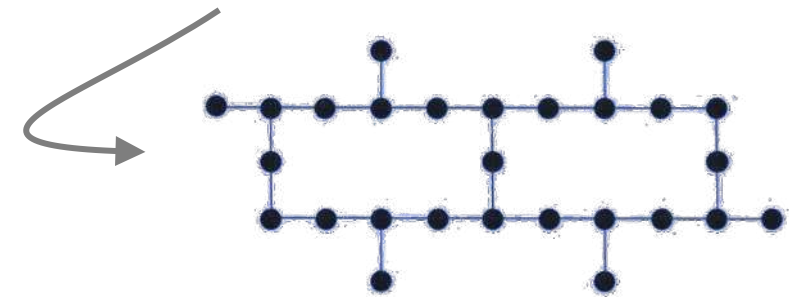
Sparse Pauli-Lindblad model



$$\Lambda(\rho) = \sum_{a=0}^{4^n-1} c_a P_a \rho P_a^\dagger$$

$$\Lambda(\rho) = \exp[\mathcal{L}](\rho)$$

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho)$$



Magic



icon: Eucalyp

Highlight: Ewout van den Berg

Notes

Dissipator for a given Pauli

$$\begin{aligned}\mathcal{D}[\rho]_P &= P \rho P^\dagger - \frac{1}{2}(P \rho P^\dagger + P^\dagger \rho P) \\ &= P \rho P^\dagger - \rho \\ &= (P \cdot P - I) \rho\end{aligned}$$

$$\begin{aligned}\hat{\mathcal{L}} &= \sum_a \gamma_a \left(\hat{\mathcal{P}}_a - \frac{1}{2} I \right) \\ &= \sum_a \gamma_a \mathcal{D}[\hat{P}_a]\end{aligned}$$

$$\begin{aligned}e^{\hat{\mathcal{L}}} &= \prod_a \exp(\gamma_a \mathcal{D}[\hat{P}_a]) \\ &= \prod_a \Lambda[\rho_a]\end{aligned}$$

$$\text{but } [\mathcal{D}[\rho_a], \mathcal{D}[\rho_b]] = 0$$

Each sub channel

$$\begin{aligned}
 \Lambda[\rho_a] &= \exp(\gamma \mathcal{D}[\rho]) \\
 &= \exp(\gamma(\hat{\rho}_a + \hat{\mathbb{I}})) \\
 &= \exp(\gamma \hat{\rho}_a) \exp(\gamma \hat{\mathbb{I}}) \exp(\cancel{\frac{-1}{2} \gamma^2 \sum \hat{\rho}_a \hat{\mathbb{I}}}) \dots \\
 &= \exp(\gamma \hat{\rho}_a) \exp(\gamma \hat{\mathbb{I}})
 \end{aligned}$$

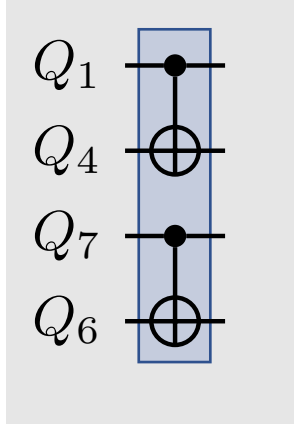
$$\begin{aligned}
 \hookrightarrow \sum_n \frac{\gamma^n \hat{\mathbb{I}}^n}{n!} &= \sum_n \frac{\gamma^n}{n!} \hat{\mathbb{I}} \\
 &= \exp(\gamma) \hat{\mathbb{I}} = \exp(\gamma) \hat{\mathbb{I}} \cdot \hat{\mathbb{I}}
 \end{aligned}$$

Each sub channel

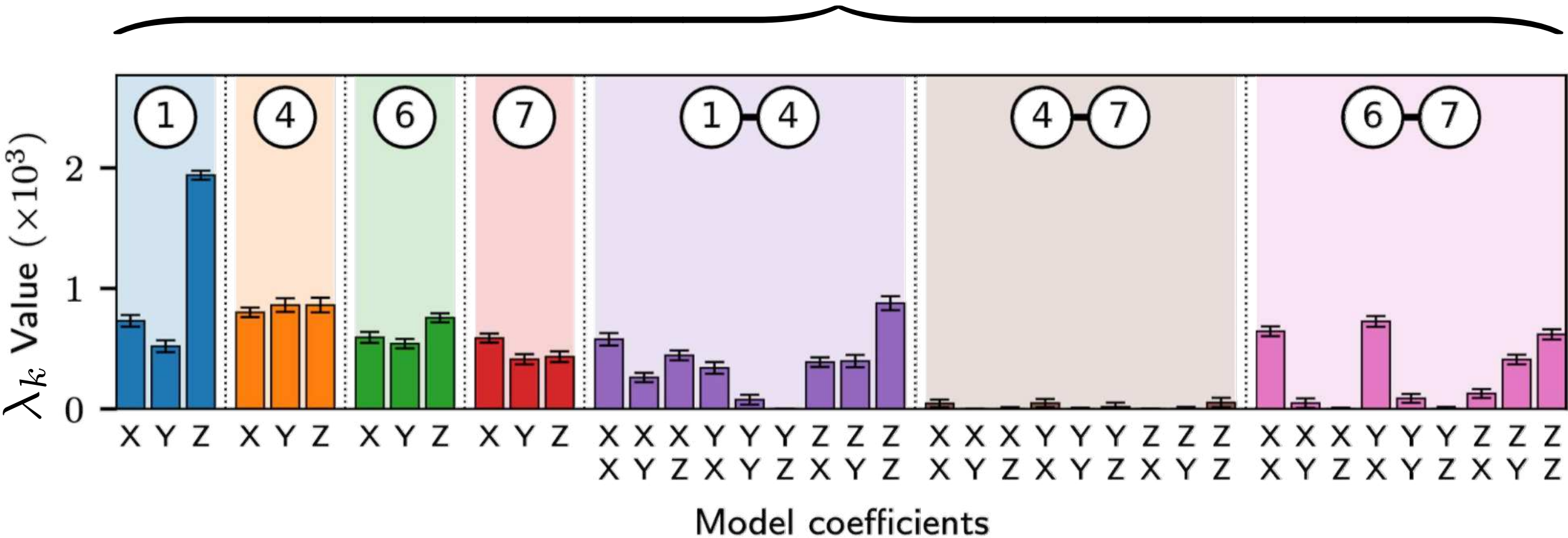
$$\begin{aligned}
 &= \exp(\gamma \hat{P}_a) \\
 &\quad \hookrightarrow \sum_{n=0}^{\infty} \frac{\gamma^n}{n!} (\hat{P}_a)^n \quad \text{note} \quad \begin{aligned} \hat{P}_a^{n-\text{even}} &= \hat{I} \\ \hat{P}_a^{n-\text{odd}} &= \hat{P}_a \end{aligned} \\
 &= \left(\sum_{\substack{n=0 \\ \text{even}}}^{\infty} \frac{\gamma^n}{n!} \right) \hat{I} + \left(\sum_{\substack{n=1 \\ \text{odd}}}^{\infty} \frac{\gamma^n}{n!} \right) \hat{P}_a \\
 &= \cosh(\gamma) \hat{I} + \sinh(\gamma) \hat{P}_a
 \end{aligned}$$

$$\Lambda[\hat{P}_a] = \cosh(\gamma_a) \hat{I} + \sinh(\gamma_a) \hat{P}_a$$

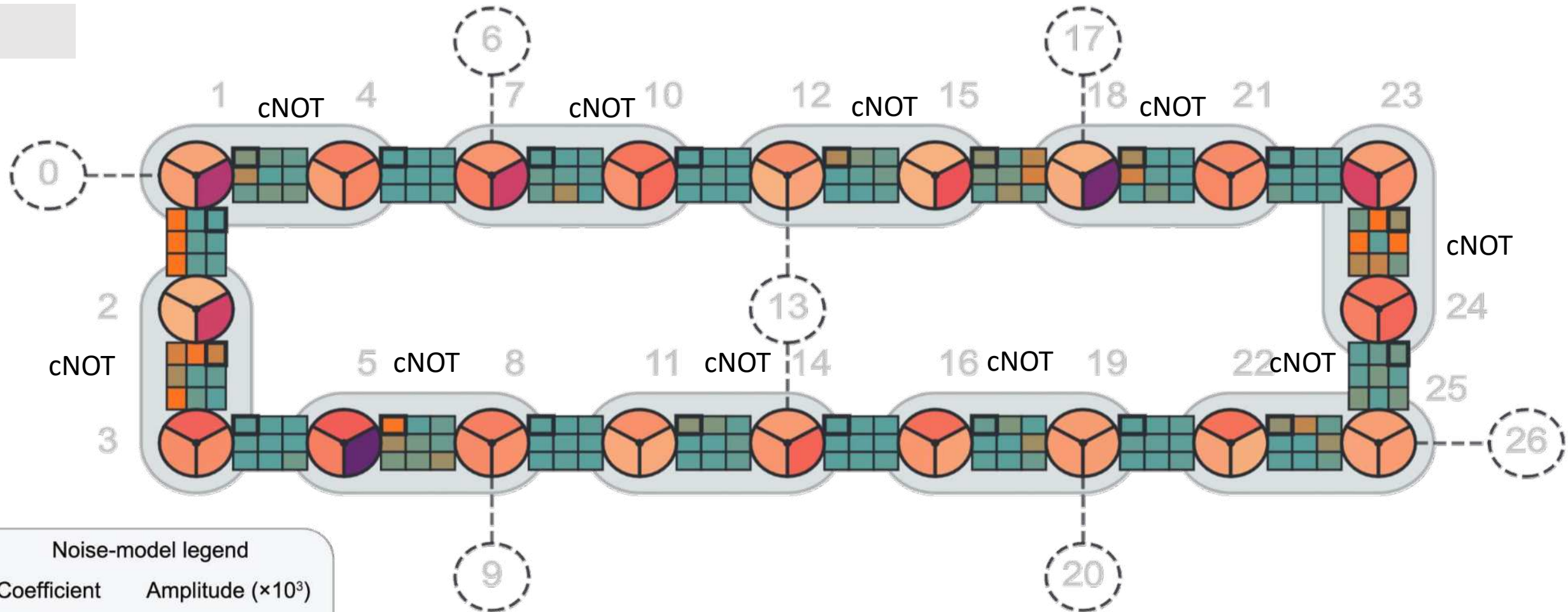
Sparse Lindblad tomogram



$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k \left(P_k \rho P_k^\dagger - \rho \right)$$



Noise tomogram for 20Q Ising-ring Trotter layer



Noise-model legend

Coefficient



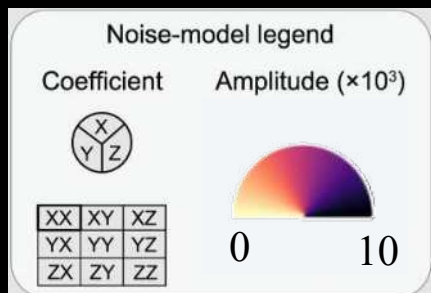
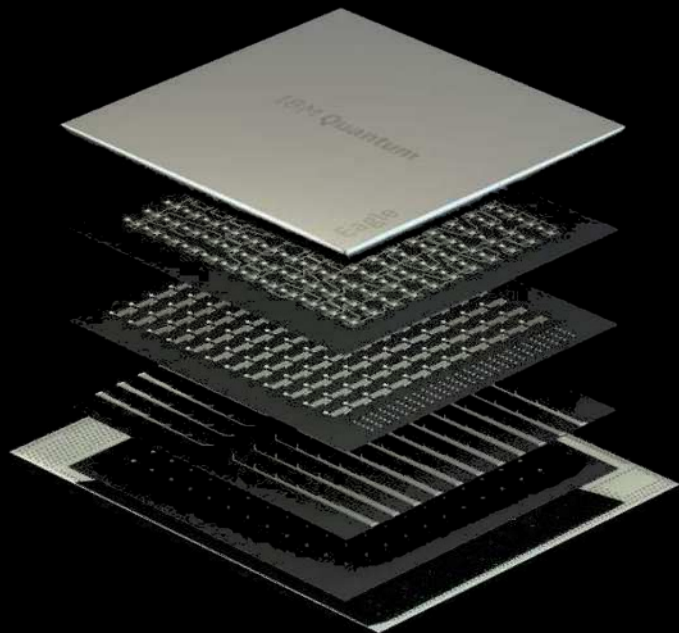
Amplitude ($\times 10^3$)



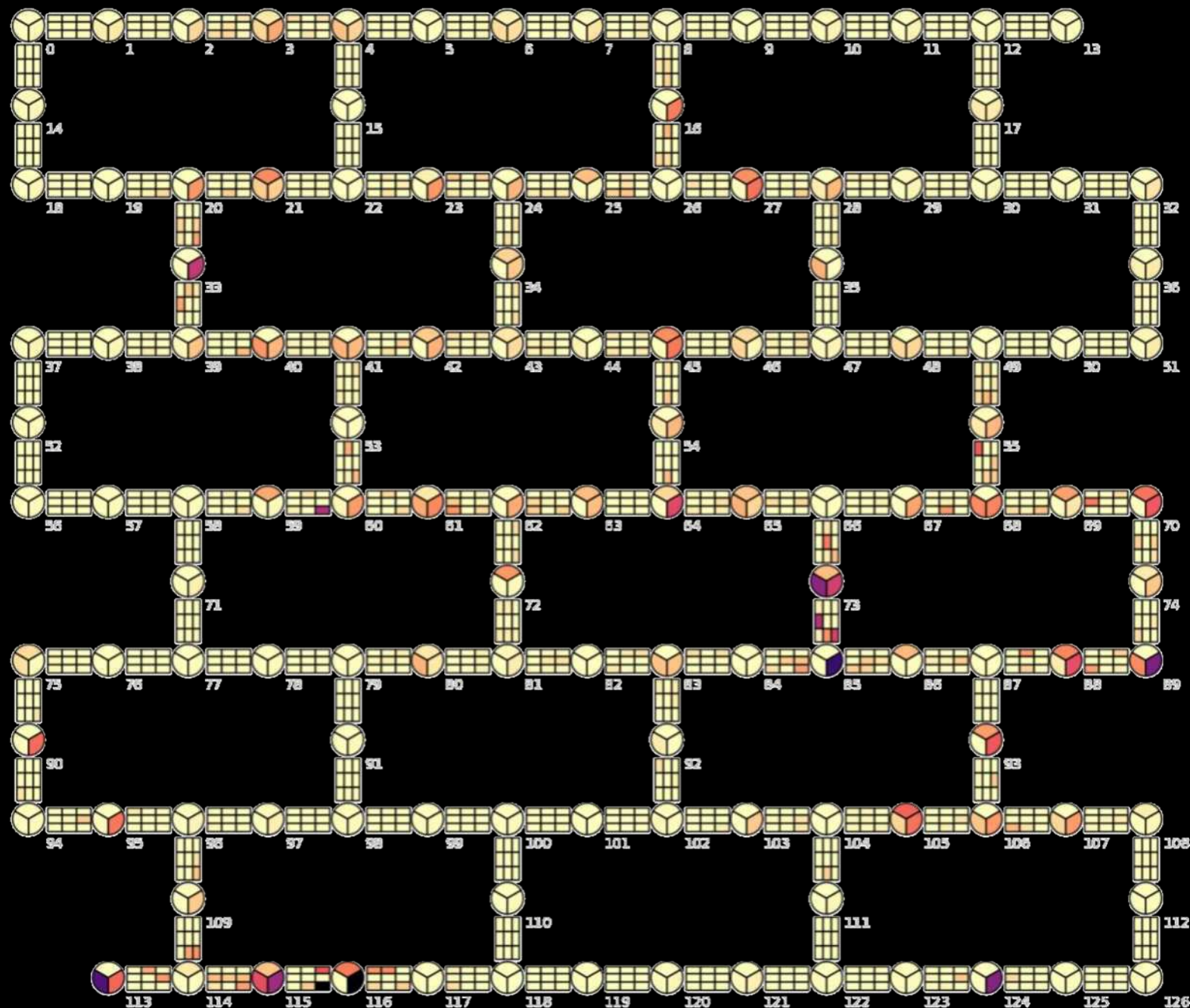
XX	XY	XZ
YX	YY	YZ
ZX	ZY	ZZ



Noise tomogram for 127Q Trotter layer



Same number of
learning circuits
as for 4Q



Simulating transverse-field Ising model time evolution with PEC



Spins

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$

J : exchange coupling between neighboring spins

h : transverse magnetic field

Average magnetization
density?

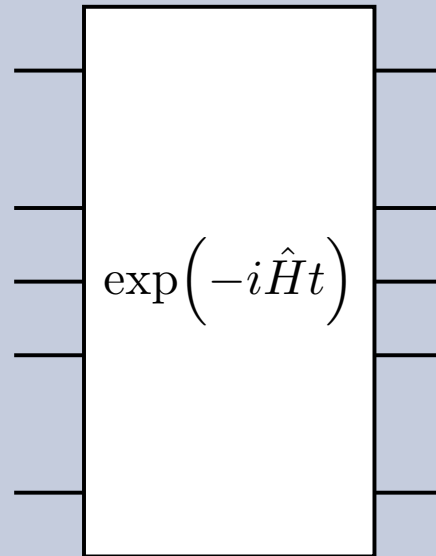
$$\vec{M} := \sum_n (\langle X \rangle_n, \langle Y \rangle_n, \langle Z \rangle_n) / N$$

Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$



Time evolution

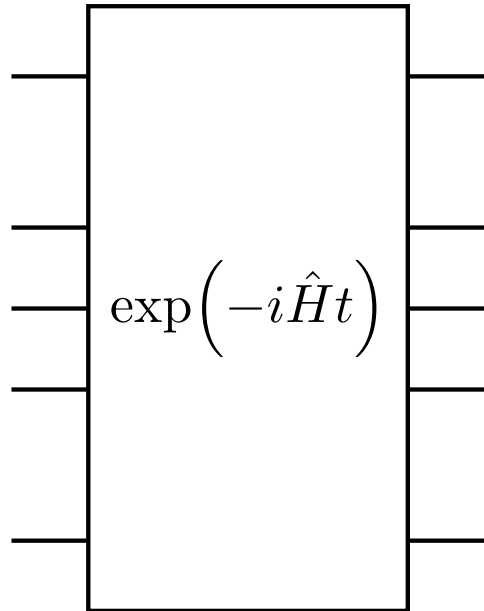


See lecture by
Frank Pollmann

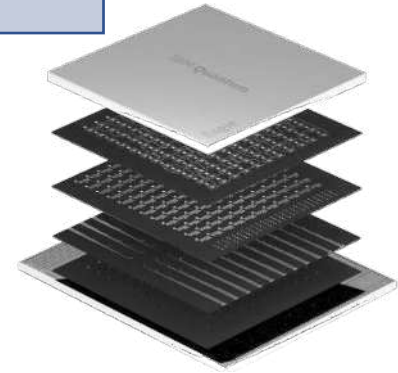
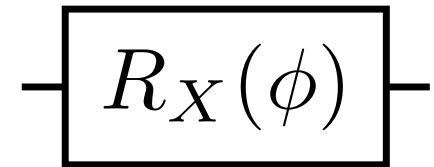
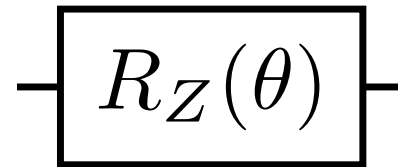
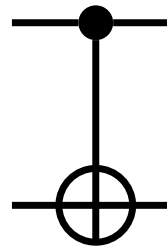


Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$

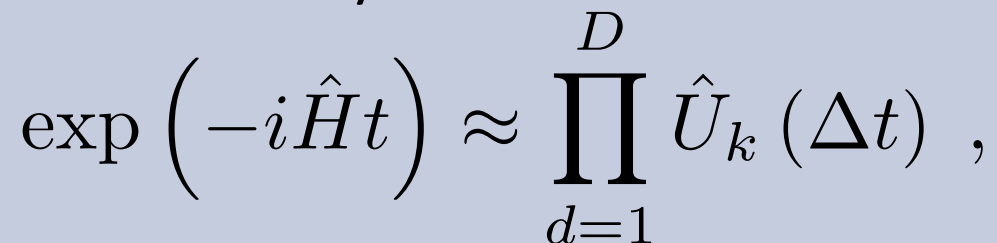


Decompose into native gates that we can actually implement on the QPU





See lecture by Frank


$$\hat{U}_1(\Delta t) = \prod_{j=1}^N e^{-i\hat{H}_j \Delta t},$$

First order expansion

$$\hat{H}_{ZZ} = \sum_j Z_j Z_{j+1}$$

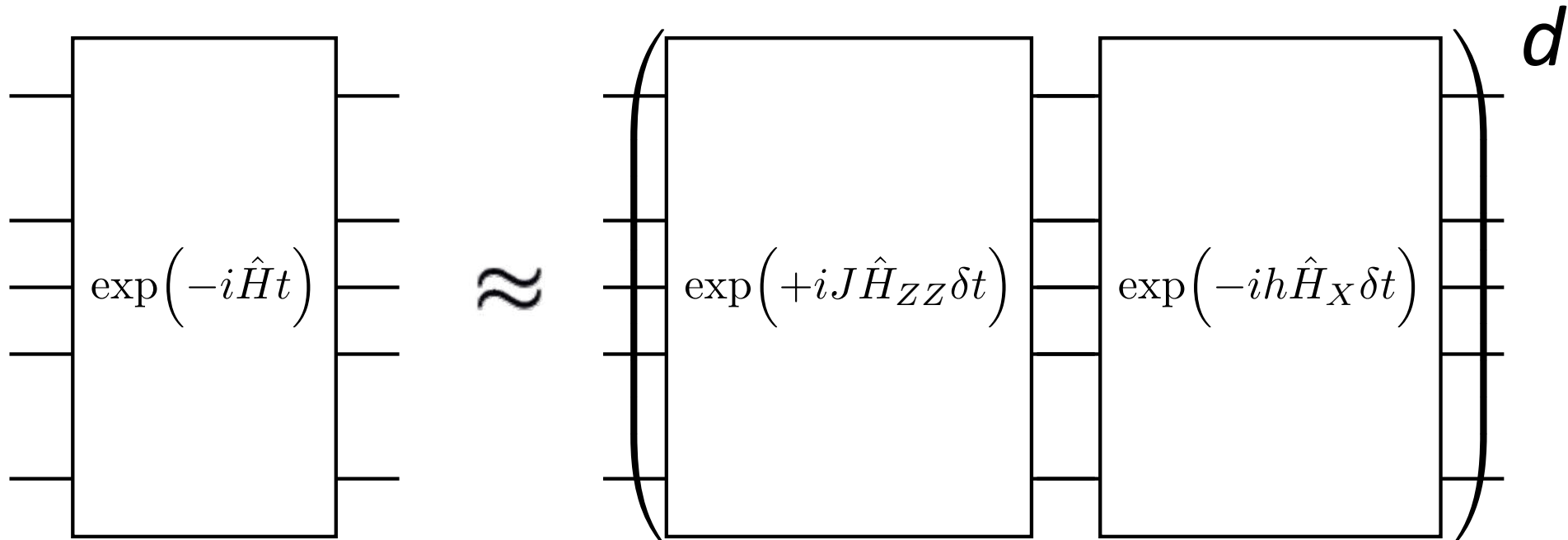
$$\hat{H}_X = \sum_j \hat{X}_j$$

First order Trotter

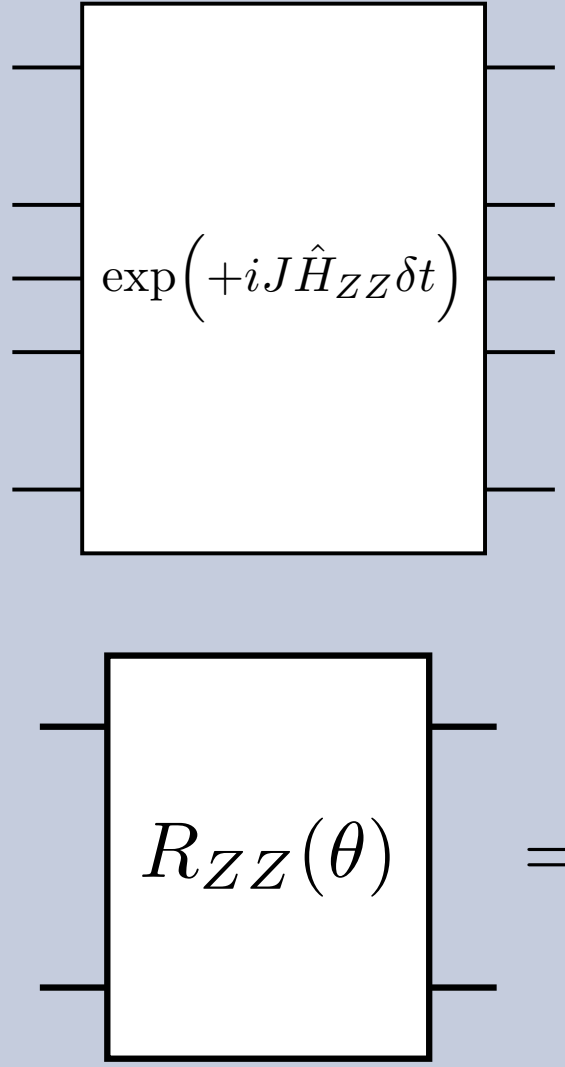
For error bounds, see [arxiv:2302.14592](https://arxiv.org/abs/2302.14592)

Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$



Step 1: Trotter circuit (30 s)



$$\exp \left(-i \frac{\theta}{2} \sum_{j=0}^{n-1} Z_j Z_{j+1} \right) = \quad (\text{use } [Z_j Z_{j+1}, Z_{j'} Z_{j'+1}] = 0)$$

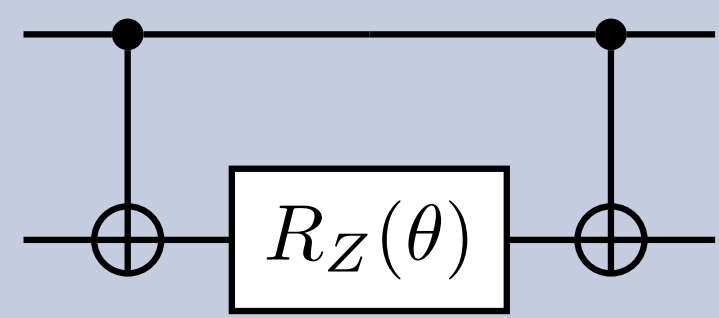
$$\theta := -2J\delta t$$

$$= \prod_{j=0}^{n-1} \exp \left(-i \frac{\theta}{2} Z_j Z_{j+1} \right)$$

$$= \prod_{j=0}^{n-1} R_{ZZ}(\theta)$$

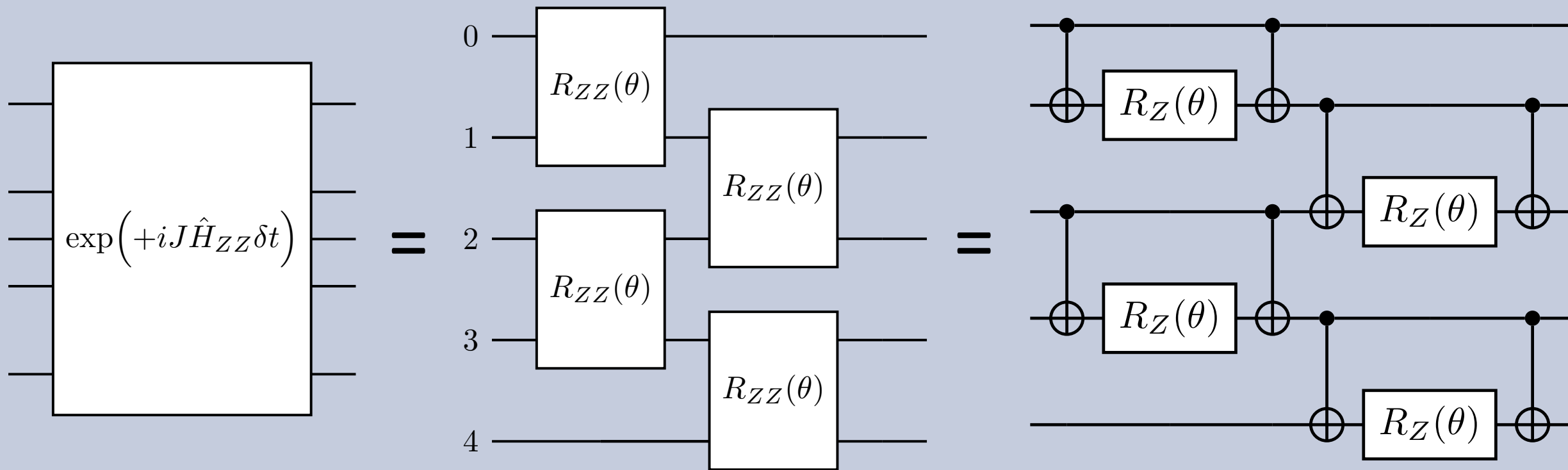
$$R_{ZZ}(\theta) = \exp \left(-i \frac{\theta}{2} ZZ \right)$$

$$= \text{cX} R_Z(\theta) \text{cX}$$



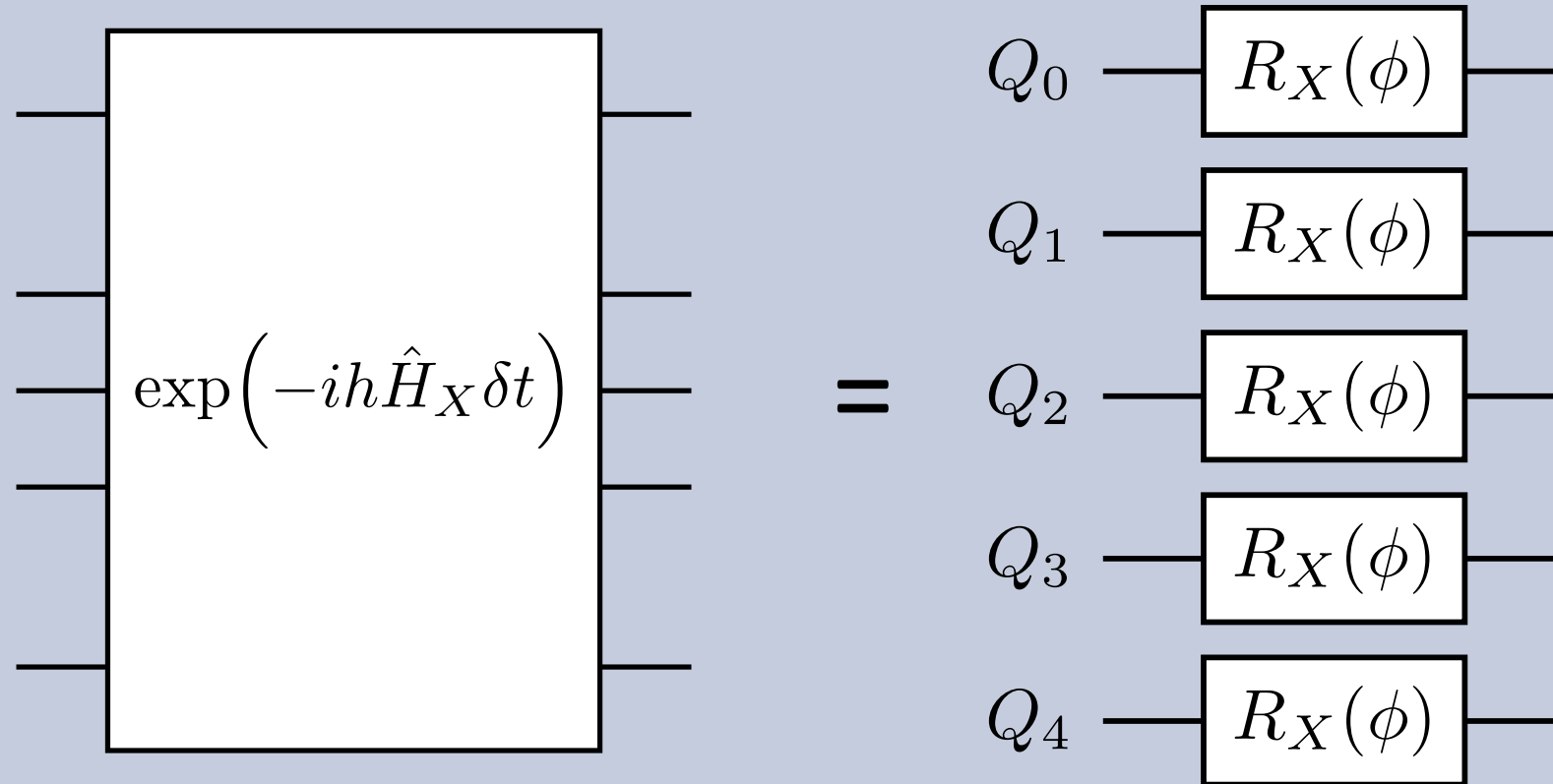


Step 1: Decompose into native gates





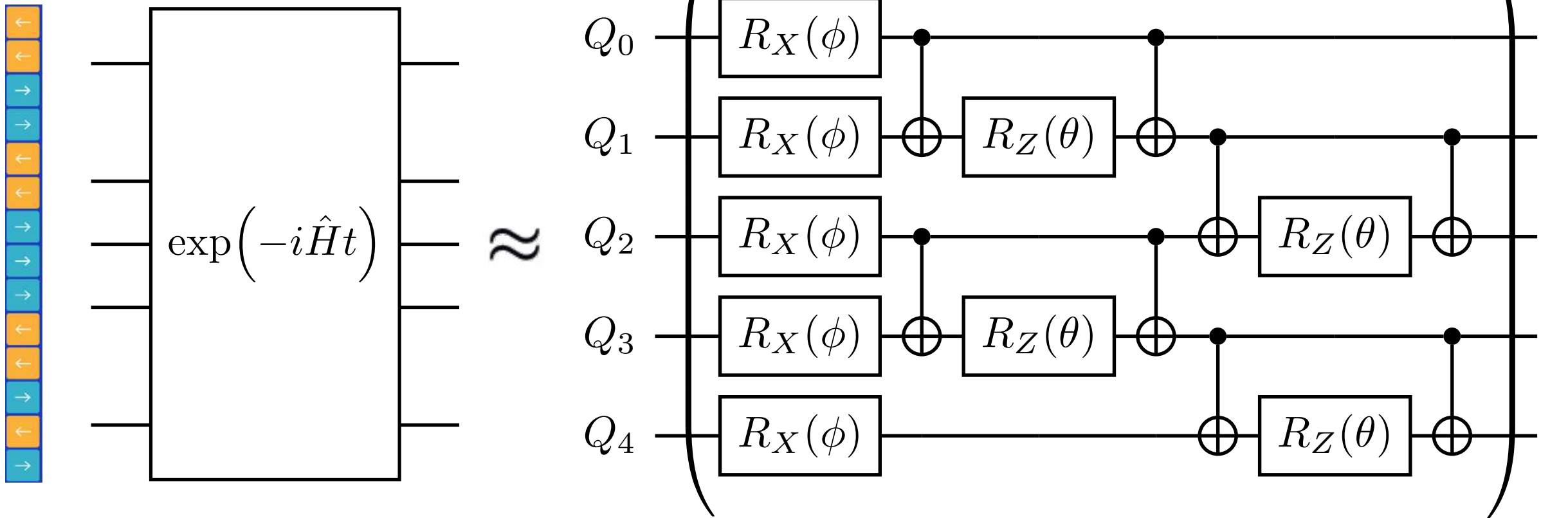
Step 1: Decompose into native gates



$$R_X(\phi) := \exp\left(-\frac{1}{2}\phi X\right)$$

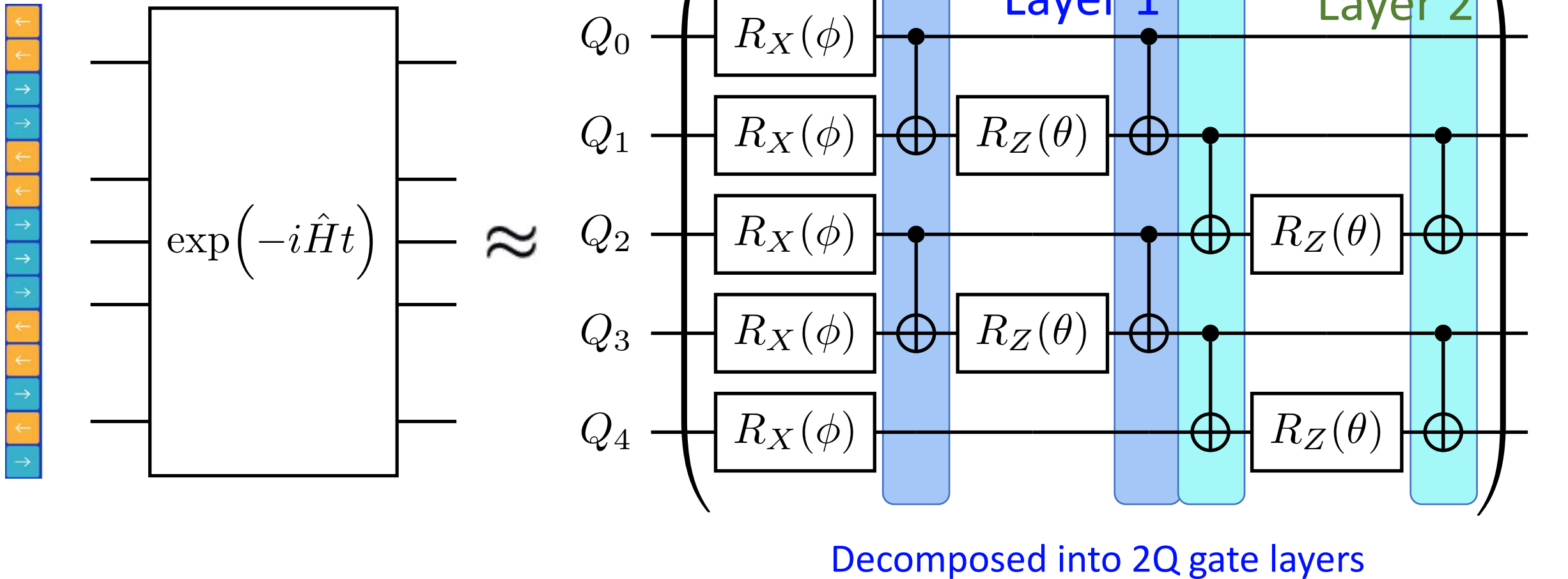
$$\phi := +2h\delta t$$

Step 1: Quantum Hamiltonian time evolution



Decomposed into quantum computer
native gates

Step 2: Decompose into layers



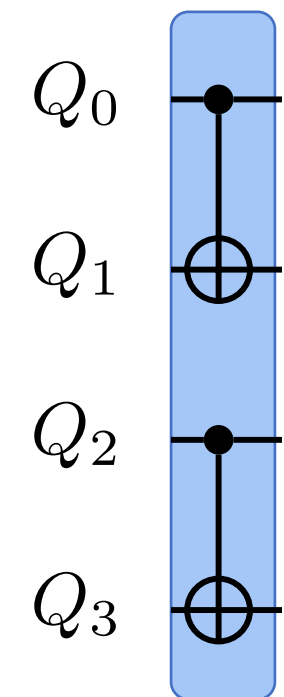
Step 2: Learn the noise on each layer

Let's make even simpler, and make it 4 qubits (instead of 5 for our first example)

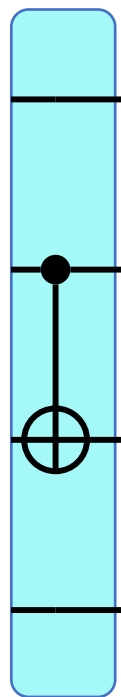
Layer 1

Layer 2

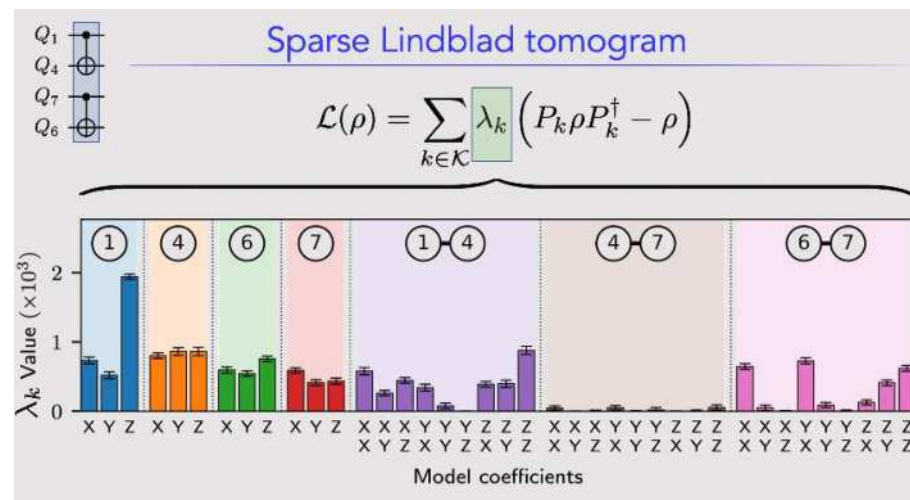
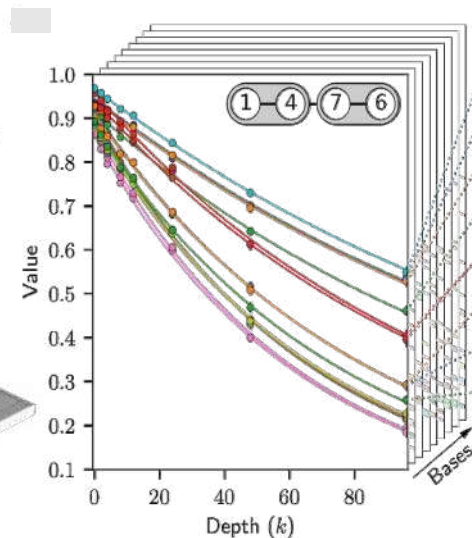
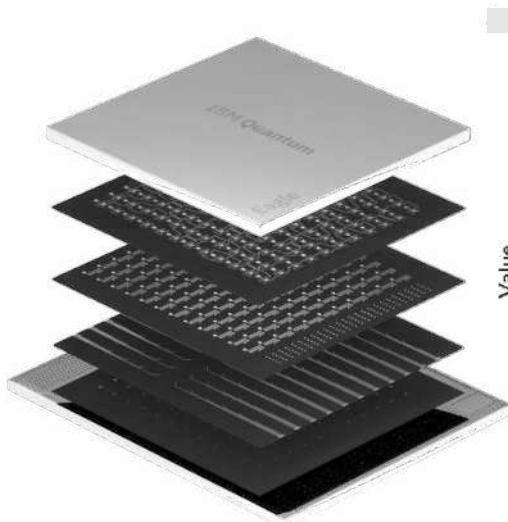
See lecture 2 for learning the noise



Λ_1



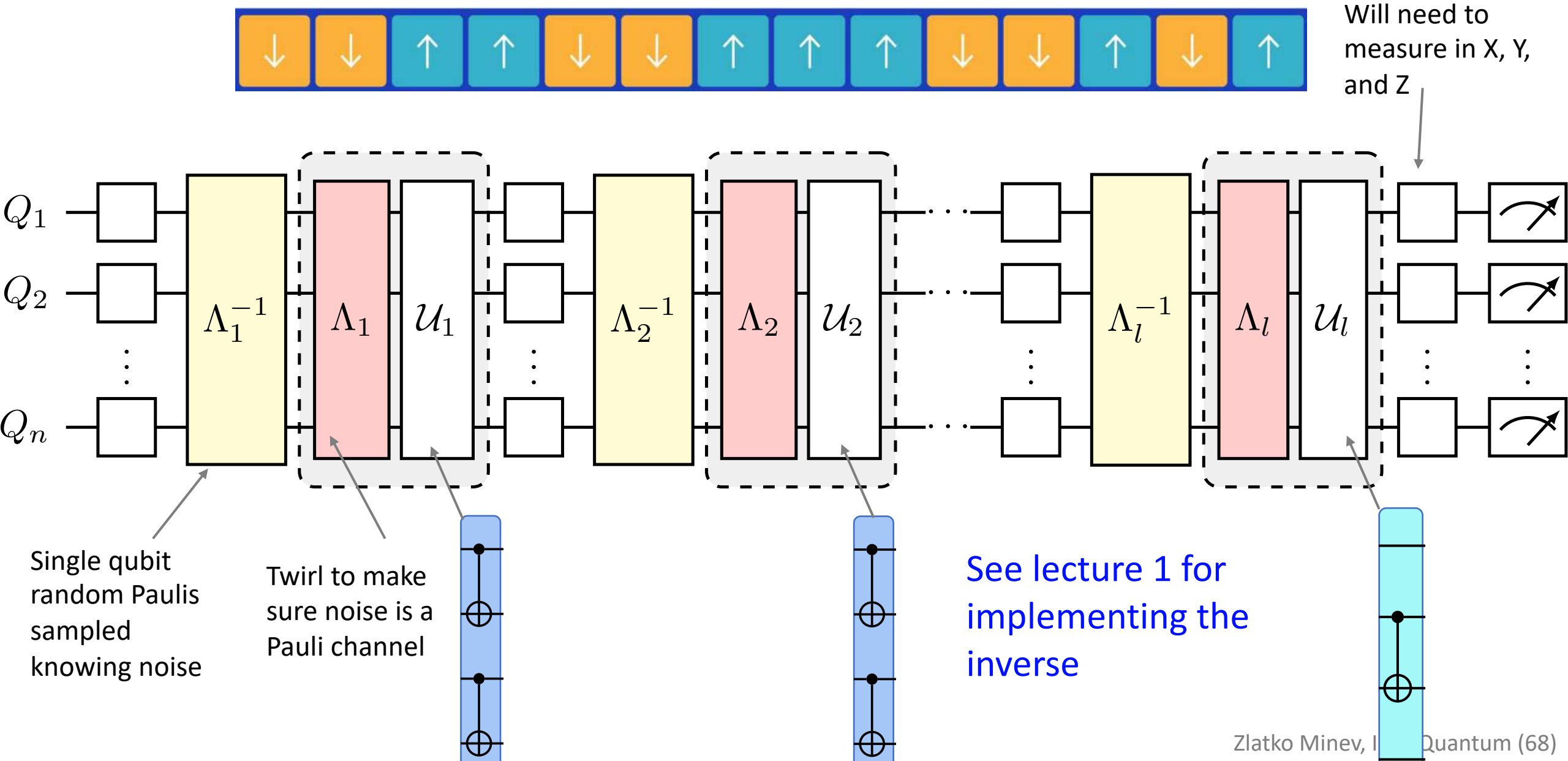
Λ_2



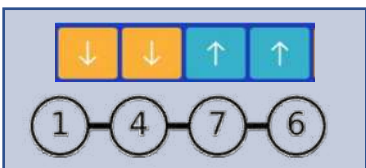
$$\gamma_1 = 1.0309 \pm 8.40 \cdot 10^{-5}$$

$$\gamma_2 = 1.0384 \pm 2.20 \cdot 10^{-4}$$

Step 3: Cancel the noise

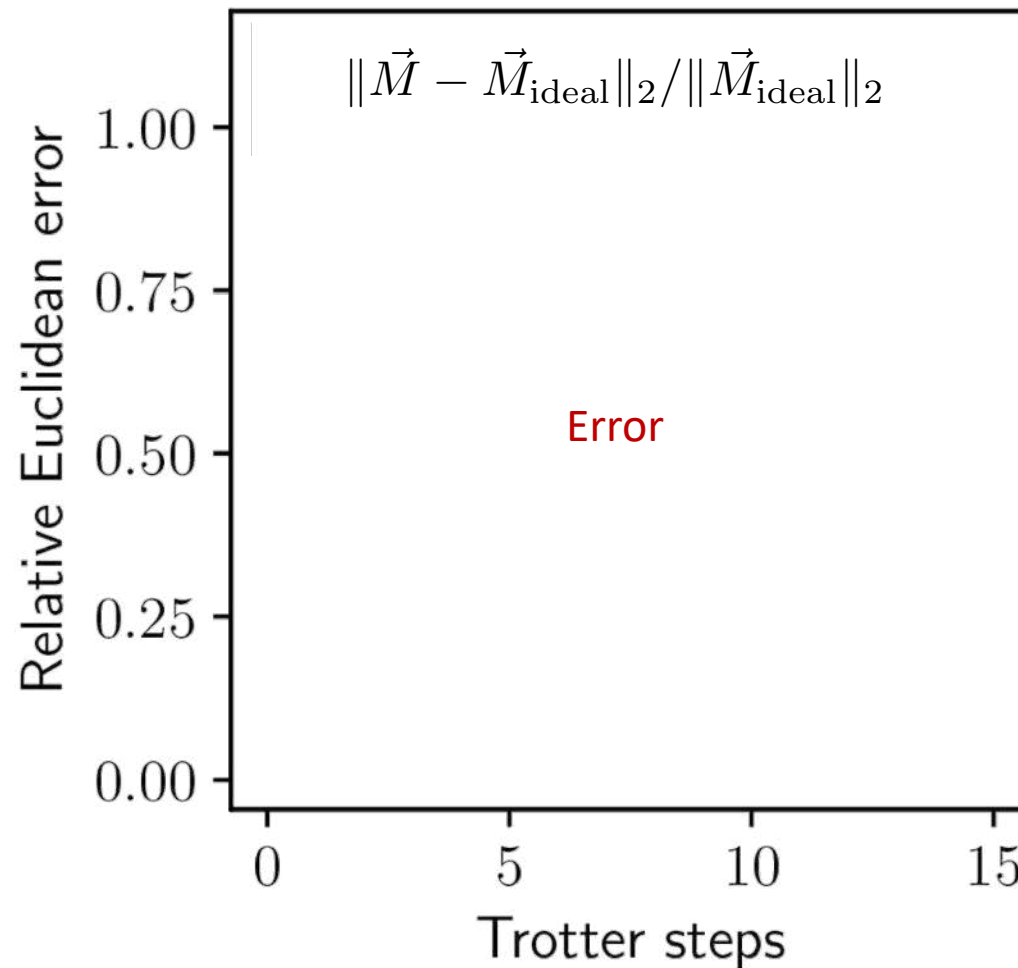
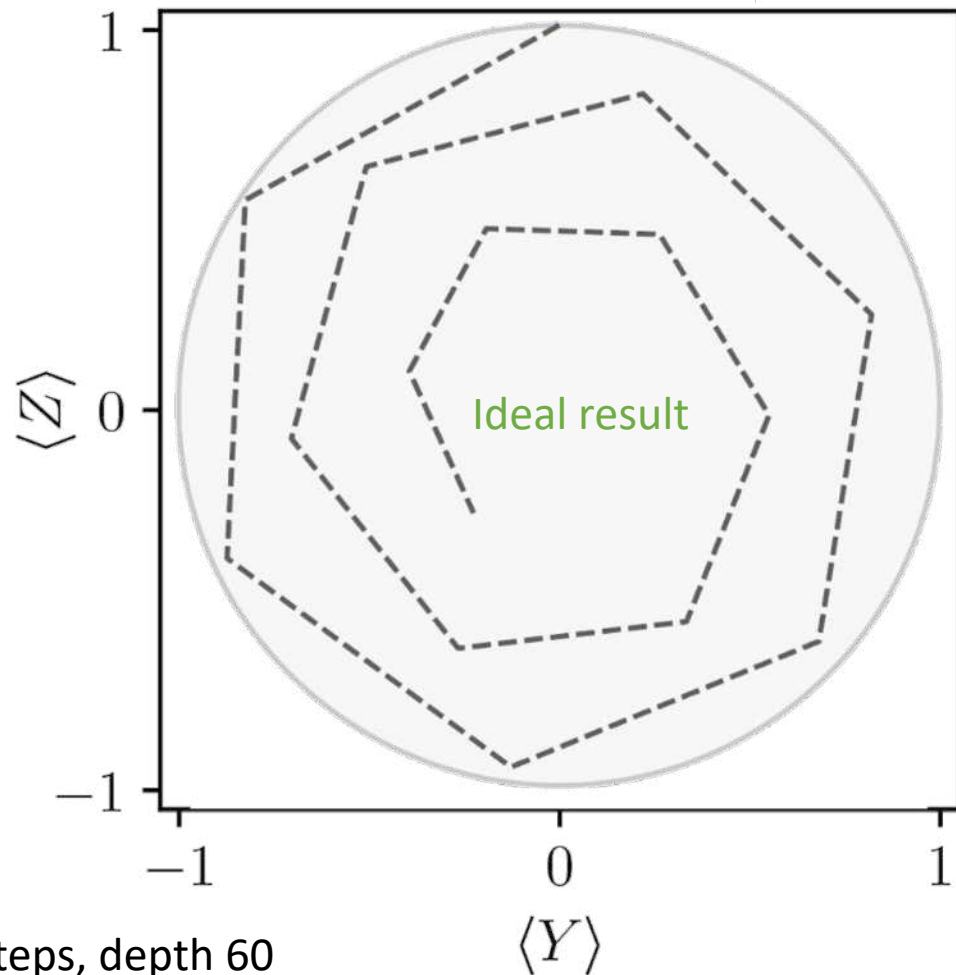


Ideal Ising model evolution



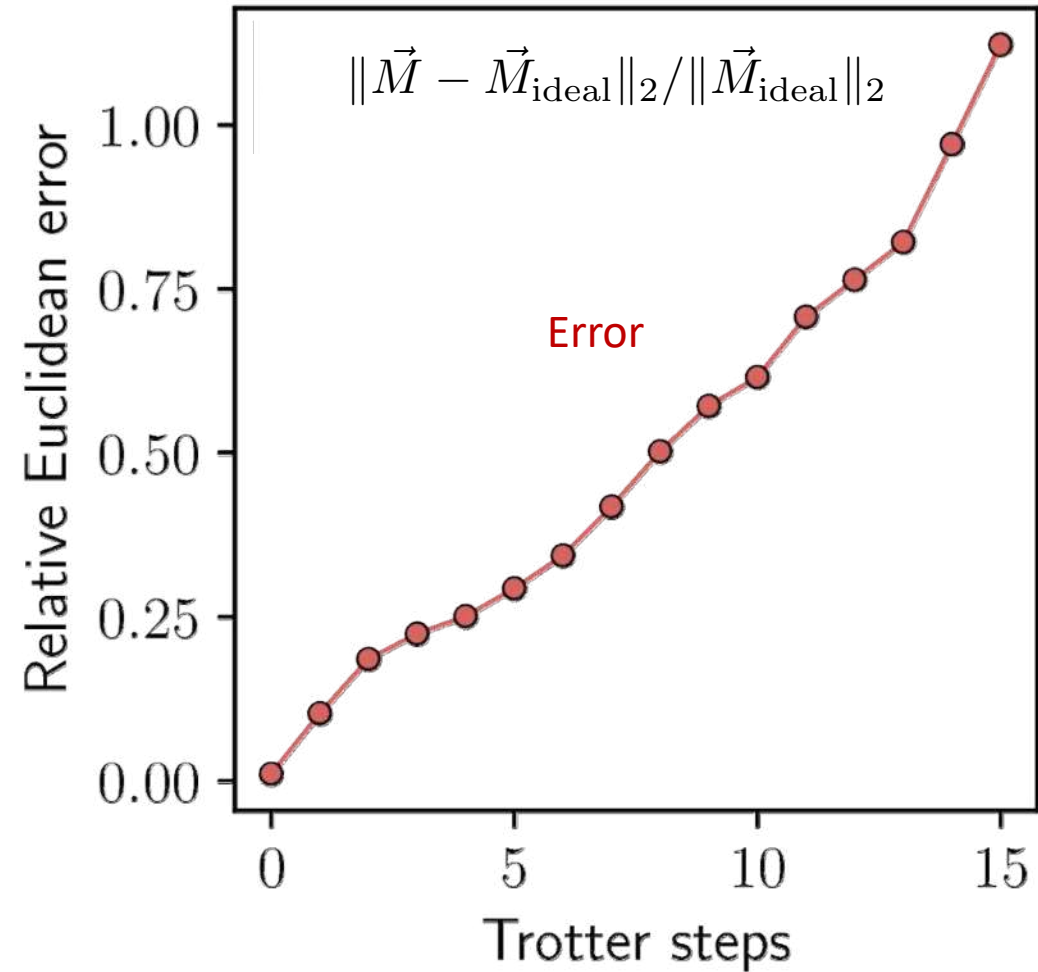
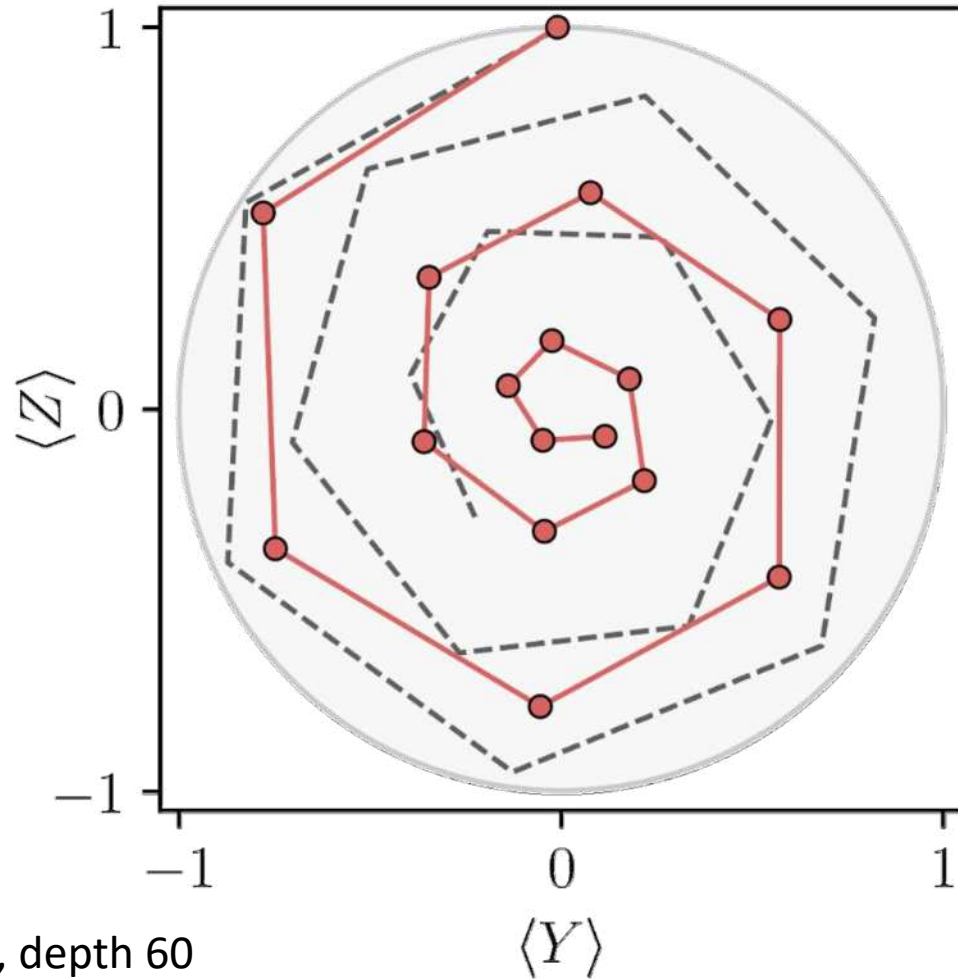
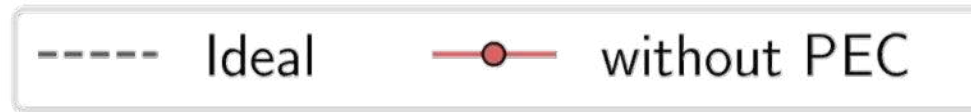
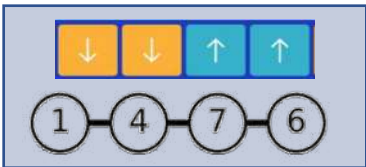
--- Ideal

$$\vec{M} := \sum_n (\langle X \rangle_n, \langle Y \rangle_n, \langle Z \rangle_n) / N$$



4Q, 15 steps, depth 60
 $h = 1, J = -0.15, \delta t = 1/4$

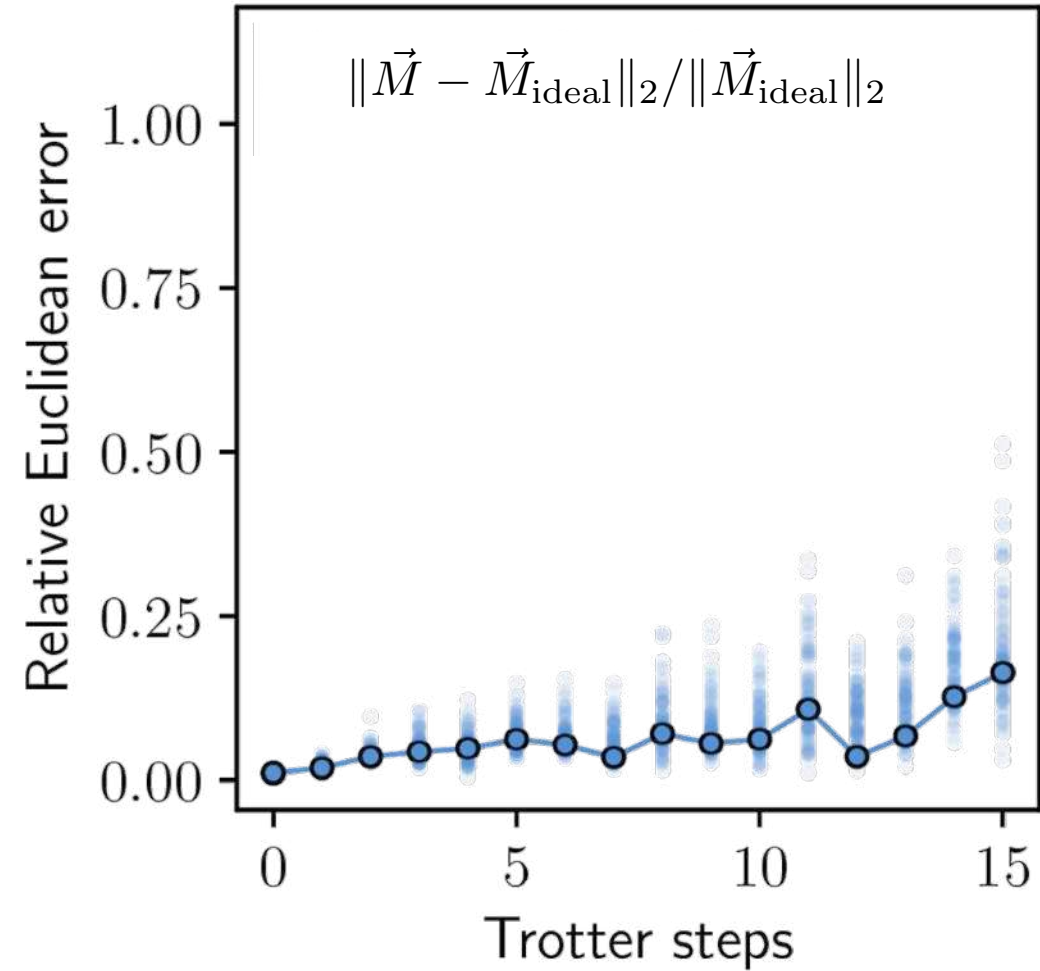
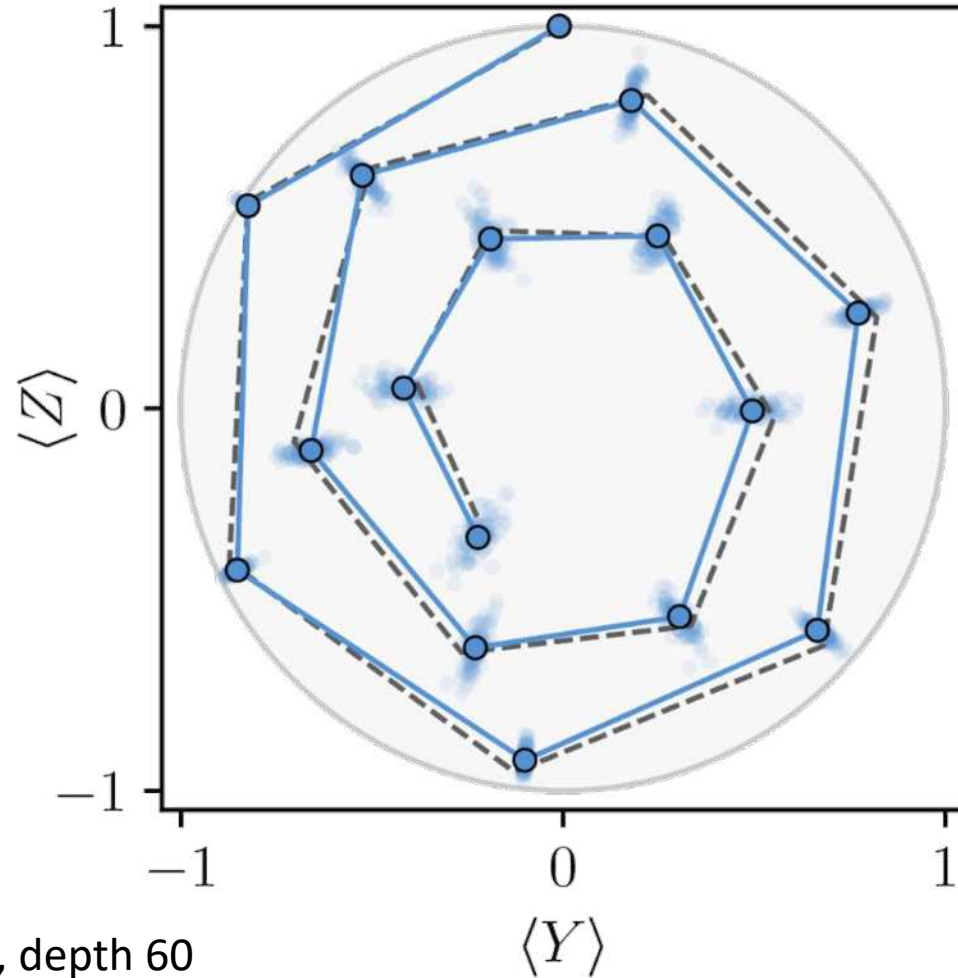
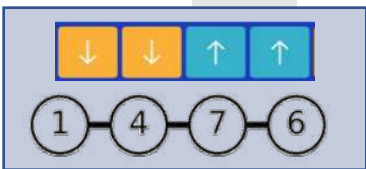
Without PEC: but with DD & twirl readout mitigation



15 steps, depth 60

$h = 1, J = -0.15, \delta t = 1/4$

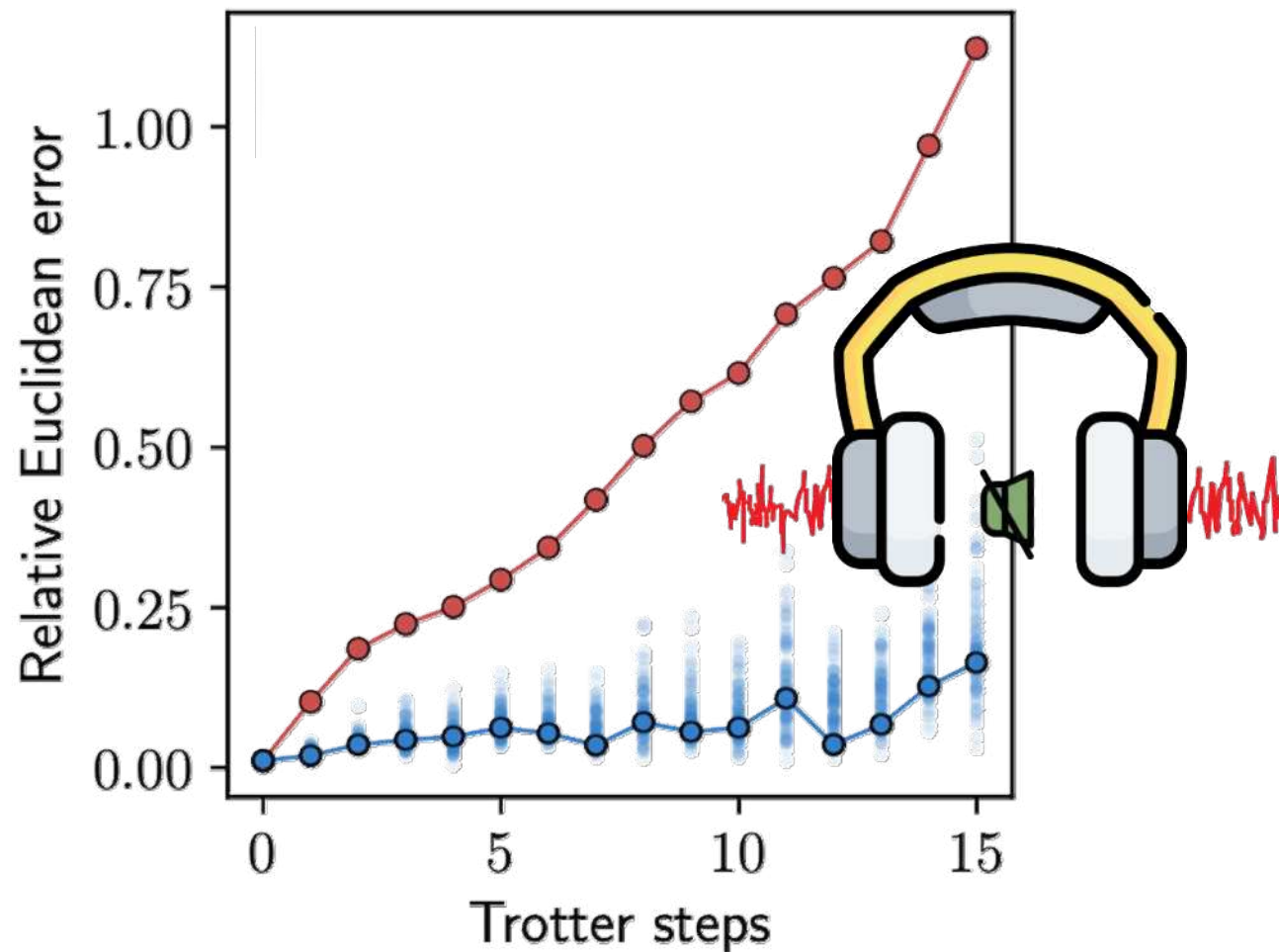
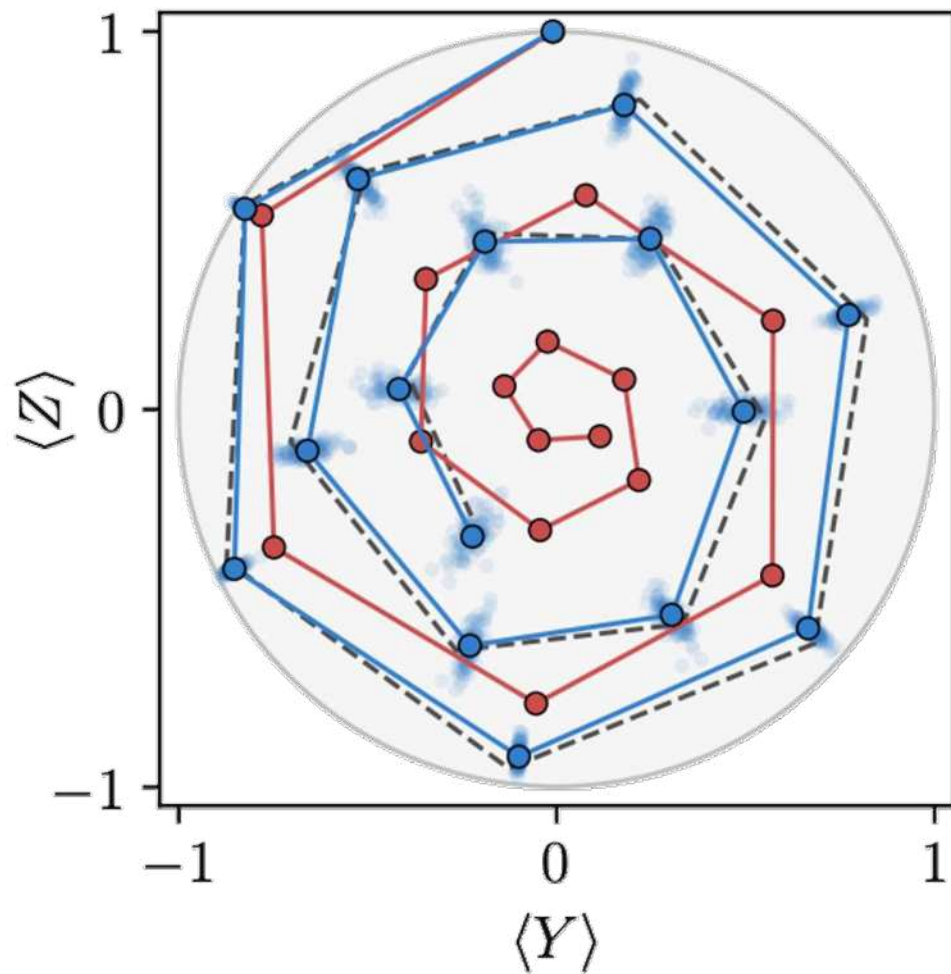
With PEC



15 steps, depth 60

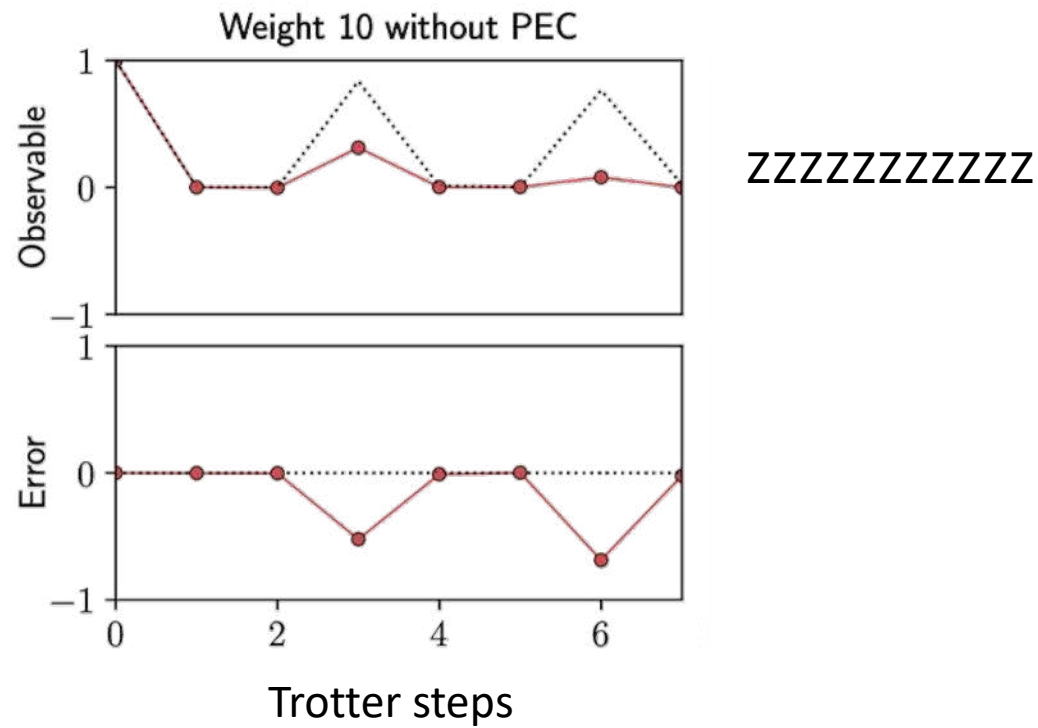
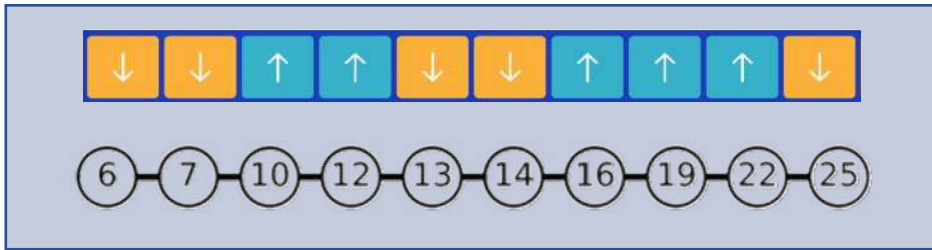
$h = 1, J = -0.15, \delta t = 1/4$

With vs. without PEC



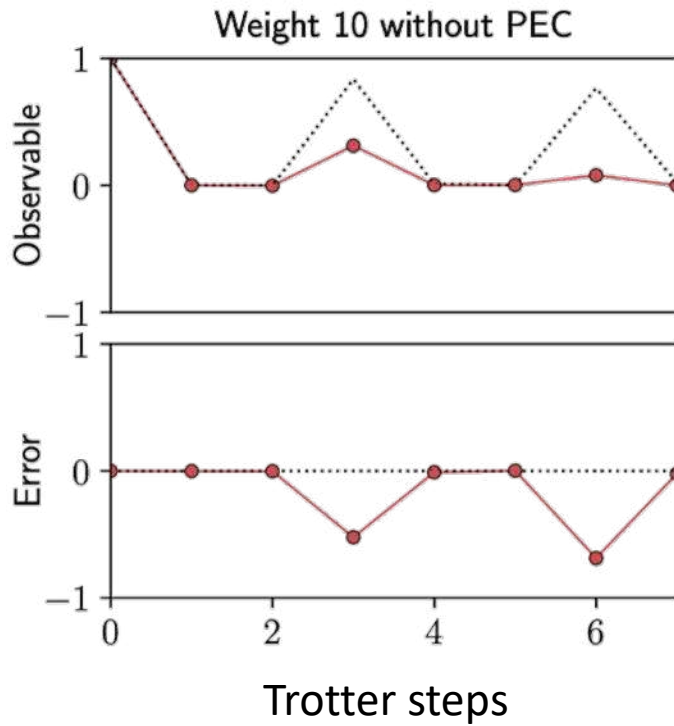
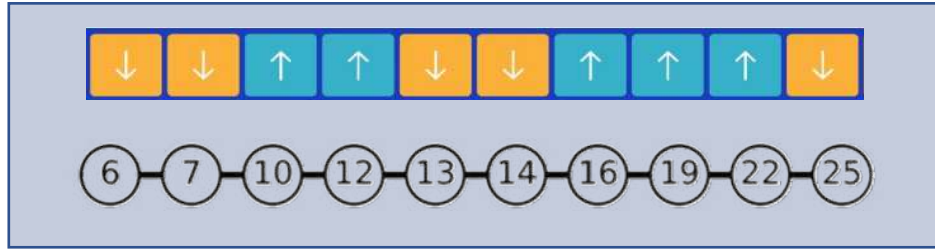
$$h = 1, J = -0.15, \delta t = 1/4$$

With vs without PEC: 10 qubit high-weight observables

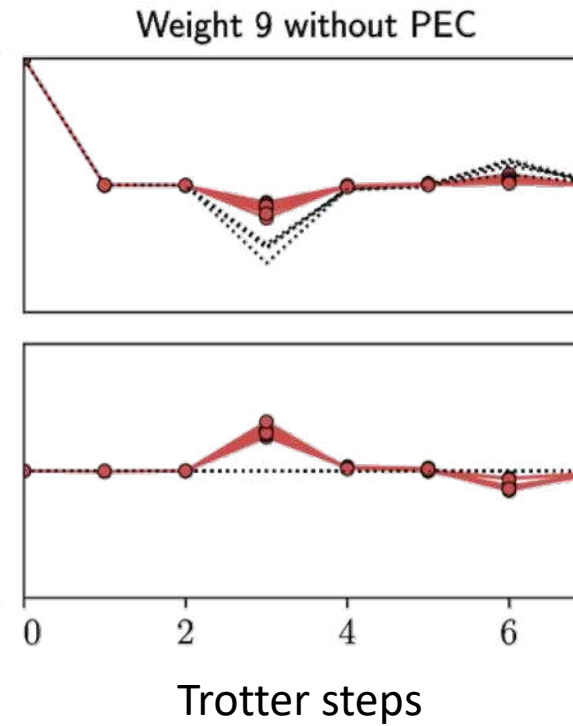


$$h = 1, J = -0.5236, \delta t = 1/4$$

With vs without PEC: 10 qubit high-weight observables

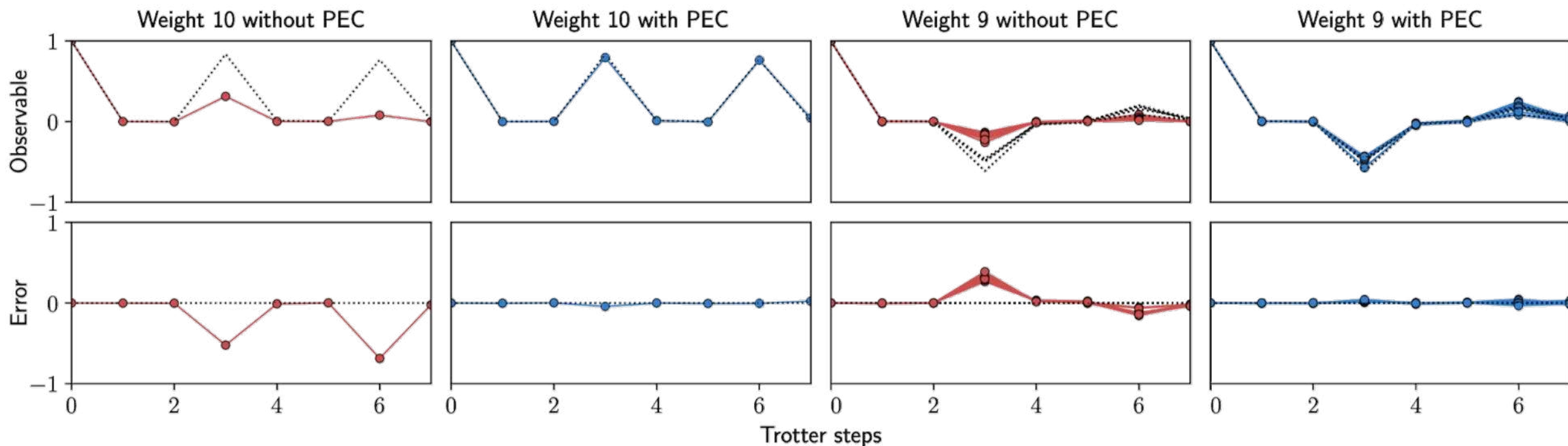
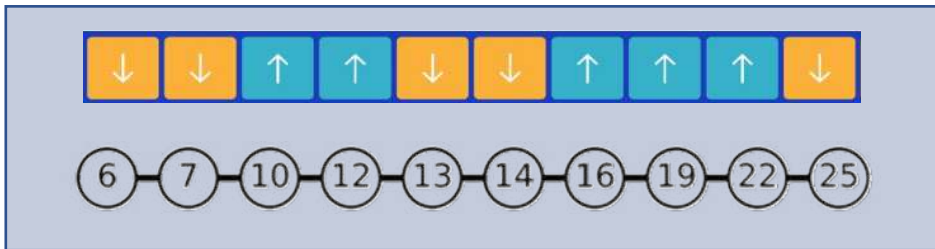


ZZZZZZZZZZZZ



IZZZZZZZZZZZZ
 ZIZZZZZZZZZZZZ
 ZZIZZZZZZZZZZZZ
 ...
 ZZZZZZZZZZZZI

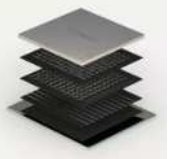
With vs without PEC: 10 qubit high-weight observables



$$\hbar = 1, J = -0.5236, \delta t = 1/4$$

Scaling and error budget

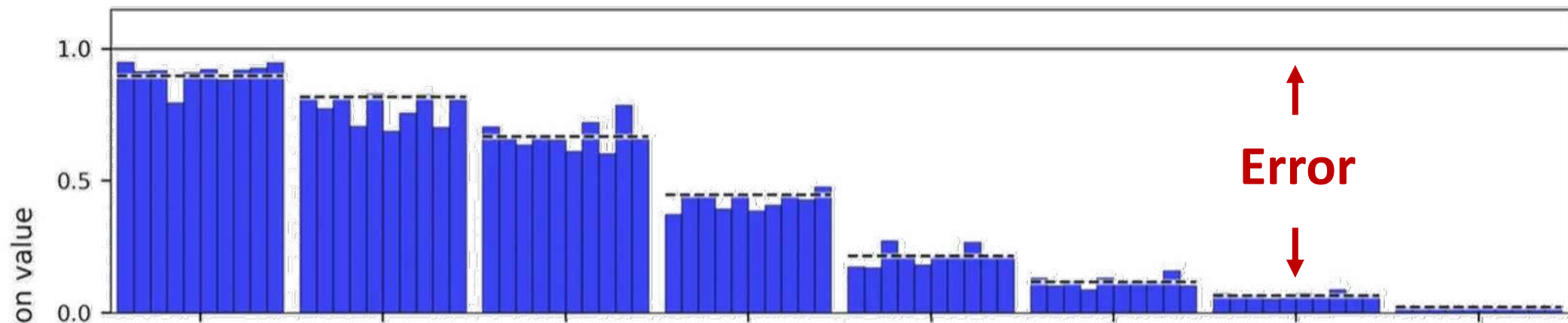




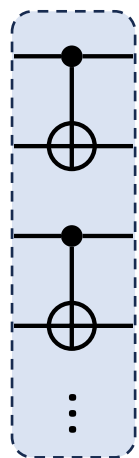
PEC on 50 qubit observables

Z stabilizers of increasing weight

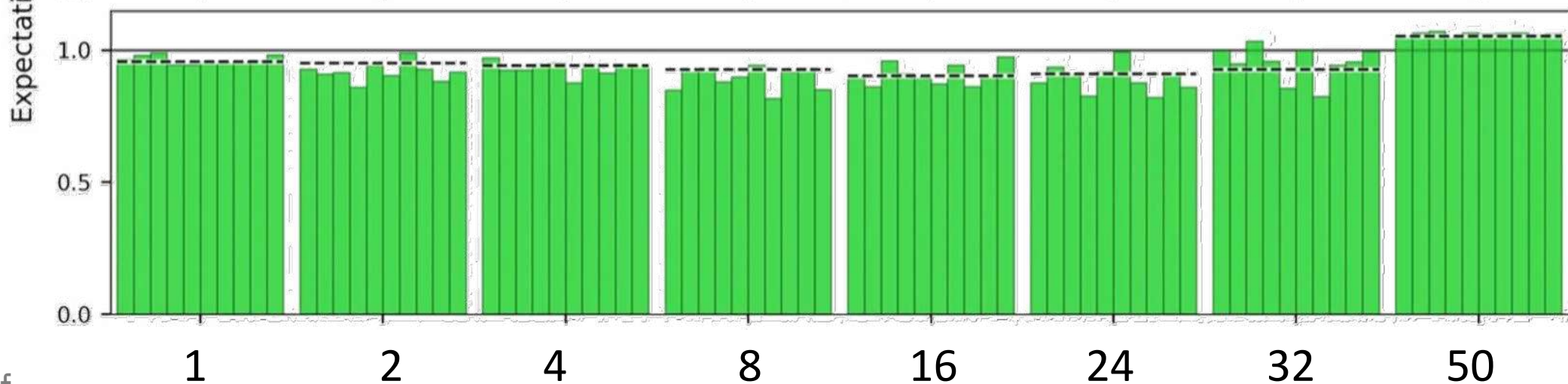
Without PEC



With PEC



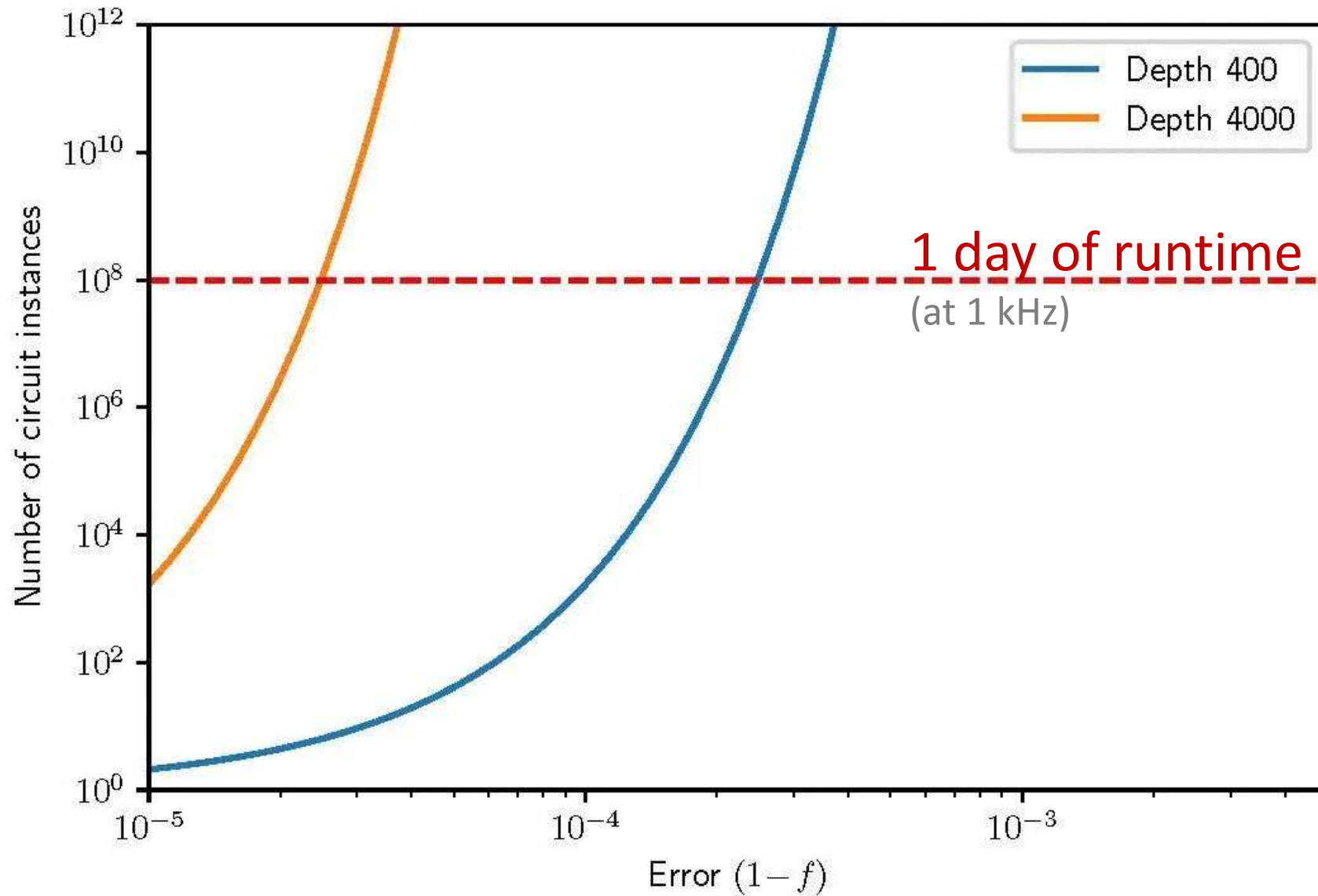
50Q
2 layers of
cNOT gates



Weight of mitigated observable

Path to 100+ qubits?

Estimating
PEC overhead
for Trotter
circuits
comprising
100 qubits



See also on speed: A. Wack, et al., Quality, speed, and scale: three key attributes to measure the performance of near-term quantum computers (2021).

Path to quantum computing

Noise-free estimators can be obtained from noisy quantum computers TODAY, at a runtime cost that is exponential in number of qubits n and circuit depth d

$$\text{Runtime} = \beta d (\bar{\gamma})^{n d} \text{ seconds}$$

d is the depth of the quantum circuit

β is a measure of the time per circuit layer operation (CLOPS) (increase by pushing **speed**)

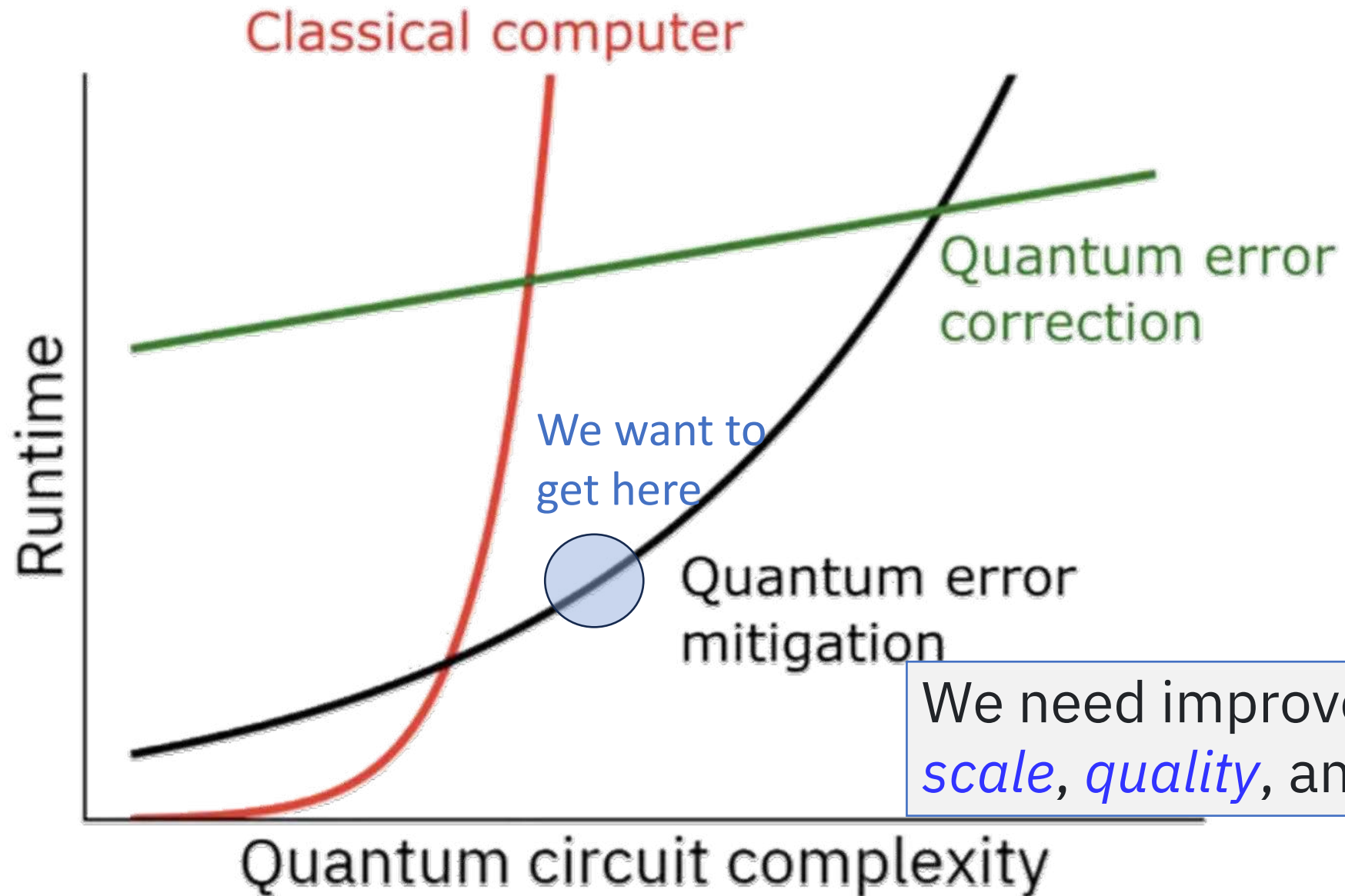
$\bar{\gamma}$ is a measure of the collective quantum noise (increasing **quality** brings it closer to 1)

n is the number of operational qubits (increase by pushing **scale**)

You can further reduce runtime using light cones and other strategies.

	Improvements	$\bar{\gamma}$
Hummingbird r2 (Brooklyn, 65Q)		1.038
Hummingbird r3 (Ithaca, 65Q)	2-3x coherence improvements over r2	1.024
Falcon r10 (Prague, 32Q)	State-of-the-art two-qubit gates, reduced crosstalk	1.012

Path to utility of quantum computers before error correction



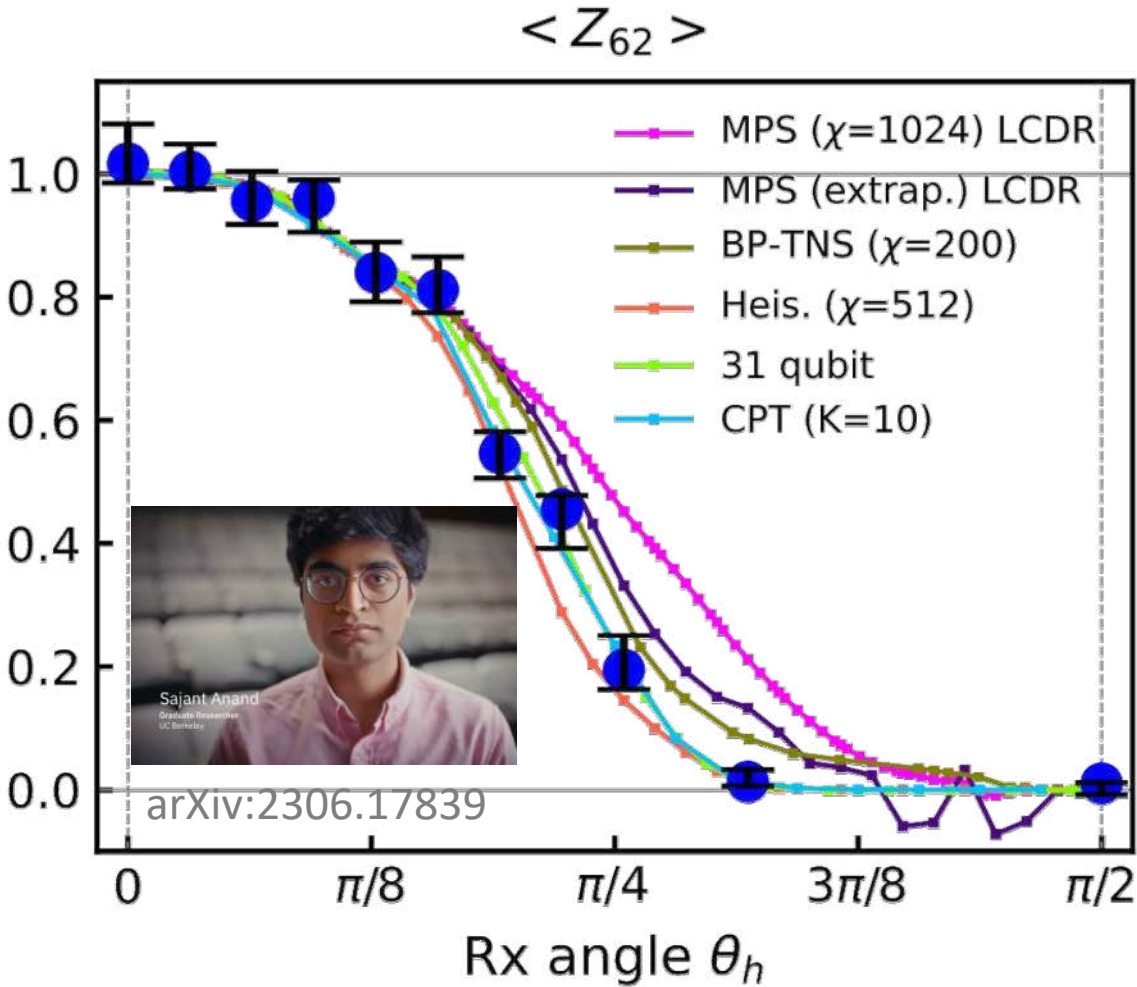
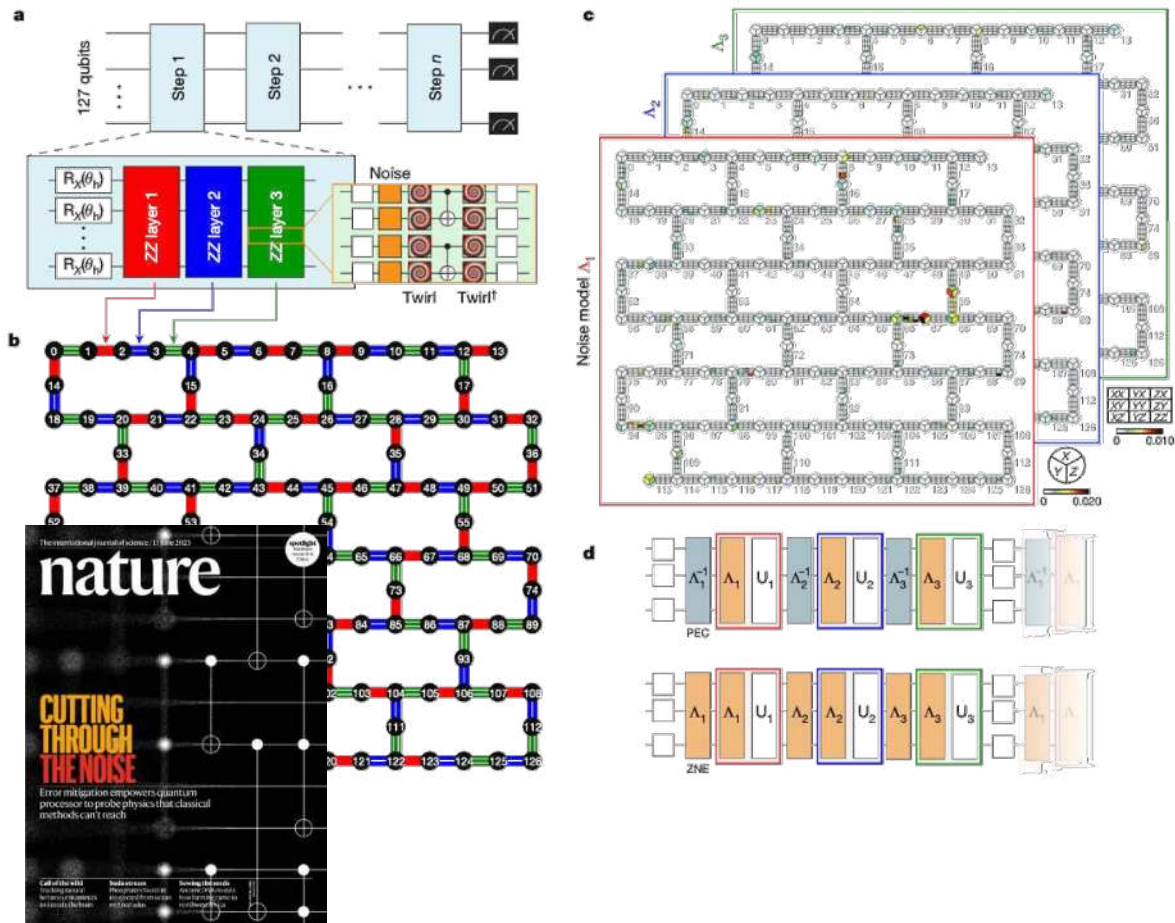
We need improvements in *scale*, *quality*, and *speed*.

PEC + ZNE for 127 qubits

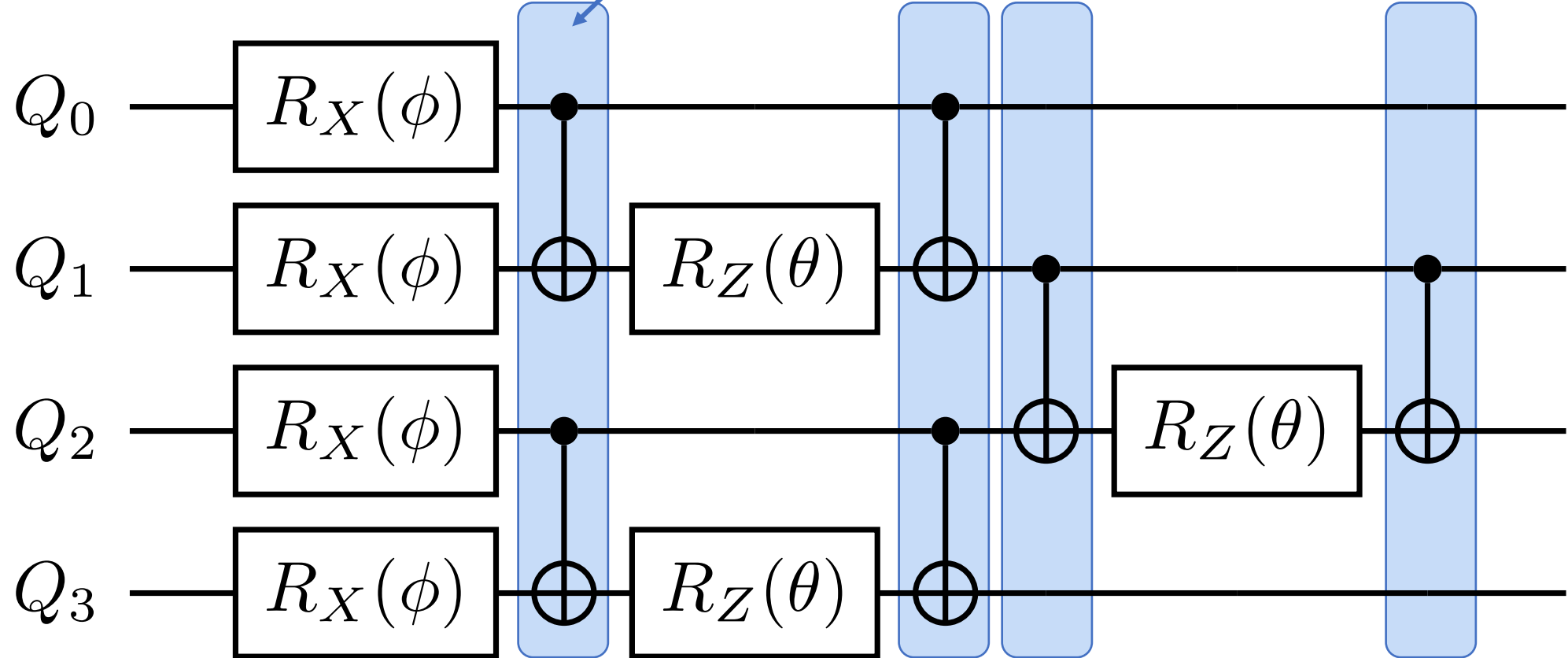
Article

Evidence for the utility of quantum computing before fault tolerance

Youngseok Kim^{1,6}, Andrew Eddins^{2,6}, Sajant Anand³, Ken Xuan Wei¹, Ewout van den Berg¹, Sami Rosenblatt¹, Hasan Nayfeh¹, Yantao Wu^{3,4}, Michael Zaletel^{3,5}, Kristan Temme¹ & Abhinav Kandala¹

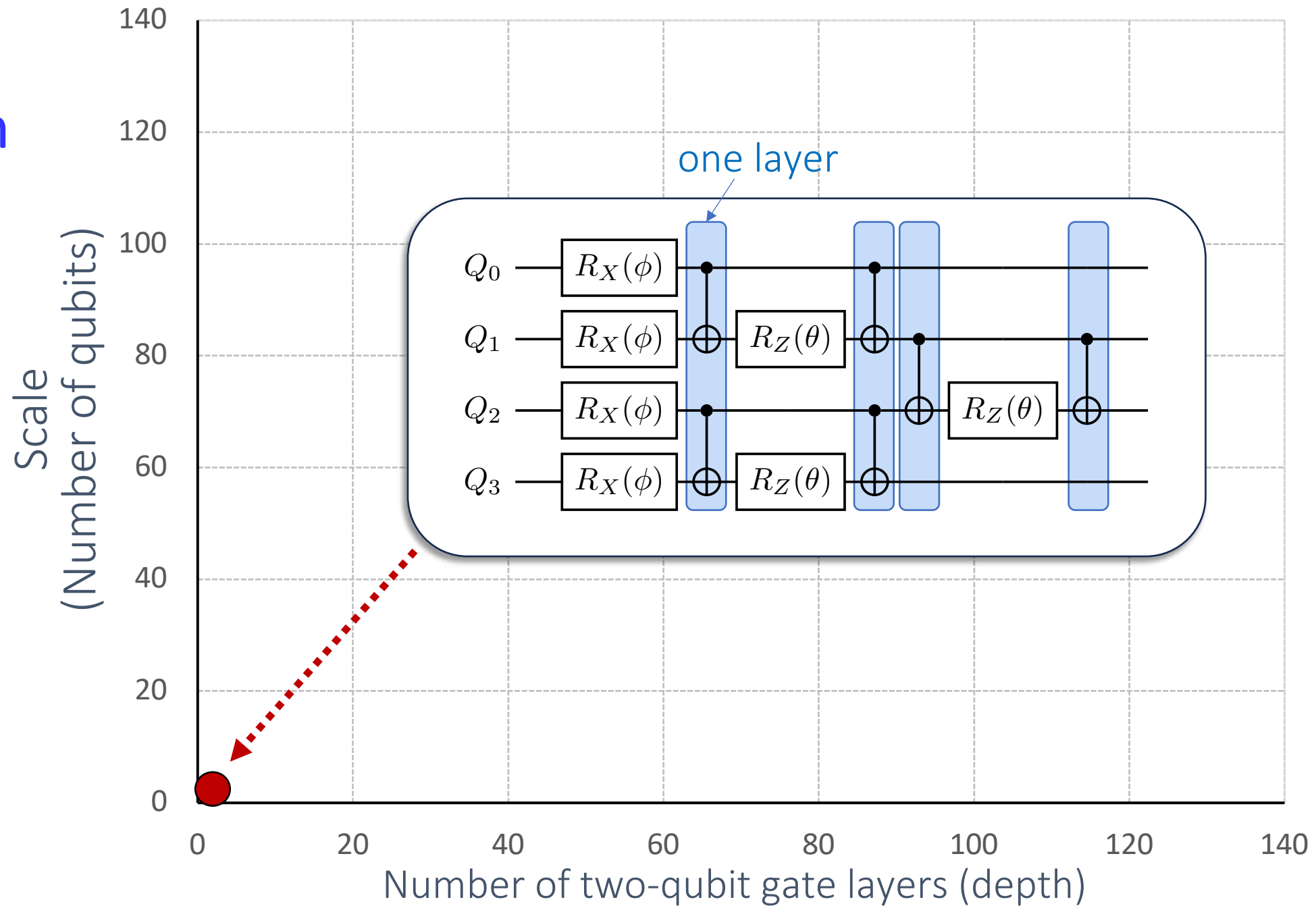


one gate layer

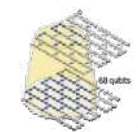


4 qubits x 4 gate layers

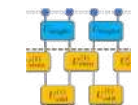
Circuit
width $\times$ depth



Some early utility-scale experiments



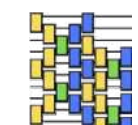
[0] Kim, Eddins, ..., Temme, Kandala
Nature 618, 500–505 (2023)



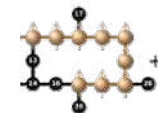
[1] Yu, Zhao, Wei.
arXiv: 2207.09994 (2022)



[2] Shtanko, Wang, Zhang, Harle, Seif,
Movassagh, Minev.
arXiv: 2307.07552 (2023)



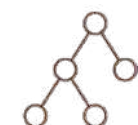
[3] Farrell, Illa, Ciavarella, Savage.
arXiv: 2308.04481 (2023)



[4] Bäumer, Tripathi, Wang, Rall,
Chen, Majumder, Seif, Minev.
arXiv: 2308.13065 (2023)



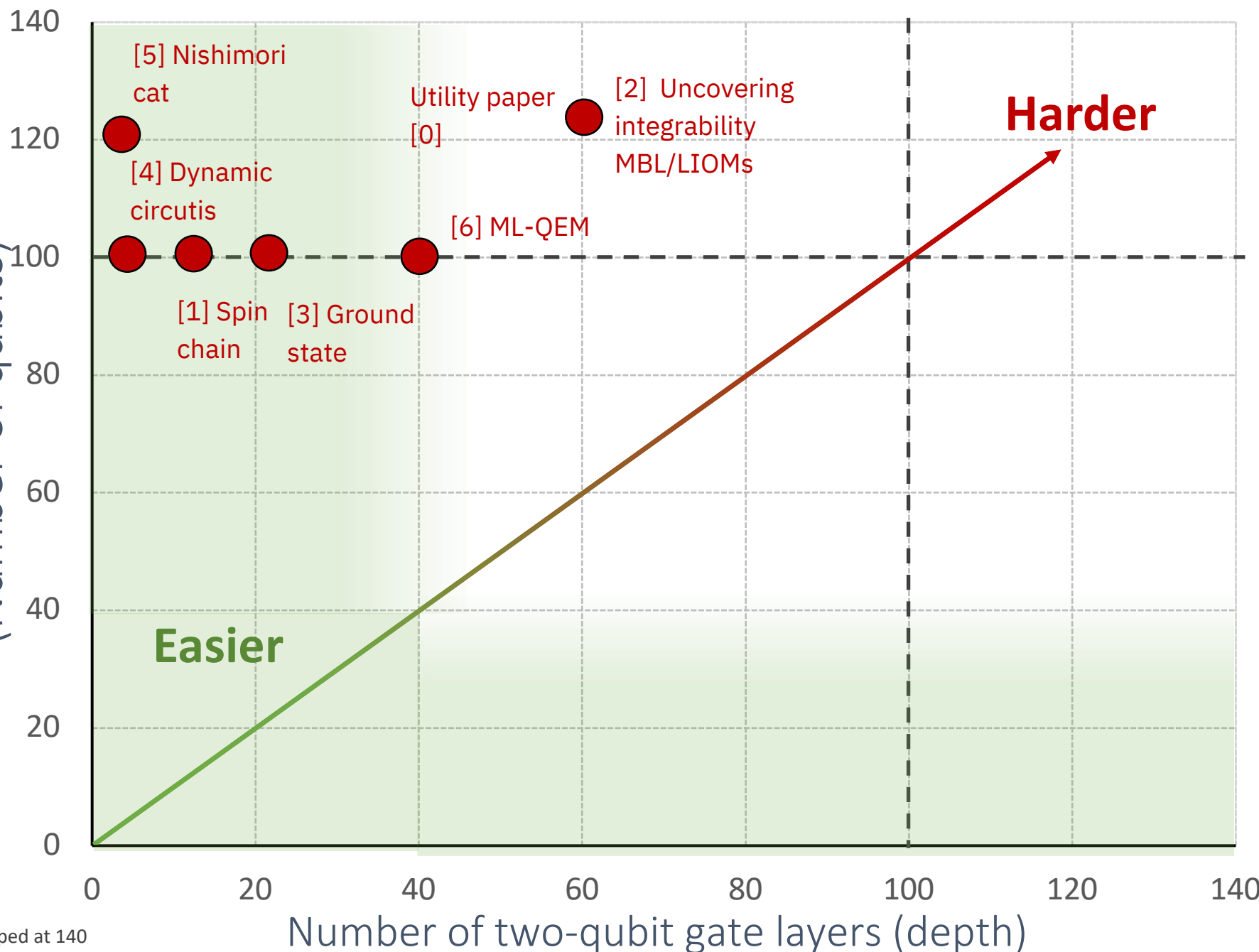
[5] Chen, Zhu, Verresen, Seif, Bäumer,
Layden, Tantivasadakarn, Zhu, Sheldon,
Vishwanath, Trebst, Kandala.
arXiv: 2309.02863 (2023)



[6] Liao, Wang, Sitdikov, Salcedo, Seif,
Minev.
arXiv: 2308.13065 (2023)

...

Scale
(Number of qubits)



Zlatko Minev, IBM Quantum (84)

* signal stops or decays more than 50% beyond this depth, capped at 140

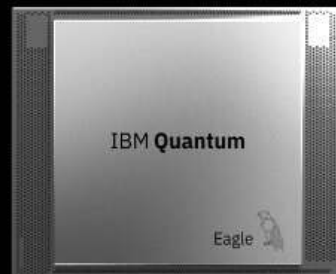
* note: quantum advantage with shallow circuits, Bravyi, Gosset, Konig, and related



2019
Falcon
27 Qubits



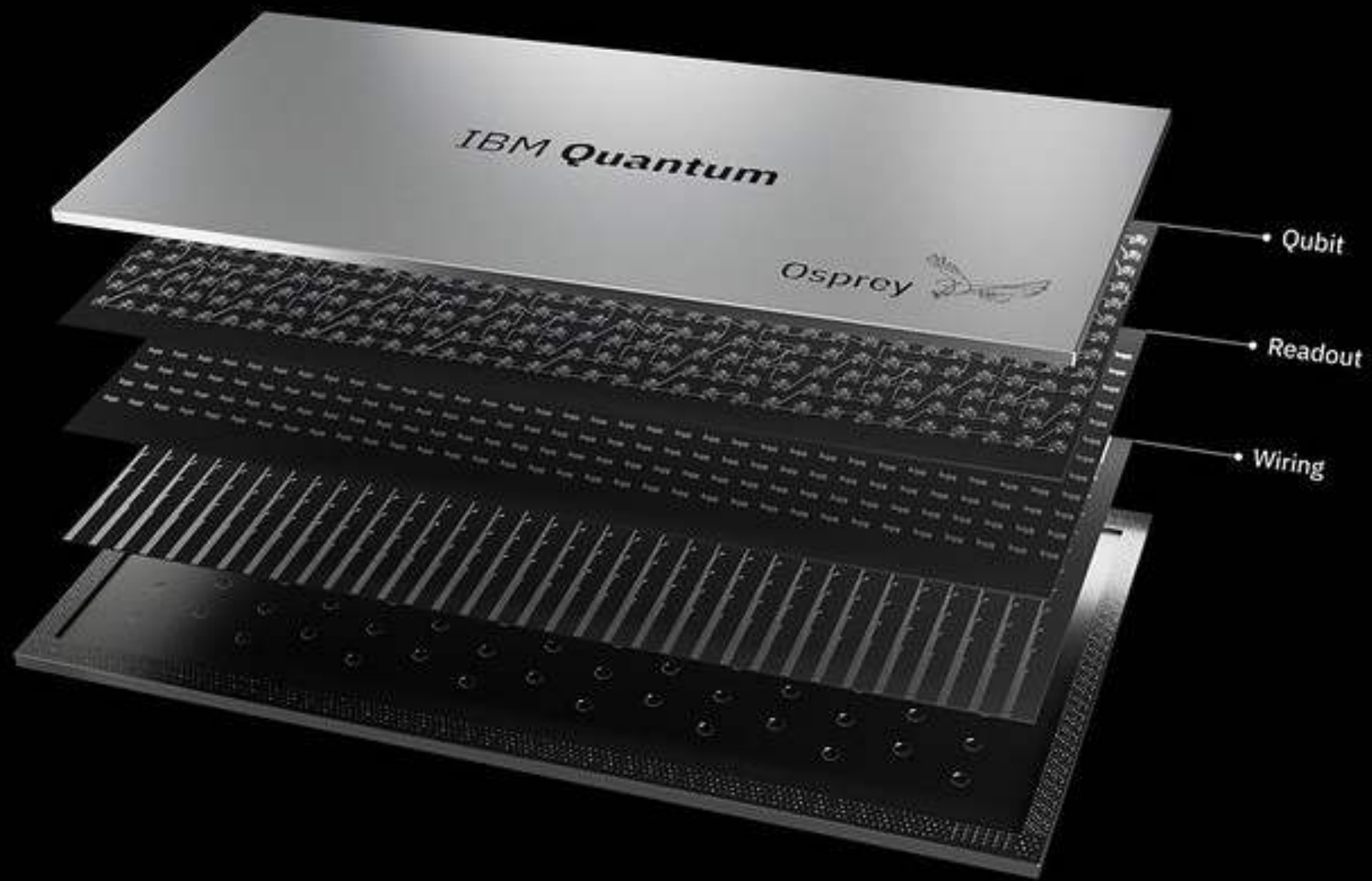
2020
Hummingbird
65 Qubits



2021
Eagle
127 Qubits



2022
Osprey
433 Qubits



Quantum Noise and Error Mitigation

Lectures 1 and 2

Big picture

Quantum computers status

Why error mitigation?

Noise in quantum computers
Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Introduction

One qubit example

Lecture 2

Learning noise

State-of-art mitigation
experiments

Hardware

Outlook



The important thing is not to stop questioning.
Curiosity has its own reason for existence.

One cannot help but be in awe when they
contemplate the mysteries of eternity, of life, of the
marvelous structure of reality.

It is enough if one tries merely to comprehend a
little of this mystery each day.

Albert Einstein

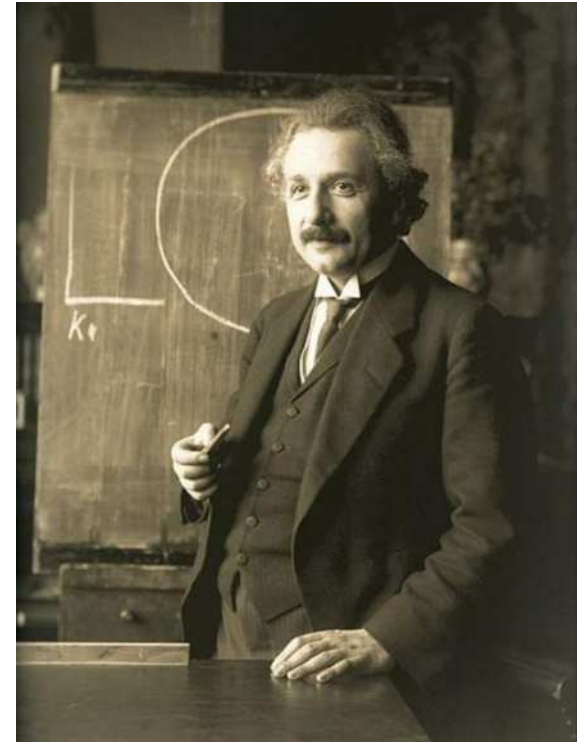


Photo: F. Schmutzer



@zlatko_minev



zlatko-minev.com

IBM Quantum

Bonus content

Comprehension questions

QUESTION 1. Which of the following is a primary goal of quantum error mitigation techniques?

- A)** To completely eliminate all errors in quantum computations, thus giving zero bias for any depth circuit within reasonable shot budget.
- B)** To reduce the effective error rate in quantum computations to acceptable levels without requiring additional qubits, but at the cost of more sampling.
- C)** To replace faulty qubits with error-free qubits.
- D)** To enhance the speed of quantum gates.

QUESTON 2. Which of the following statements best describes the concept of probabilistic error cancellation in quantum computing?

- A)** It involves using classical error correction codes to detect and correct errors in quantum circuits.
- B)** It refers to the use of redundant qubits to store quantum information, thus preventing errors from occurring.
- C)** It is a technique to estimate and mitigate the impact of noise in near-term quantum computations by applying the noise channel on average.
- D)** It uses quantum error correction codes, such as the surface code, to identify and correct errors directly on quantum bits.

Scaling PEC to n qubits and larger circuits (ADVANCED – OPTINAL)

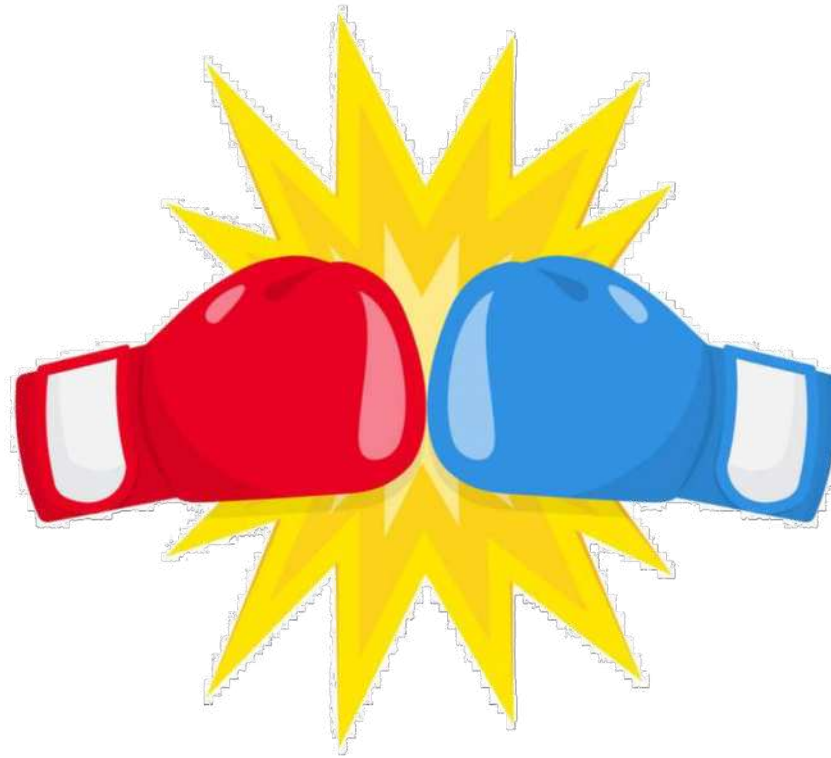


image: makc76



Language of errors and error mitigation

$$\frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}$$

@formulas_for_your_comfort

Notation: Super bra ket

$$\rho \in L(\mathcal{H})$$

$$\hat{A}, \hat{B} \in L(\mathcal{H})$$

$$\mathcal{S} \in L(L(\mathcal{H}))$$

$$|\hat{A}\rangle\rangle \leftrightarrow \hat{A}$$

$$\mathcal{S}|\hat{A}\rangle\rangle \leftrightarrow \mathcal{S}(\hat{A})$$

Vectorization isomorphism

$$\langle\langle \hat{A} | \hat{B} \rangle\rangle \leftrightarrow \langle \hat{A}, \hat{B} \rangle$$

$$= \text{Tr}(\hat{B} \hat{A}^\dagger)$$

$$\begin{aligned} \langle\langle \hat{A} | \mathcal{S} | \hat{B} \rangle\rangle &= \langle\langle \hat{A} | \mathcal{S}(\hat{B}) \rangle\rangle \\ &= \langle \hat{A}, \mathcal{S}(\hat{B}) \rangle \\ &= \text{Tr}(\hat{A}^\dagger \mathcal{S}(\hat{B})) \end{aligned}$$

$$\begin{aligned} \langle\langle \hat{A} | \cdot \rangle\rangle &\leftrightarrow \langle \hat{A}, \cdot \rangle \\ &= \text{Tr}(\hat{A}^\dagger \cdot) \end{aligned}$$

$$\begin{aligned} |\hat{A}\rangle\rangle \langle\langle \hat{B} | &\leftrightarrow \hat{A} \langle \hat{B}, \cdot \rangle \\ &\leftrightarrow \hat{A} \text{Tr}(\hat{B}^\dagger \cdot) \end{aligned}$$

Familiar ideas revisited with super notation

$$\langle \hat{P}_a | \hat{P}_b \rangle = \text{Tr} \left(\hat{P}_a^\dagger \hat{P}_b \right) = d \delta_{ab}$$

$$\Gamma = \{I, X, Y, Z\}^{\otimes n}$$

$$|\Gamma| = 4^n = d^2, \quad d = 2^n$$

$$\mathcal{I} = \sum_{a \in \Gamma} \frac{|\hat{P}_a\rangle \langle \hat{P}_a|}{\langle \hat{P}_a | \hat{P}_a \rangle}$$

$$a, b \in \Gamma$$

$$\Lambda = \mathcal{I} \Lambda \mathcal{I} = \sum_{a, b \in \Gamma} \frac{\langle \hat{P}_a | \Lambda | \hat{P}_b \rangle}{\langle \hat{P}_a | \hat{P}_a \rangle \langle \hat{P}_b | \hat{P}_b \rangle} |\hat{P}_a\rangle \langle \hat{P}_b| = \frac{1}{d} \sum_{a, b \in \Gamma} \Lambda_{ab} |\hat{P}_a\rangle \langle \hat{P}_b| ,$$

Stochastic Pauli channel

$$\Lambda : L(\mathcal{H}) \rightarrow L(\mathcal{H}) \qquad 0 \leq p_a \leq 1 ,$$

$$\Lambda(\rho) = \sum_{a \in \Gamma} p_a P_a \rho P_a , \qquad \sum_{a \in \Gamma} p_a = 1 ,$$

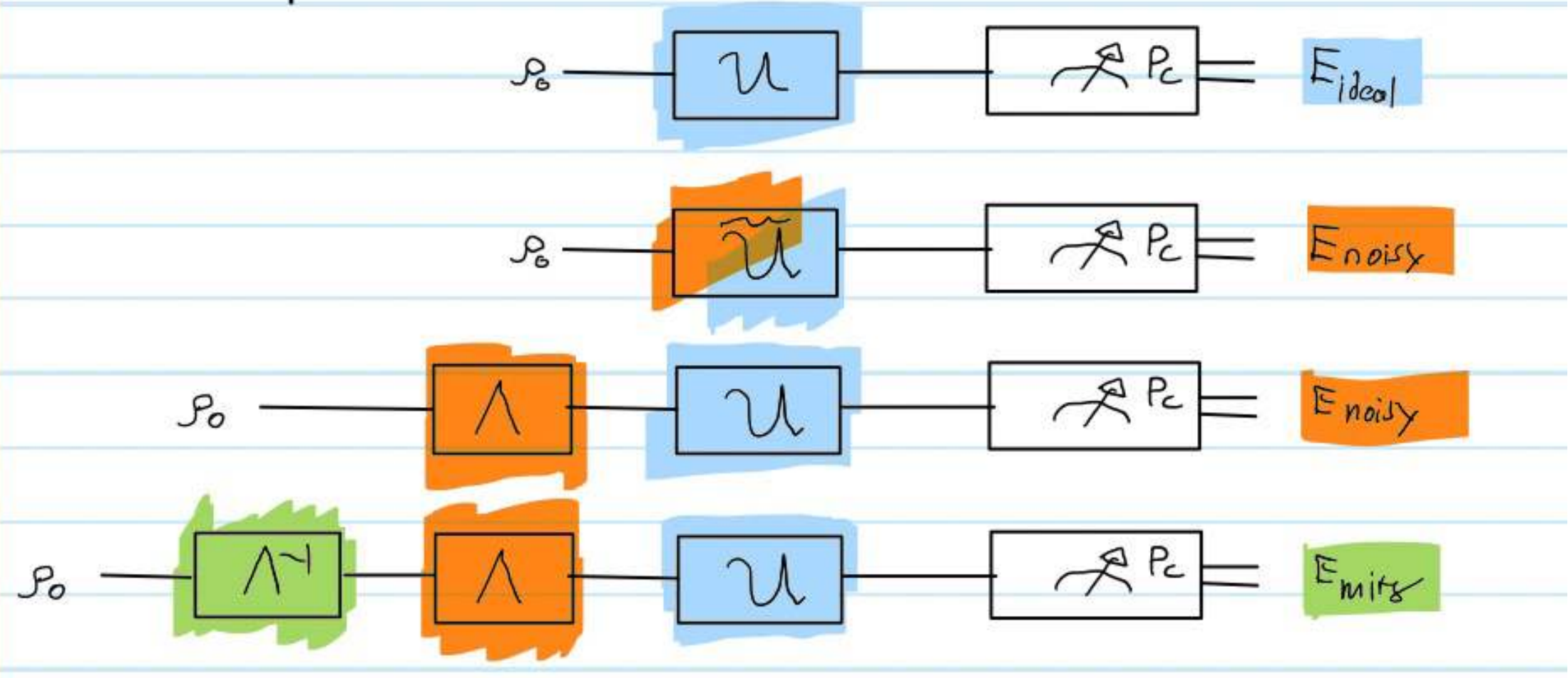
$$\Lambda(\rho) = \frac{1}{\sqrt{|\Gamma|}} \sum_{b \in \Gamma} f_b \operatorname{Tr}(P_b \rho) P_b = \sum_{b \in \Gamma} f_b \frac{|P_b\rangle\rangle \langle\langle P_b|}{\langle\langle P_b | P_b \rangle\rangle} ,$$

$$\Lambda(P_b) = f_b P_b , \quad \forall a \in \Gamma ,$$

$$-1 \leq f_b \leq 1 \quad f_I = 1 ,$$

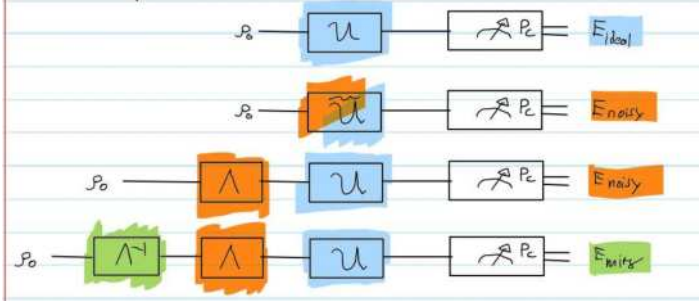
Probabilistic error cancellation: Derivation

From 1 Step



Probabilistic error cancellation: Derivation

From 1 Step



Channel definitions

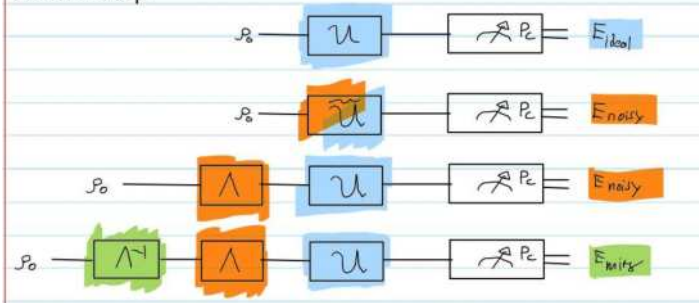
$$U = U \cdot U^\dagger$$

$$\Lambda \approx \sum_a f_a |P_a\rangle\langle P_a|$$

$$= \sum_b c_b P_a$$

Probabilistic error cancellation: Derivation

From 1 Step



Channel definitions

$$U = U \cdot U^\dagger$$

$$\langle \hat{P}_C \rangle(\cdot) = \langle \langle P_C | \cdot \rangle \rangle \quad \text{Need}$$

$$\begin{aligned} \Lambda &= \sum_a f_a |P_a\rangle \langle P_a| \\ &= \sum_b c_b P_a \end{aligned}$$

$$-1 \leq f_a \leq 1$$

$$c_b \geq 0, \sum_b c_b = 1$$

$$c_b = \frac{1}{2^n} \sum_a (-1)^{\langle a, b \rangle_{sp}} f_a$$

$$\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b$$

$$\begin{aligned} \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle \langle P_a| \\ &= \sum_b c_b^{\text{inv}} P_b \end{aligned}$$

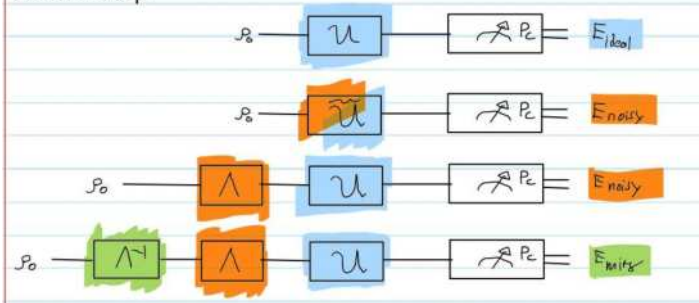
$$\vec{c}_b^{\text{inv}} = W \vec{f}_a^{-1}$$

$$c_b^{\text{inv}} = \frac{1}{2^n} \sum_a (-1)^{\langle a, b \rangle_{sp}} f_a$$

$$c_b^{\text{inv}} \in \mathbb{R}$$

Probabilistic error cancellation: Derivation

From 1 Step



Channel definitions

$$U = U \cdot U^\dagger$$

$$\langle \hat{P}_C \rangle(\cdot) = \langle P_C | \cdot \rangle \quad \text{Need}$$

$$\Lambda \approx \sum_a f_a |P_a\rangle \langle P_a|$$

$$-1 \leq f_a \leq 1$$

$$= \sum_b c_b P_a$$

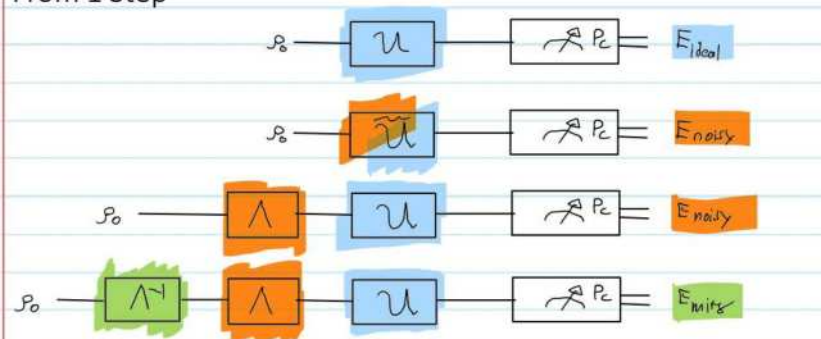
$$c_b \geq 0, \sum_b c_b = 1$$

$$c_b = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a$$

$$\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b$$

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |P_a\rangle\langle P_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle\langle P_a| \\
 &= \sum_b c_b^{-1} P_b
 \end{aligned}$$

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{f_a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b \\
 \vec{c}_b^{-1} &= W \vec{f}_a^{-1} \\
 c_b^{-1} &= \frac{1}{2^n} \sum_a (-1)^{c_a \cdot b} f_a \\
 c_b^{-1} &\in \mathbb{R}
 \end{aligned}$$

Need to introduce

- Super-op. based notation
- Pauli channels $\mathbb{P}$ & $\mathbb{P}^\dagger$
- Super-op. notation
- W
- Pauli class representation
- W is $2^n \times 2^n$
- $C \in \mathbb{R}^n$

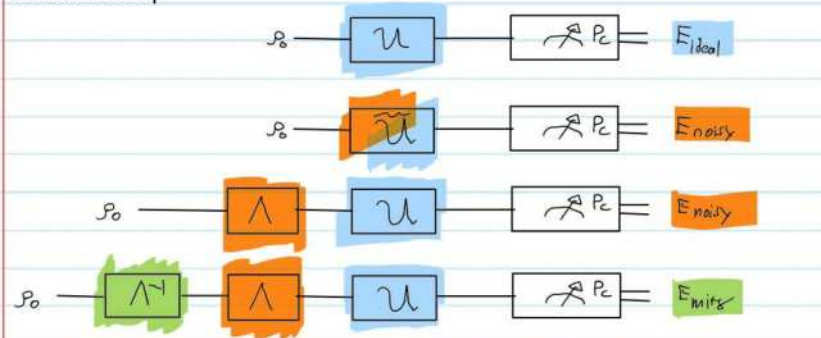
$$E_{ideal} = \langle \hat{P}_c \rangle_U$$

$$= \langle P_c | U | P_0 \rangle$$

ideal exp value with noiseless unitary

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |p_a\rangle\langle p_a| \\
 &= \sum_b c_b P_a \\
 \Lambda^\dagger &= \sum_a f_a^{-1} |p_a\rangle\langle p_a| \\
 &= \sum_b c_b^\dagger P_b
 \end{aligned}$$

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b \\
 \vec{c}_b^\dagger &= W \vec{f}_a^\dagger \\
 c_b^\dagger &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^\dagger \\
 c_b^\dagger &\in \mathbb{R}
 \end{aligned}$$

Need to introduce:
 • Super-op. bra-ket notation
 • Pauli matrices $\sigma_x, \sigma_y, \sigma_z$
 • Super-op notation
 • Vectors
 • Pauli trace representation
 • SG
 • CNOT

$$E_{\text{ideal}} := \langle \hat{P}_c \rangle_U$$

$$= \langle\langle P_c | U | p_0 \rangle\rangle$$

ideal exp value with noiseless unitary

$$E_{\text{noisy}} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

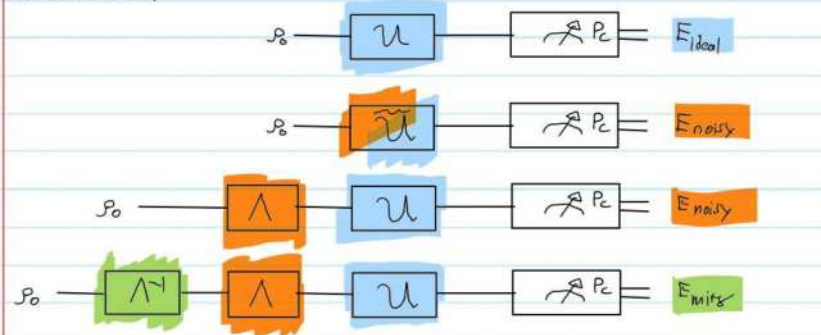
$$= \langle\langle P_c | U \Lambda | p_0 \rangle\rangle$$

$$= \langle\langle P_c | \tilde{U} | p_0 \rangle\rangle$$

noisy-gate expectation value

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |p_a\rangle\langle p_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |p_a\rangle\langle p_a| \\
 &= \sum_b c_b^{-1} P_b
 \end{aligned}$$

$\langle \hat{P}_c \rangle(\cdot) = \langle P_c | \cdot$
 $-1 \leq f_a \leq 1$
 $c_b \geq 0, \sum_b c_b = 1$
 $c_b = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a$
 $\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b$
 $\vec{c}_b^{-1} = W \vec{f}_a^{-1}$
 $c_b^{-1} = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1}$
 $c_b^{-1} \in \mathbb{R}$

Need to introduce:
 • Super-op. based notation
 • Pauli elements P_i & PTM
 • Super-op notation
 • Pauli trace representation
 • SG to $\vec{c}_b$
 • $c_b \in \mathbb{R}$

$$E_{ideal} := \langle \hat{P}_c \rangle_U$$

$$= \langle\langle P_c | U | p_0 \rangle\rangle$$

ideal exp value with noiseless unitary

$$E_{noisy} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

$$= \langle\langle P_c | U \Lambda | p_0 \rangle\rangle$$

$$= \langle\langle P_c | \tilde{U} | p_0 \rangle\rangle$$

noisy-gate expectation value

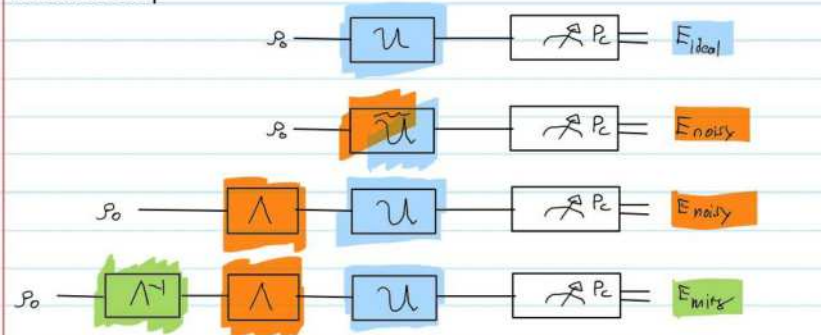
$$E_{mitg} := \langle\langle P_c | U \Lambda \Lambda^{-1} | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_a f_a^{-1} |p_a\rangle\langle p_a| \right) | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_b c_b^{-1} P_b \right) | p_0 \rangle\rangle$$

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |p_a\rangle\langle p_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |p_a\rangle\langle p_a| \\
 &= \sum_b c_b^{-1} P_b
 \end{aligned}$$

Need to introduce:

- Super-op. based notation
- Pauli channel / PTM
- super-op notation
- Pauli class representation
- SG
- CG

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b \\
 \vec{c}_b^{-1} &= W \vec{f}_a^{-1} \\
 c_b^{-1} &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1} \\
 c_b^{-1} &\in \mathbb{R}
 \end{aligned}$$

$$E_{ideal} := \langle \hat{P}_c \rangle_U$$

$$= \langle\langle P_c | U | p_0 \rangle\rangle$$

ideal exp value with noiseless unitary

$$E_{noisy} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

$$= \langle\langle P_c | U \Lambda | p_0 \rangle\rangle$$

$$= \langle\langle P_c | \tilde{U} | p_0 \rangle\rangle$$

noisy-gate expectation value

$$E_{mitg} := \langle\langle P_c | U \Lambda \Lambda^{-1} | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_a f_a^{-1} |p_a\rangle\langle p_a| \right) | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_b c_b^{-1} P_b \right) | p_0 \rangle\rangle$$

$$= \sum_b c_b^{-1} \langle\langle P_c | U \Lambda P_b | p_0 \rangle\rangle$$

sum of trajectories with weight $\hat{c}_b^{-1} \in \mathbb{R}$

$$= \underbrace{\sum_b c_b^{-1}}_{\text{classical post process}} \underbrace{\langle P_c \rangle(\tilde{U}, b)}_{\text{quantum}}$$



$$E_{mitg} := \langle \langle P_c | U \Lambda U^\dagger | \rho_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \left(\sum_a f_a^{-1} | \rho_a \rangle \langle P_a | \right) | \rho_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \left(\sum_b c_b^{inv} P_b \right) | \rho_0 \rangle \rangle$$

$$= \sum_b c_b^{inv} \langle \langle P_c | U \Lambda P_b | \rho_0 \rangle \rangle$$

sum of trajectories
with weight $c_b^{inv} \in \mathbb{R}$

$$= \underbrace{\sum_b c_b^{inv}}_{\text{classical post process}} \underbrace{\langle P_c \rangle(\tilde{u}, b)}_{\text{quantum}}$$

classical
post process



Quantum circuit we can execute on the quantum device and find exp. value from.

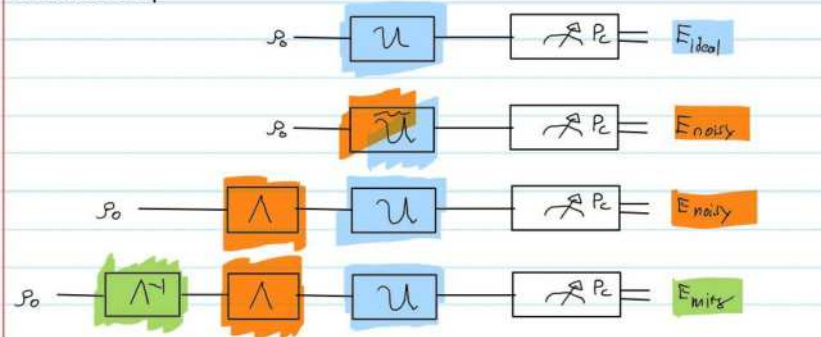
$\therefore$ To find noise-free val all we have to do is to compute exp-val of all 4^n b-modified circuits! This would give us ideal exp value.

However, $| \{ b \} | = 4^n$ grows exponentially, hence, infeasible.

but what if we could sample from it to approximate full sum. But... can't sample directly from c_b^{inv} which

does not form a valid prob-distribution. let's solve:

From 1 Step



Channel definitions

$U = U \cdot U^\dagger$

$\Lambda = \sum_a f_a | \rho_a \rangle \langle P_a |$
 $= \sum_b c_b P_b$

$\Lambda^{-1} = \sum_a f_a^{-1} | \rho_a \rangle \langle P_a |$
 $= \sum_b c_b^{inv} P_b$

$\langle \hat{P}_c \rangle(\cdot) = \langle \langle P_c | \cdot \rangle \rangle$

$-1 \leq f_a \leq 1$

$c_b \geq 0, \sum_b c_b = 1$

$c_b = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a$

$\vec{c}_b^{inv} = W^{-1} \vec{f}_a$

$\vec{f}_a = W \vec{c}_b$

$\vec{c}_b^{inv} = W \vec{f}_a^{-1}$

$c_b^{inv} = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a$

$c_b^{inv} \in \mathbb{R}$

Need to introduce:
 • Super-op. for each unitary
 • Pauli circuit (PTAT)
 • Super-op. for each
 • Unit. class. representation
 • SQ. to 4^n
 • Class.

Quasi probability distribution

c_b^{inv} can be outside $[0,1]$

$\sum_b c_b^{inv} =: \gamma \geq 1$ generally, for Λ not unitary

eg b4 th chapter

$$\Lambda = (1-p)[I + pX \cdot X]$$

$$\Lambda^{-1} = \left(1 + \frac{p}{1-2p}\right)I + \frac{-p}{1-2p}X \cdot X$$

$$c_{\uparrow}^{inv} = 1 + \frac{p}{1-2p}$$

$b = (0,0)$
choice vector

$$c_{\downarrow}^{inv} = -\frac{p}{1-2p}$$

$b = (1,0)$

Turn into probability

$$c_b^{inv} = \underbrace{\text{sgn}(c_b^{inv})}_{\text{sign} \in \{-1, +1\}} \underbrace{\frac{|c_b^{inv}|}{\gamma}}_{\substack{\text{prob} \in [0,1] \\ \text{scale}}} \gamma$$

$$\tilde{c}_b^{inv} := \frac{|c_b^{inv}|}{\|c_b^{inv}\|_1} \rightarrow \|c_b\|_1 = \sum_b |c_b^{inv}| \Leftrightarrow L_1 \text{ norm}$$

$$= \frac{|c_b^{inv}|}{\gamma}$$

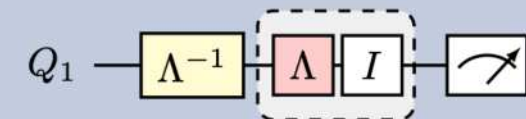


probability

$$P_I = |1 - q|/\gamma$$

$$P_X = |q|/\gamma$$

$$P_I + P_X = 1$$



||

sign

$$I \quad s_I = \text{sign}(1-q)$$

+

$$X \quad s_X = \text{sign}(q)$$

E mitigated

$$E_{\text{mitg}} = \sum_b c_b^{\text{inv}} \langle \hat{P}_c \rangle(\tilde{u}, b)$$

$$= \sum_b \text{sgn}(c_b^{\text{inv}}) \frac{|c_b^{\text{inv}}|}{\gamma} \gamma \langle \hat{P}_c \rangle(\tilde{u}, b)$$

$$= \gamma \sum_b \underbrace{\text{sgn}(c_b^{\text{inv}})}_{\text{scale}} \underbrace{\bar{c}_b^{\text{inv}}}_{\text{classical post-processing}} \underbrace{\langle \hat{P}_c \rangle(\tilde{u}, b)}_{\text{valid QC circuit can run & find value on HW}}$$

scale classical post-processing. valid QC circuit can run & find value on HW

In this form, the decomposition of the error error mitigated expectation value is simply a sum over expectation values of Pauli-gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, c bar b inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.

c_b^{inv} can be outside $[0,1]$

$\frac{1}{\gamma} c_b^{\text{inv}} =: \gamma \geq 1$ generally, for Λ not unitary

eg b & b chapter $\Lambda = (1-p)I + pX \cdot X$
 $\Lambda^\dagger = (1+p/2p)I + \frac{-p}{1-2p} X \cdot X$

$c_b^{\text{inv}} = 1 + \frac{p}{1-2p}$ $c_x^{\text{inv}} = \frac{-p}{1-2p}$
 $b=(0,0)$ class prob $b=(1,0)$

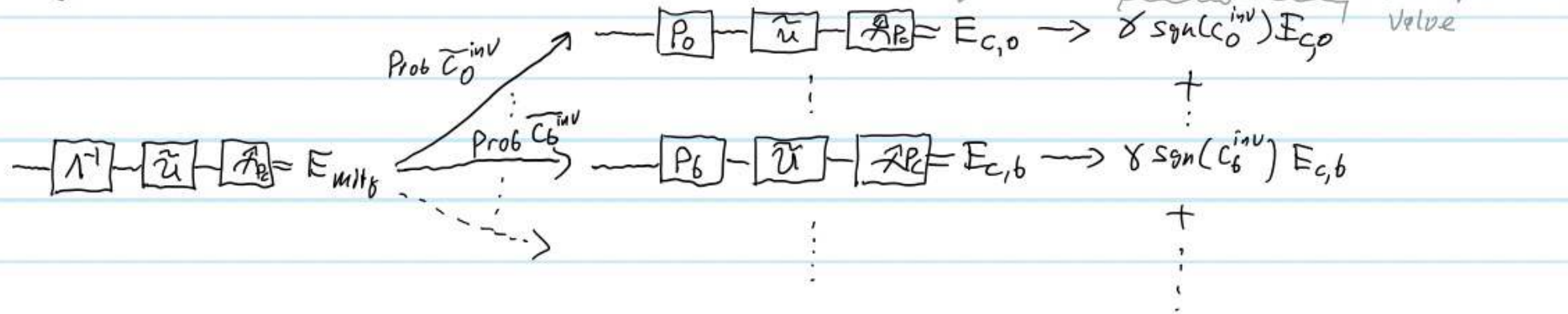
Turn into probability

$$c_b^{\text{inv}} = \underbrace{\text{sgn}(c_b^{\text{inv}})}_{\text{sign} \in \{-1, +1\}} \underbrace{\frac{|c_b^{\text{inv}}|}{\gamma}}_{\substack{\text{prob} \in [0,1] \\ \text{scale}}} \gamma$$

$$\bar{c}_b^{\text{inv}} := \frac{|c_b^{\text{inv}}|}{\|c_b^{\text{inv}}\|_1} \xrightarrow{\text{abs inv}} \|c_b\|_1 = \sum_b |c_b^{\text{inv}}| \Leftrightarrow L_1 \text{ norm}$$

$$= \frac{|c_b^{\text{inv}}|}{\gamma}$$

In other words



Quasi-Probability Distribution

c_b^{inv} can be outside $\{0,1\}$
 $\sum_b c_b^{inv} = 1$ generally, for Λ not unitary
 ex: Hadamard $\Lambda = \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z)$
 $c_b^{inv} = \frac{1}{\sqrt{2}}$ for $b=0,1$ $c_b^{inv} = -\frac{1}{\sqrt{2}}$ for $b=2,3$

Turn into probability
 $C_b^{inv} = \text{sgn}(c_b^{inv}) |c_b^{inv}| \gamma$
 $\text{sgn}(c_b^{inv}) \in \{1,-1\}$ $|c_b^{inv}| \in [0,1]$ $\gamma \in [0,1]$ $\text{Prob}(b) \in [0,1]$ $\text{sgn}(c_b^{inv}) \in \{0,1\}$

$\tilde{c}_b^{inv} = \frac{1}{\gamma} |c_b^{inv}| \gamma = |c_b^{inv}| \gamma = \text{Prob}(b) \in [0,1]$
 $= \frac{|c_b^{inv}|}{\gamma}$

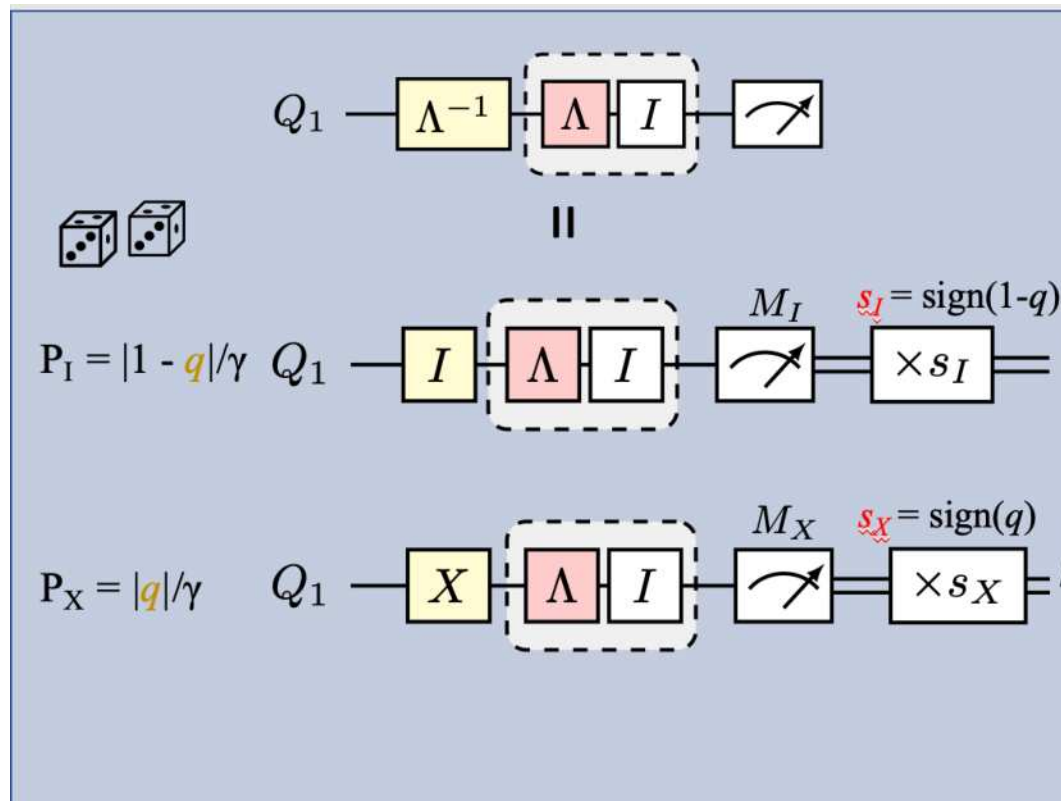
$$E_{Mitg} = \sum_b C_b^{inv} \langle \tilde{U}, b \rangle$$

$$= \sum_b \text{sgn}(c_b^{inv}) \frac{|c_b^{inv}|}{\gamma} \langle \tilde{U}, b \rangle$$

$$= \gamma \sum_b \text{sgn}(c_b^{inv}) \tilde{c}_b^{inv} \langle \tilde{U}, b \rangle$$

scale classical prob. $\tilde{c}_b^{inv}$ by γ to get $\text{Prob}(b)$ $\text{sgn}(c_b^{inv})$ from $\tilde{c}_b^{inv}$ $\text{sgn}(c_b^{inv}) = \text{sgn}(\tilde{c}_b^{inv})$

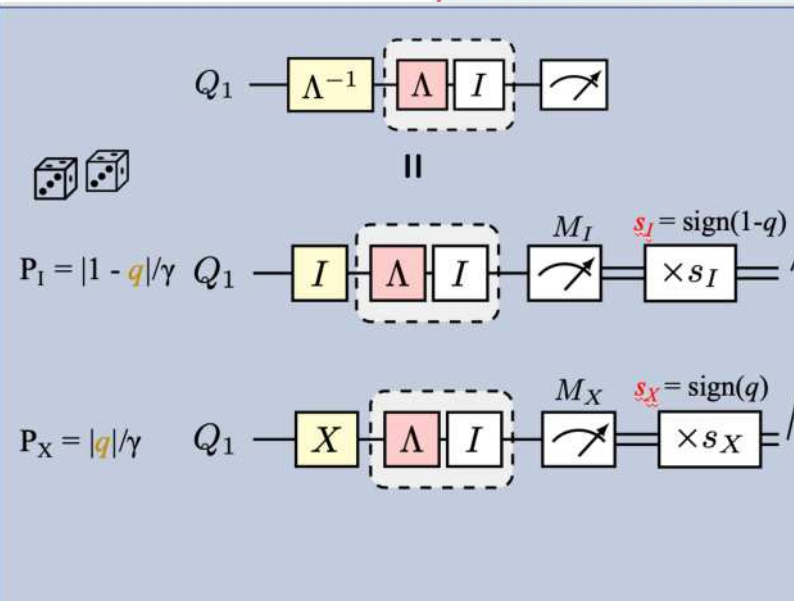
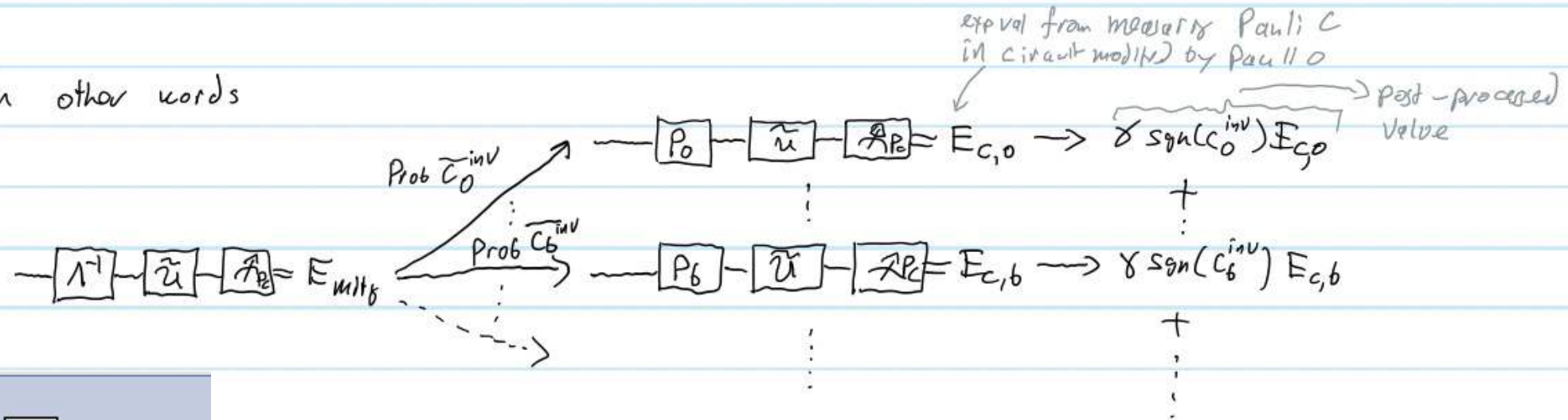
In this form, the decomposition of the error error mitigated expectation value is simply a sum over expectation values of Pauli gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, γ bar inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.



Estimator

post processing.

In other words



$$E_{c,mitg} = \sum_b \gamma \text{sgn}(C_b^{inv}) E_{c,b}$$

mitigated value for Pauli C obtained from the quasi-prob distribution

From above, we know this is an unbiased estimator,
but what about the error and sampling

mitigated value for Pauli C obtained from the quasi-prob distribution

From above, we know this is an unbiased estimator,
but what about the error and sampling

Estimator, Sampling, and Error Bounds

Sample circuit of the form

$$\left\{ \underbrace{\bar{C}_b^{\text{inv}}}_{\text{prob}} : \underbrace{-[P_b] - [X] - [XP_b]}_{\text{Pauli observable single-shot outcome ie e-val}} = Y \in \{-1, +1\} \right\} \rightarrow X = \text{sgn}(\bar{C}_b^{\text{inv}}) Y$$

post processed value X
 $X \in \{-1, +1\}$

Let's say we sample M instances, randomly sample a single value for b and obtain one-shot value on the QC, ie one random instance of $Y = 1$ or $Y = -1$, which we then post-process.

The results are thus the classical random variables

$$\{X_1, X_2, \dots, X_M\} \quad \text{or} \quad \{X_m : m = 1, \dots, M\}$$

Quasi-Probability Distribution

C_b^{inv} can be outside $\{0, 1\}$

$\bar{C}_b^{\text{inv}} = \text{sgn}(C_b^{\text{inv}})$ generally for A not unitary

eg. for the circuit $A = (H \otimes I) P_{12} (H \otimes I)$

$$C_b^{\text{inv}} = \frac{1}{2} \frac{P_{12}}{P_{12} + I} \quad C_b^{\text{inv}} = \frac{1}{2} \frac{P_{12}}{P_{12} + I}$$

Turn into probability

$$C_b^{\text{inv}} = \text{sgn}(C_b^{\text{inv}}) \frac{|C_b^{\text{inv}}|}{\sum_{b \in \{0,1\}} |C_b^{\text{inv}}|} \gamma$$

$$\bar{C}_b^{\text{inv}} = \frac{1}{2} \frac{P_{12}}{P_{12} + I} \rightarrow \text{sgn}(C_b^{\text{inv}}) = \frac{1}{2} \frac{P_{12}}{P_{12} + I} \rightarrow L, \text{ now}$$

$$E_{\text{avg}} = \sum_b \bar{C}_b^{\text{inv}} \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

$$= \sum_b \text{sgn}(C_b^{\text{inv}}) \frac{|C_b^{\text{inv}}|}{\sum_b |C_b^{\text{inv}}|} \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

$$= \sum_b \text{sgn}(C_b^{\text{inv}}) \bar{C}_b^{\text{inv}} \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

scale classical prob. value by inverse of sum of probs

In this form, the decomposition of the error mitigated expectation value is simply a sum over expectation values of Pauli-gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, $\bar{C}_b^{\text{inv}}$ inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.

Probabilistic error cancellation: Derivation

where each $X_m \in \{1, -1\}$ and is distributed to model.

Bernoulli distribution with some probability which can be any valid value and can vary from shot-to-shot m .

Our mitigation estimator is then for M shots:

$$E_M := \gamma \frac{1}{M} \sum_{m=1}^M X_m = \frac{1}{M} \sum_{m=1}^M \gamma \text{sgn}(c_{b_m}^{\text{inv}}) \gamma_{m,b}(\tilde{U}, b_m)$$

rand outcome of m-th shot for bth circ.
↑
noisy circuit
γ_{m,b}($\tilde{U}, b_m$)
↓
Pauli chosen for m-th shot

There are now 2 random processes:

b_m : which pauli b we pick for shot m

γ_{b_m} : which outcome ± 1 we get for b_m circuit of shot m

Unbiased Estimator of the Ideal, noise-free circuit expectation

$$\mathbb{E}[E_M] = \frac{1}{M} \sum_{m=1}^M \mathbb{E}[\gamma X_m] \quad \text{iid rand vars}$$

Unbiased estimator

$$\mathbb{E}[F_M] = \frac{1}{M} \sum_{m=1}^M \mathbb{E}[\gamma X_m] \quad \text{iid rand vars}$$

$$= \mathbb{E}[\gamma X_m] \quad \text{no } X_m \text{ is different}$$

$$= \mathbb{E}[\gamma \text{sgn}(c_{b_m}^{\text{inv}}) Y_{b_m}(\tilde{u}, b_m)] \quad \text{where rand var is } b_m \text{ now, not just } m, \text{ so}$$

$$\approx \mathbb{E}_{b_m}[\gamma \text{sgn}(c_b^{\text{inv}}) Y_b(\tilde{u}, b)] \quad \text{drop notation} \quad \text{Prob}[b] = \bar{c}_b^{\text{inv}}$$

$$= \sum_b \mathbb{E}_Y[\gamma \text{sgn}(c_b^{\text{inv}}) Y_b] \text{Prob}[b]$$

$$= \sum_b \underbrace{\gamma \text{sgn}(c_b^{\text{inv}})}_{\text{post-process of outcome}} \underbrace{\bar{c}_b^{\text{inv}}}_{\text{sample prob}} \underbrace{\mathbb{E}_Y[Y_b]}_{\text{rand outcome}} \longrightarrow$$

$$\approx \sum_b \gamma \text{sgn}(c_b^{\text{inv}}) \frac{\bar{c}_b^{\text{inv}}}{\gamma} \ll P_C | \mathcal{U} \wedge \mathcal{P}_b | \mathcal{P}_0 \gg$$

$$\text{note: } \langle \hat{P}_C \rangle(\tilde{u}, b) = \ll P_C | \mathcal{U} \wedge \mathcal{P}_b | \mathcal{P}_0 \gg \\ = \mathbb{E}[Y_b]$$

$\uparrow$ Rand variable ± 1 for output of the b -th pauli circuit.

$\therefore$ for some classical func of b $f(b)$ which does not depend on the value Y_b but only on the label b :

$$\mathbb{E}[f(b) Y_b] = f(b) \langle \hat{P}_C \rangle(\tilde{u}, b) \quad \text{latko Minev, IBM Quantum (113)}$$

Probabilistic error cancellation: Derivation

$$= \langle \langle P_c | U \Lambda \left(\sum_b c_b^{\text{inv}} P_b \right) | P_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \Lambda^{-1} | P_0 \rangle \rangle$$

$$= \langle \langle P_c | U | P_0 \rangle \rangle$$

$$= \langle \hat{P}_c \rangle (U_{\text{ideal}}) \quad \text{without noise!}$$

unbiased estimator of the true noise-free, ideal value of the circuit



the value y_b but only on the label b :

$$\mathbb{E}[f(b) y_b] = f(b) \langle \hat{P}_c \rangle (U, b)$$

(Optional step) Variance

Variance of E_M

$$\underbrace{V_{\{b_m, x_m\}}}_{\text{with respect to}} [E_M] = \frac{\gamma^2}{M^2} \sum_{m=1}^M V[x_m]$$

$$= \frac{\gamma^2}{M} V[x(U, b)]$$

x_m iid
can drop subscript m
and emphasize b and value x

note the same variance is just
rescaled by γ^2 due to γ

Generalizing: Raveling trajectories with quasiprobabilities

Channel we *want*
to implement

CPTP operation we
can implement

$$\mathcal{C}(\cdot) = \sum_i a_i \mathcal{F}_i(\cdot)$$

Real coefficients, turn
into *quasi-probability*

Putting the following techniques all on the same footing

Technique

Prob. error cancelation (PEC)

Circuit cutting (knitting) of gates

Circuit cutting of wires

Classical sim. algorithms (QP)

Channel $\mathcal{C}$

noise inverse

non-local gate

large unitary

unitary